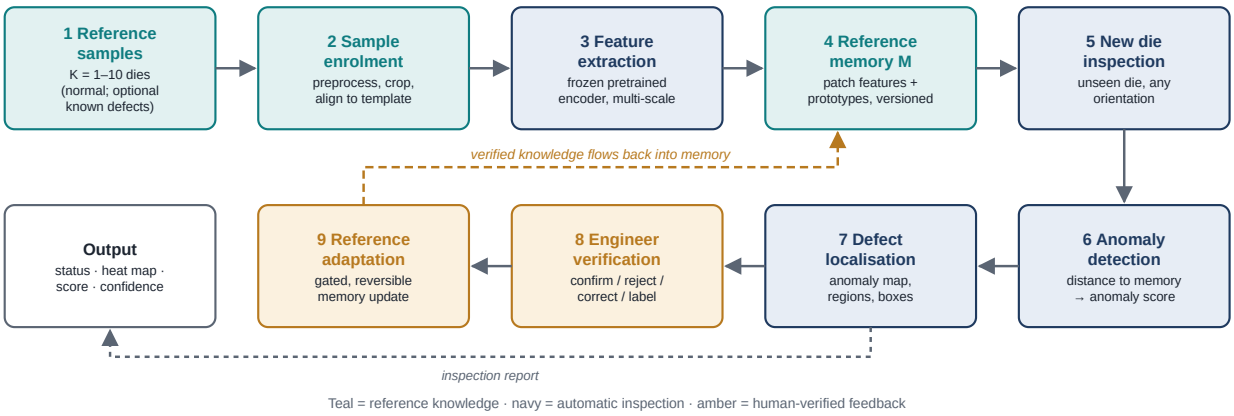
RESEARCH TECHNICAL DOSSIER · EXPLORATORY PROPOSAL

Sample-Guided Semiconductor Wafer Die Inspection: Few-Shot, Rotation-Robust Anomaly Detection, Localisation and Verified Feedback Adaptation

Learn from a few reference dies, inspect unseen dies, locate abnormal regions, separate known from unknown defects, and improve safely from engineer-verified feedback.

Diagram 1 · Complete research concept



Teal = reference knowledge · navy = automatic inspection · amber = human-verified feedback

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Status: proposed study. No experiments have been run. Every die image in this document is a synthetic illustration.

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2 Abstract

Semiconductor dies are inspected to find defects such as particles, scratches, bridges and missing structures. Deep-learning inspectors usually need many labelled defect images for every product, yet new products and process steps start with few labels, and new defect types keep appearing. This dossier proposes and plans a **sample-guided die inspection system**. An engineer gives the system a small set of reference dies (K = 1–10, normally defect-free). A frozen, pretrained visual encoder converts the references into a **reference memory** of patch features. Every new die is geometrically aligned to the references, compared patch by patch with the memory, and given an anomaly score, a pixel-level anomaly map and a status: *normal*, *known defect* or *unknown anomaly*. Engineer-verified decisions update the memory through a gated, versioned and reversible procedure. Model weights are not retrained.

A literature review of 34 papers (2010–2026, weighted to 2024–2026) shows that the building blocks exist separately: memory-based anomaly detection (PatchCore, AnomalyDINO), registration-based and rotation-invariant few-shot detection (RegAD, GraphCore), open-set recognition at wafer-map level, continual learning on fab SEM data, and a 2026 preprint on human-in-the-loop memory correction. We found no published study that evaluates their combination on semiconductor die imagery with **defect-preserving alignment**, **pixel-level known/unknown triage** and **noisy, verified feedback**. We present that combination as a set of testable hypotheses, not as an established result. The dataset audit shows that no public dataset contains patterned dies with pixel masks *and* wafer coordinates. The plan therefore uses Carinthia-S (SEM, expert masks), the IEEE DataPort Wafer Surface Defect set (single-die optical images), MVTec AD (method validation), WM-811K (wafer-level patterns only) and a clearly labelled synthetic patterned-die benchmark. The dossier provides the mathematical formulation, eight experiments with leak-free splits, metrics, baselines, ablations, a 40-week roadmap, a risk register and a runnable toy prototype.

Keywords: few-shot anomaly detection; semiconductor die inspection; reference memory; image registration; rotation robustness; defect localisation; open-set recognition; human-in-the-loop; continual learning.

3 Executive Summary

> **THE IDEA IN ONE SENTENCE**
>
> Give the AI a few sample dies, let it remember what they look like, show it new dies in any orientation, let it point at what looks wrong and say whether it is a known or unknown defect, and let verified engineer feedback improve its memory safely over time.

Table 1. Summary of the proposal at a glance.

Problem	Supervised defect inspectors need many labelled examples per product and defect type; new products, rotated or shifted dies and new defect types break them; image-level labels do not say *where* a defect is.
Proposed approach	Reference memory of frozen pretrained patch features built from K = 1–10 reference dies; geometric alignment that warps the *references*, not the inspected die; nearest-neighbour anomaly maps; three-way triage (normal / known / unknown); gated, versioned memory updates from engineer feedback.
What already exists	Memory-based anomaly detection [1, 2]; strong few-shot variants [3, 4, 5]; registration and rotation-invariant few-shot methods [6, 7]; wafer-map open-set recognition [8]; continual learning on SEM data [9]; training-free memory correction from human feedback on MVTec AD [10].
Defensible gap	No reviewed study evaluates, on semiconductor die imagery: (i) whether alignment or invariance *preserves* real defects; (ii) pixel-level known/unknown triage from normal references plus a small defect memory; (iii) feedback updates under noisy or wrong feedback with quarantine and rollback; (iv) same-wafer vs cross-wafer vs cross-type transfer as separate, explicit splits.
Recommended core contributions	**C2** defect-preserving alignment with a defect preservation ratio (DPR); **C3** reference-guided pixel-level unknown-anomaly triage; **C5** verified feedback adaptation with safeguards, if time allows. C1 is the shared evaluation protocol. C4 (wafer-level context) is an optional extension.
Data reality	No public patterned-die dataset with masks and wafer coordinates was found. The plan combines Carinthia-S (4,591 SEM images with expert masks), the 500-image Wafer Surface Defect die set (boxes), MVTec AD, WM-811K (wafer maps only) and a synthetic patterned-die benchmark, which is always reported separately.
Smallest meaningful prototype	PatchCore-style memory with a WRN-50 or DINOv2 encoder, K-shot enrolment, rotation sweep with and without alignment, DPR test, and leave-one-class-out unknown detection on Carinthia-S. This runs on a single consumer GPU, or slowly on a CPU.
What to implement first	(1) Reproduce PatchCore on MVTec AD. (2) K-shot enrolment curves. (3) Rotation sweep plus "warp the reference" alignment. (4) Carinthia-S localisation. (5) Unknown triage. (6) Feedback simulation.

The document is written for two readers at once. A student new to the field should be able to follow every step, from "what is a wafer?" to the equations and the code skeleton. A faculty member should be able to judge the scientific contribution, the evidence behind each claim and the feasibility of the plan. Published evidence is marked EVIDENCE, and claims of this proposal that still need testing are marked HYPOTHESIS. Synthetic illustrations carry the label SYNTHETIC in their captions.

4 Research Idea Explained Simply

4.1 The story, without jargon

Imagine a quality inspector who has to check thousands of identical chips. Nobody gives them a catalogue of every possible defect. They are shown a few good chips and told: "this is what correct looks like; tell me when something is different, and show me where."

A good human inspector does four things. First, they **study the good samples** carefully and remember the patterns: where the pads are, how the lines run, what the texture looks like. Second, when a new chip arrives, they **turn it to the same orientation** as the samples before comparing. Third, they **compare region by region** and point at places that do not match anything they remember. Fourth, when a supervisor says "that one was fine, it is just a normal variation" or "yes, that is a scratch", they **update what they remember**, carefully, so the same mistake is not repeated.

This research builds an AI system that does the same four things:

1. **Sample-guided learning.** The engineer gives K reference dies (for example 1, 2, 5 or 10). The system stores a compact description of every small region of those dies in a *reference memory*. It does not need thousands of labelled defects.
2. **Geometric alignment.** Each new die may be rotated (0°, 90°, 180°, 270° or any angle), shifted, scaled or lit differently. The system estimates that change and brings the references into the same pose as the new die, without resampling the new die itself, so that tiny defects are not blurred away.
3. **Comparison, scoring and localisation.** Every region of the new die looks for its closest match in the memory. Regions with no close match get a high *anomaly score*. Together these scores form a heat map that shows where the die looks abnormal, how large the area is and how different it is.
4. **Known, unknown and feedback.** A suspicious region is compared with a small library of known defect types. If it matches one clearly, it gets that label. If it matches none, it is reported as an *unknown anomaly*, not forced into the wrong class. An engineer then confirms, rejects, corrects or labels the result, and verified decisions update the memory through safety checks.

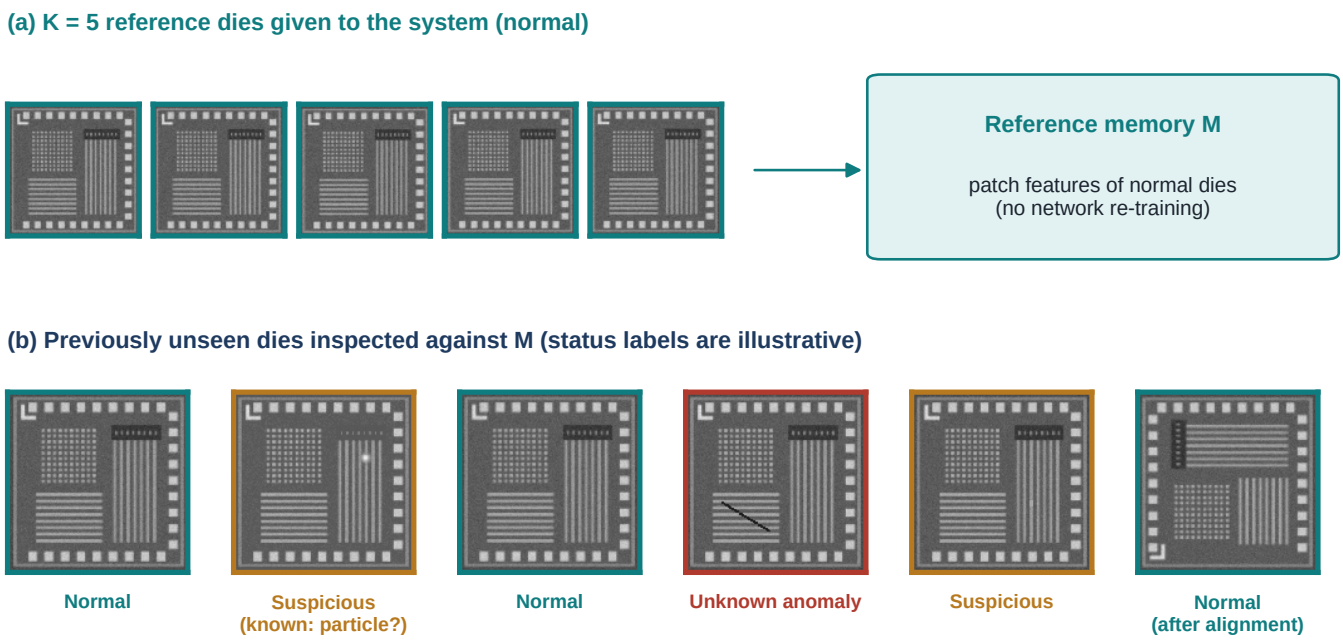


Figure 1. The core idea. (a) A few normal reference dies are turned into a reference memory, with no network re-training. (b) New dies that were not in the reference set are compared with that memory; each receives a status. SYNTHETIC Stylised die renderings; status labels are illustrative, not model outputs.

4.2 What is new, and what is not

None of the individual ideas above is new. Comparing a die with a reference die is the oldest approach in wafer inspection [11, 12, 13]; memory-based anomaly detection with pretrained features is well established [2, 1]; recent methods reach strong results from a single normal image on standard benchmarks [3]. What this research proposes is to **combine and test** these ideas on semiconductor die imagery under conditions that published work leaves out. Does alignment destroy small defects? Can unknown defects be localised at pixel level and kept apart from known ones? Do references from one wafer work on another wafer or another product? Does feedback really help, or does it only memorise the corrected samples? Section 16 sets out exactly which parts are established and which are open.

4.3 The key AI ideas, from zero

The semiconductor concepts (wafer, die, defects, wafer maps, microscopy) are explained in Section 5. If you are completely new to chip manufacturing, read Section 5 first. The cards below explain the machine-learning concepts 10–19 of the research brief. Each follows the same pattern: a simple explanation, a semiconductor example, the technical explanation, and why it matters here.

Concept 10 Why reference-based learning is useful	
SIMPLE	Instead of teaching the system every way a die can be wrong, you show it what right looks like and ask it to spot differences. It is easier to collect good examples than to collect every kind of mistake.
EXAMPLE	A fab starts a new product. On day one there are hundreds of good dies but only a handful of reviewed defects. A reference-based inspector can start working after enrolling a few good dies; a supervised classifier cannot.
TECHNICAL	Reference-based (one-class) methods model the distribution of normal data, p(normal appearance), and flag low-likelihood or high-distance samples. Classical die-to-die inspection subtracts neighbouring dies [11, 13]; modern versions compare deep features [2, 1].
WHY IT MATTERS	It removes the need for large labelled defect sets, which do not exist for new products, and it can, in principle, detect defect types that nobody has seen before.

Concept 11 Few-shot learning	
SIMPLE	Learning from just a few examples, like a person who recognises a new kind of fruit after seeing it two or three times.
EXAMPLE	An engineer marks 2 good dies of a new design. The system must then inspect the remaining 300 dies on the wafer.
TECHNICAL	In K-shot learning only K labelled examples per class (or K normal samples, in few-shot anomaly detection) are available at deployment time. Methods either compare against stored examples (metric learning: prototypical, matching and Siamese networks [14, 15, 16]) or rely on strong pretrained features with nearest-neighbour matching [3], or train a model that generalises across categories [6, 17].
WHY IT MATTERS	This is the core of "give the AI a few sample dies". Experiment A measures how performance changes with K = 1, 2, 5, 10.

Concept 12 Anomaly detection

SIMPLE	Finding things that do not fit the usual pattern, without being told in advance what the unusual things look like.
EXAMPLE	A die with a particle on a metal line looks different from all good dies at that spot, even if no particle was ever shown to the system.
TECHNICAL	Given normal training data only, learn a score $A(x)$ that is low for normal samples and high for anomalies. Families include memory/kNN methods (SPADE, PatchCore), density models (PaDiM, FastFlow), reconstruction and synthetic-anomaly methods (DRAEM), and student–teacher discrepancy (Reverse Distillation, EfficientAD) [2, 1, 18, 19, 20, 21, 22].
WHY IT MATTERS	It is the detection engine of the proposed system. Its main weakness is that "different" is not always "defective": noise, focus or harmless process variation can also look different, so thresholds and verification are essential.

Concept 13 Defect localisation

SIMPLE	Not just saying "this die is bad", but drawing where the problem is.
EXAMPLE	The system highlights a 6 × 2 µm region on the third metal line from the left, so the review engineer can drive the SEM straight to it.
TECHNICAL	Outputs can be a continuous anomaly map (one score per pixel or patch), a thresholded binary mask, bounding boxes around connected regions, or instance masks. Evaluation uses pixel AUROC, AUPRO, Dice/IoU or box IoU, depending on what ground truth exists [23].
WHY IT MATTERS	Localisation is what makes inspection results actionable and checkable. It also lets us measure defect size and test whether alignment weakens small defects.

Concept 14 Geometric registration

SIMPLE	Lining up two pictures so that the same structure is at the same place in both, like laying a transparent sheet over a map and turning it until the roads match.
EXAMPLE	A die image was captured with a 2° stage rotation and a 15-pixel offset. Before comparison, registration estimates this rotation and offset so a via in the reference falls on the corresponding via in the target.
TECHNICAL	Estimate a transform T (rigid, similarity, affine or non-rigid) that maximises similarity between images. Methods include intensity-based (ECC [24], phase correlation [25], Fourier–Mellin [26]), feature-based (SIFT/ORB + RANSAC [27, 28, 29]) and learned (spatial transformers [30], RegAD [6]). A survey is given in [31].
WHY IT MATTERS	Without alignment, a rotated normal die looks abnormal everywhere (Section 10, Diagram 5). With careless alignment, interpolation can blur or cut off the very defects we want to find. Contribution C2 addresses exactly this trade-off.

Concept 15 Rotation robustness

SIMPLE	Giving the same right answer whether the die is upright, sideways or upside down.
EXAMPLE	Dies are loaded onto the tool in different orientations, or images come from tools with different coordinate conventions, so the same die appears at 0°, 90°, 180° or 270°, plus small residual angles.
TECHNICAL	Two strategies exist: *normalise* the pose before comparison (registration), or make the features *invariant* to rotation (rotation augmentation of the memory, or invariant features such as GraphCore's [7]). Invariance can be harmful when orientation itself carries meaning: a horizontal line is normal while a vertical one in the same place is a defect.
WHY IT MATTERS	Few-shot memories cover few orientations, so rotation is a main cause of false alarms. RQ2 compares alignment, augmentation and invariance.

Concept 16 Open-set recognition

SIMPLE	A classifier that is allowed to say "I don't know this one" instead of forcing every input into one of the classes it was taught.
EXAMPLE	The system knows "particle", "scratch" and "bridge". A new residue defect appears after a chemistry change. It should be reported as *unknown*, not labelled "particle".
TECHNICAL	Closed-set classifiers partition the whole feature space among known classes. Open-set recognition bounds each class region and rejects samples that fall outside all of them [32, 33]. Open-set supervised anomaly detection keeps a separate path for unseen anomalies [34].
WHY IT MATTERS	Forcing a new defect into a wrong class hides it from engineers. Contribution C3 and Experiment E test unknown-defect handling.

Concept 17 Continual learning

SIMPLE	Learning new things over time without forgetting old ones.
EXAMPLE	After a recipe change, the normal appearance of a layer shifts slightly. The system should accept the new normal appearance without losing the ability to recognise the old scratch type.
TECHNICAL	Updating network weights on new data alone causes *catastrophic forgetting*. Remedies include regularising important weights (EWC [35]), replaying stored old examples, and expanding memory or prompts (UCAD [36]). Fab studies show fine-tuning forgets earlier SEM defect classes quickly [9].
WHY IT MATTERS	The feedback loop is a form of continual learning. Here it is done by editing a memory rather than retraining weights, which makes every change measurable and reversible.

Concept 18 Reference memory

SIMPLE	The system's notebook of what normal dies (and a few known defects) look like, region by region.
EXAMPLE	For each of 5 reference dies, the memory stores about 1,000 small region descriptions: corners of pads, via grids, line edges and background texture.
TECHNICAL	A set M of feature vectors extracted by a frozen encoder from reference images, often reduced by coreset subsampling [1]. New patches are scored by their distance to the nearest memory entries; a k-nearest-neighbour index (e.g. FAISS [37]) makes this fast.
WHY IT MATTERS	The memory is what "learning from samples" means in this project: enrolment and feedback change the memory, not the network. That is cheap, fast, auditable and reversible.

Concept 19 Human-in-the-loop learning

SIMPLE	People check the AI's answers, and their checked answers are used to make the AI better.
EXAMPLE	A review engineer opens a flagged die, sees it is a known harmless pattern near the seal ring, and clicks "Reject (false alarm)". That decision can teach the memory that this pattern is normal.
TECHNICAL	Human feedback is used to correct labels, select informative samples (active learning [38]) or edit the model directly. Training-free memory correction of a PatchCore model from expert feedback was recently studied on MVTec AD with simulated feedback [10].
WHY IT MATTERS	Fab engineers already review flagged defects. Using their decisions is natural, but wrong feedback can poison the memory. Contribution C5 studies safeguards: quarantine, gates, second review and rollback.

5 Semiconductor and Wafer Background

This section explains the manufacturing side of the problem from zero, as concepts 1–9 of the research brief. It ends with the most important distinction in this dossier: **die-surface images** and **wafer maps** are different kinds of data and answer different questions.

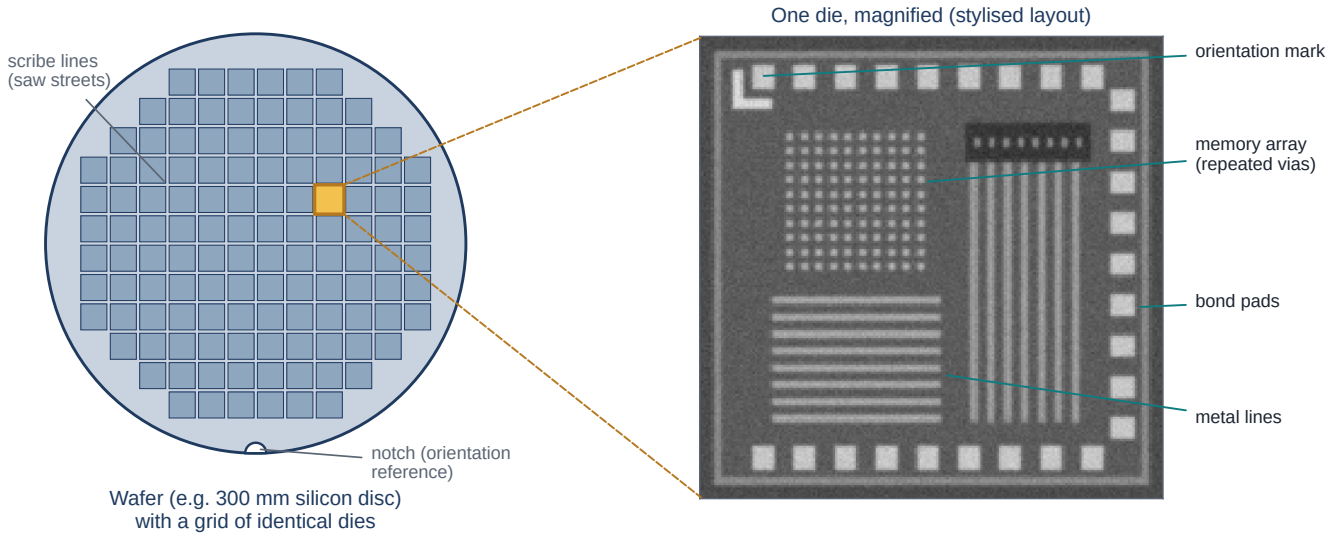


Figure 2. A wafer is a disc carrying a grid of identical dies separated by scribe lines; the notch marks the wafer's orientation. Right: one die, magnified, with typical structures (bond pads, a repeated via array, metal lines, an orientation mark). SYNTHETIC Stylised drawing; real dies contain millions of much smaller structures.

Concept 1 What is a semiconductor wafer?

SIMPLE	A thin, very flat, mirror-polished disc of silicon on which hundreds of identical chips are built at the same time, layer by layer.
EXAMPLE	A 300 mm silicon wafer passes through hundreds of process steps (deposition, lithography, etching, implantation, polishing) over several weeks. At the end it may carry several hundred copies of the same processor die.
TECHNICAL	Wafers are typically 150, 200 or 300 mm in diameter and under 1 mm thick. A notch or flat marks the crystal orientation, so tools can align the wafer. Each process *layer* adds patterned material. Inspection is performed after critical steps, such as after development (ADI) and after etch (AEI) [9].
WHY IT MATTERS	Everything on one wafer shares the same process history, so dies of the same wafer are expected to look alike. This is what makes same-wafer reference inspection plausible (Scenario 1).

Concept 2 What is a die?

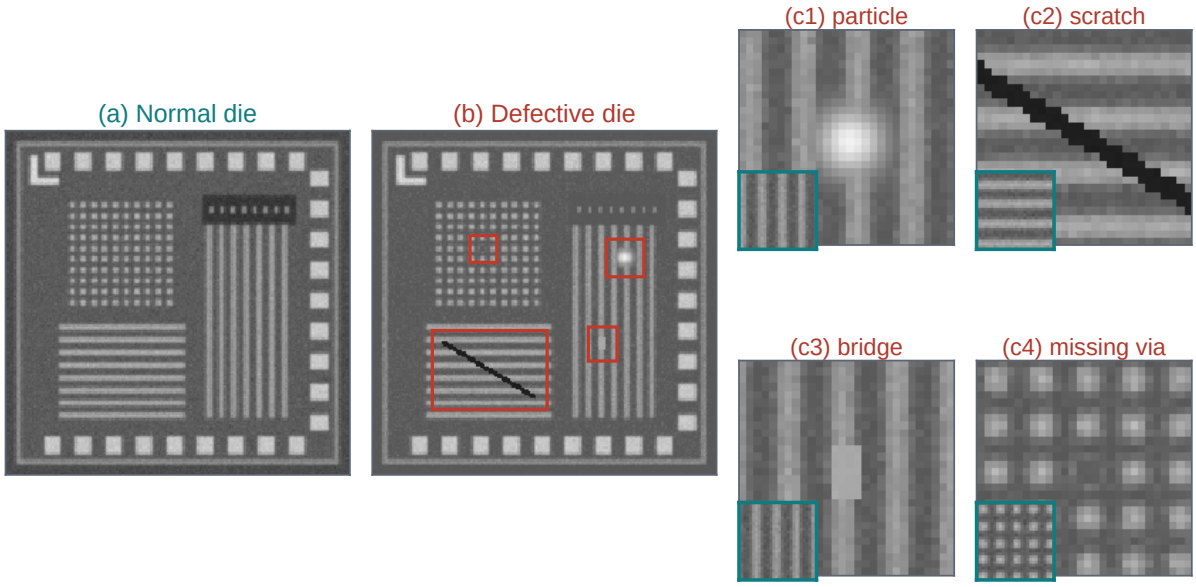
SIMPLE	One rectangular chip on the wafer. After manufacturing, the wafer is sawn along the gaps (scribe lines) and each die becomes an individual chip.
EXAMPLE	A 10 × 10 mm die on a 300 mm wafer gives roughly 600 dies. Each has the same design: the same pads, memory arrays and wiring at the same positions.
TECHNICAL	Dies are laid out on a regular grid with a fixed step (pitch) and are indexed by (row, column). Because all dies of a product share one design, any die can serve as a reference for another. This is the basis of die-to-die inspection [13, 11].
WHY IT MATTERS	Die identity and position (i, j) link the image-level analysis (inside one die) to the wafer-level analysis (across dies). That link is Stage 9 of the method.

Concept 3 What is a normal die?

SIMPLE	A die whose appearance matches the intended design, apart from harmless variation such as slight noise, brightness differences or tiny process fluctuations.
EXAMPLE	Two good dies photographed one after the other never give identical pixels: line widths differ by a few nanometres, the illumination changes slightly and the sensor adds noise. Both are still "normal".
TECHNICAL	"Normal" is a *distribution*, not a single image. A reference memory must capture enough of this natural variation, or it will raise false alarms on good dies. With K = 1 the system sees only one sample of that distribution, which is why calibration at low K is hard (Section 18.6).
WHY IT MATTERS	The quality of the reference set sets the false-alarm rate. A defective die accidentally used as a reference makes that defect "normal". This is why memory quality control and feedback gating exist.

Concept 4 What is a defective die?

SIMPLE	A die with a physical imperfection that may stop it working or shorten its life.
EXAMPLE	A particle lands on two neighbouring metal lines during deposition and later creates a short circuit (a *bridge*). Or a line is interrupted (an *open*), or a via is missing.
TECHNICAL	Defects are classified by type (particle, scratch, bridge, open, residue, stain, missing or extra pattern), by size relative to the design rule, and by whether they are *killer* defects (cause electrical failure) or *nuisance* defects (visible but harmless). Visual anomaly is not the same as electrical failure.
WHY IT MATTERS	The system detects *visual* anomalies. Whether an anomaly is a killer defect requires engineering judgement or electrical test. This is one reason for the human verification step.



large panel: defective crop · small inset: same place on the normal die

Synthetic conceptual illustration — not real SEM / optical data

Figure 3. (a) A normal die and (b) a defective die of the same design, with defect regions boxed. (c1–c4) Magnified crops of four defect types: particle, scratch, bridge between lines, missing via. Each crop has a small inset of the same location on the normal die. Comparing the large crop with its inset is exactly what a reference-based inspector does. SYNTHETIC

Concept 5 What are semiconductor surface defects?

SIMPLE	Unwanted marks, particles, missing or extra material, or damage visible on the die surface.
EXAMPLE	The Carinthia SEM dataset groups real SEM defect images from one production layer into six defect classes [39]; the Wafer Surface Defect dataset uses particle, scratch, stain and liquid residual [40].
TECHNICAL	Surface defects differ in scale (from a few nanometres to millimetres), contrast (bright particles, dark scratches, low-contrast stains) and context (random vs systematic). Systematic defects repeat at the same design location on many dies; a die-to-die comparison cannot see them if every reference has them too.
WHY IT MATTERS	Defect size relative to the encoder's patch size decides whether a defect is detectable at all (Section 18.4), and systematic defects are a known blind spot of reference methods (Section 26).

Concept 6 What is wafer-map analysis?

SIMPLE	Looking at a map of the whole wafer, where each die is a coloured square showing whether it passed or failed, and asking whether the failing dies form a pattern.
EXAMPLE	Failing dies concentrated in a ring near the wafer edge point to an edge-related process problem; a straight line of failing dies suggests a scratch from wafer handling.
TECHNICAL	A wafer map (wafer bin map) is a 2-D grid of categorical values per die. The WM-811K dataset contains 811,457 such maps, of which 172,950 are labelled into nine pattern classes (Center, Donut, Edge-Loc, Edge-Ring, Loc, Near-full, Random, Scratch, None) [41]. Methods include feature-based classifiers, CNNs, few-shot learning [42] and open-set recognition [8].
WHY IT MATTERS	Wafer maps answer "*which* dies fail and in what spatial pattern?", not "*where* on the die is the defect?". They are useful context (Contribution C4), but they cannot evaluate pixel localisation.

Concept 7 What is microscopic die-image inspection?

SIMPLE	Taking magnified photographs of each die, with light or with an electron beam, and checking them for defects.
EXAMPLE	An optical tool scans the wafer quickly and flags candidate locations; an SEM then takes high-resolution images of each candidate for review and classification.
TECHNICAL	Optical inspection is fast with resolution limited by the wavelength of light; scanning electron microscopy (SEM) reaches nanometre resolution but is slow and adds its own noise, charging and contrast effects. SEM-based automatic defect inspection has been reviewed systematically: reference-based detection dominated until about 2020, after which deep learning became more common, especially for classification [11].
WHY IT MATTERS	The proposed method works on die images (optical or SEM). Its claims are tested on SEM (Carinthia-S), optical die images (Wafer Surface Defect) and synthetic patterned dies.

Concept 8 Why is defect detection difficult?	
SIMPLE	Defects are rare, tiny, varied and often look similar to harmless variation. Good dies are never pixel-identical, and the images vary with tool settings.
EXAMPLE	A 3-pixel particle on a 2,000 × 2,000 pixel image must be found, while slight focus changes across the image must *not* be flagged.
TECHNICAL	Main difficulties: (1) extreme class imbalance; (2) small objects relative to the field of view; (3) nuisance variation from noise, illumination, focus and process drift; (4) geometric variation from stage rotation, offsets and scale; (5) periodic patterns that make alignment ambiguous; (6) open-ended defect types; (7) domain shift between tools, layers and products [43].
WHY IT MATTERS	Every stage of the method answers one of these difficulties: preprocessing (3), alignment (4, 5), multi-scale features (2), reference memory (1, 6), open-set triage (6), feedback (7).

Concept 9 Why are defect labels expensive or limited?	
SIMPLE	Labelling a defect needs an expert, an expensive microscope and time, and new defect types keep appearing.
EXAMPLE	To draw a pixel mask of a bridge in an SEM image, an engineer must recognise it, confirm it, and trace it carefully. Many fabs therefore label only boxes or image-level classes [44].
TECHNICAL	Label cost scales with annotation detail (image label < box < pixel mask) and with expert scarcity. Defects of interest are rare by design. Fab data is confidential, so few public datasets exist. Active learning [38] and box-supervised segmentation [44] try to reduce cost.
WHY IT MATTERS	This is the core motivation for few-shot, normal-reference learning, and it limits the experiments we can run (Section 17).

5.1 Two different kinds of data: die images and wafer maps

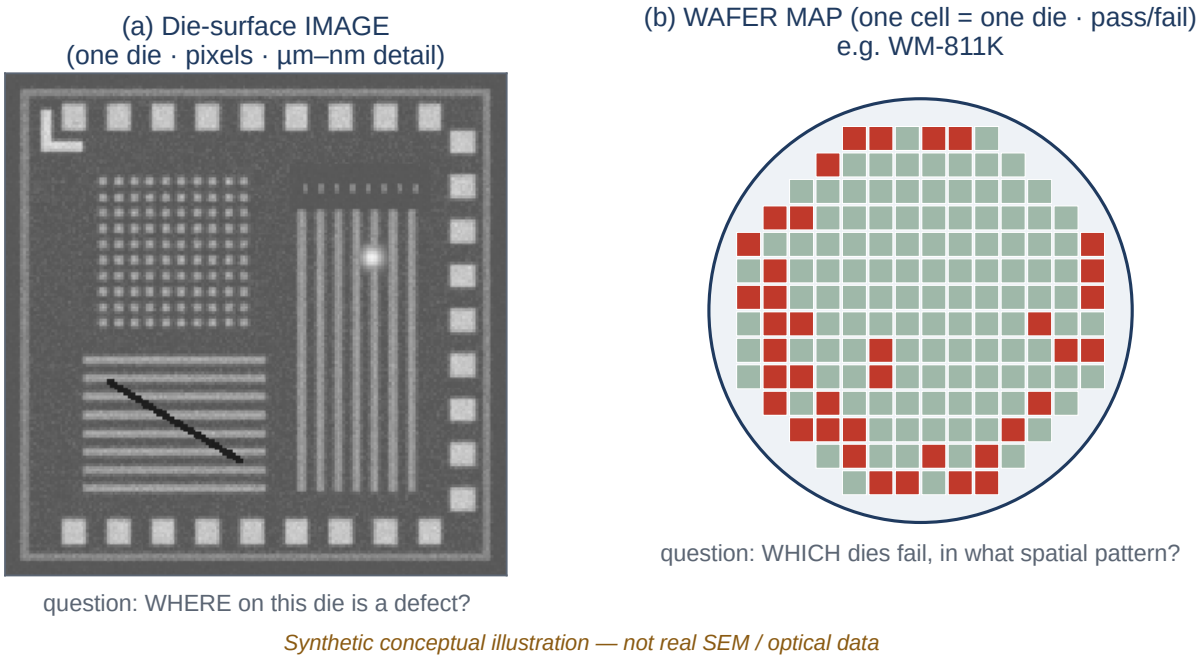


Figure 4. The two data types used in this dossier, which must never be confused. (a) A die-surface image contains pixels and shows *where* on one die something is wrong. (b) A wafer map has one cell per die (pass/fail or bin code) and shows *which* dies fail and in what spatial pattern. WM-811K and MixedWM38 are wafer-map datasets; Carinthia-S and Wafer Surface Defect are image datasets. SYNTHETIC

Table 2. Die-surface images versus wafer maps.

Property	Die-surface image	Wafer map
Unit	Pixels inside one die	One cell per die on the wafer
Question answered	Is there a defect on this die, where, how big?	Which dies fail, and do the failures form a spatial pattern?
Typical source	Optical inspection or SEM review images	Electrical test (bin codes) or aggregated inspection results
Example public data	Carinthia / Carinthia-S (SEM), Wafer Surface Defect (optical)	WM-811K, MixedWM38
Role in this research	Main object of study: detection, localisation, rotation, unknowns, feedback	Optional context: aggregate per-die scores, study spatial clusters (C4)

6 Problem Statement

6.1 Informal statement

Given a small set of reference die images from one product (and, optionally, a few verified defect examples), build an inspector that, for any new die of the same product, possibly from another wafer and in any orientation, outputs: whether the die is normal, contains a known defect type or contains an unknown anomaly; where the abnormal regions are; how large and how severe they are; and how confident the system is. It must improve over time from engineer-verified feedback without retraining the network and without degrading on previously handled cases.

6.2 Formal problem definition (Artifact 1)

FORMAL DEFINITION

Let $R = \{r_1, \ldots, r_K\}$ be K normal reference images of die type d, and $D = \{(x_j, c_j)\}$ an optional small set of verified defect images with class labels $c_j \in \mathscr{C}_{\text{known}}$. A target die image $x \in \mathbb{R}^{H \times W}$ is observed as $x = g(T(x^*))$, where x^* is the die in canonical pose, T an unknown geometric transform (rotation θ, translation t, scale s) and g an unknown photometric change. The task is to learn, **without updating encoder weights**, a function

$$F(x; R, D) \rightarrow (\ S \in \mathbb{R}^{H \times W},\ A \in \mathbb{R},\ y \in \{\text{normal}, \text{known:}c, \text{unknown}\},\ u \in [0, 1]\ )$$

where S is a pixel anomaly map in the coordinates of x, A an image anomaly score, y the triage decision and u a confidence. The function must (i) be invariant in its decisions to T and g within stated ranges, (ii) never assign a region outside all known classes to a known class with high confidence, and (iii) admit an update operator $M_{t+1} = U(M_t,$ verified feedback) that does not reduce performance on a held-out set. Performance is evaluated separately for target dies from the **same wafer**, **other wafers** of the same design and **other die types**.

6.3 Scope and non-goals

- **In scope:** visual anomaly detection and localisation on die images; geometric and photometric robustness; known/unknown triage; memory-based feedback adaptation; optional wafer-level aggregation.
- **Not in scope:** root-cause analysis (the system marks *where* a die looks abnormal, not *why*); electrical failure prediction; production deployment, user interfaces or tool integration; real-time tool control.

7 Research Motivation

Label scarcity is structural, not temporary. Each new product, layer or recipe starts with few reviewed defects, and fab data rarely leaves the fab [11]. Methods that need thousands of labelled defect images per product arrive too late in the product's life to matter.

The industry already thinks in references. Die-to-die and die-to-standard-reference comparison are established industrial practice [13], and reference-based algorithms were the dominant approach in SEM inspection until about 2020 [11]. A reference-memory approach with deep features continues this tradition and keeps its strengths: it is interpretable ("this region does not match any reference region") and needs no defect labels.

Strong few-shot results exist, but not on dies. Training-free patch matching with DINOv2 features reached 96.6% image AUROC from a single normal image on MVTec AD [3]. Whether such numbers survive the conditions of die inspection is unknown: periodic layouts, SEM noise, tiny defects, arbitrary rotations and cross-wafer drift. There is even a semiconductor counter-argument: one group argues that external-reference matching is brittle for IC images because layouts vary and alignment is hard, and proposes reference-free intrinsic normality instead [45]. This tension is worth testing directly.

Rotation and alignment are under-studied for defect preservation. Few-shot methods either learn alignment [6] or become rotation-invariant [7]. Both are evaluated on detection accuracy, not on whether the alignment step itself *weakens* small defects. For inspection this matters: a missed defect is far more costly than a false alarm.

Unknown defects and feedback are where inspection systems fail in practice. Closed-set classifiers force new defect types into old classes. Continual updates forget [9]. A 2026 preprint showed that expert memory corrections can help a PatchCore model, but with clean, simulated feedback on natural-object images, and warned that naive evaluation is inflated by memorisation [10]. Real engineers make mistakes, so safe adaptation under imperfect feedback is an open and practical question.

8 Proposed System Overview

Diagram 1 · Complete research concept

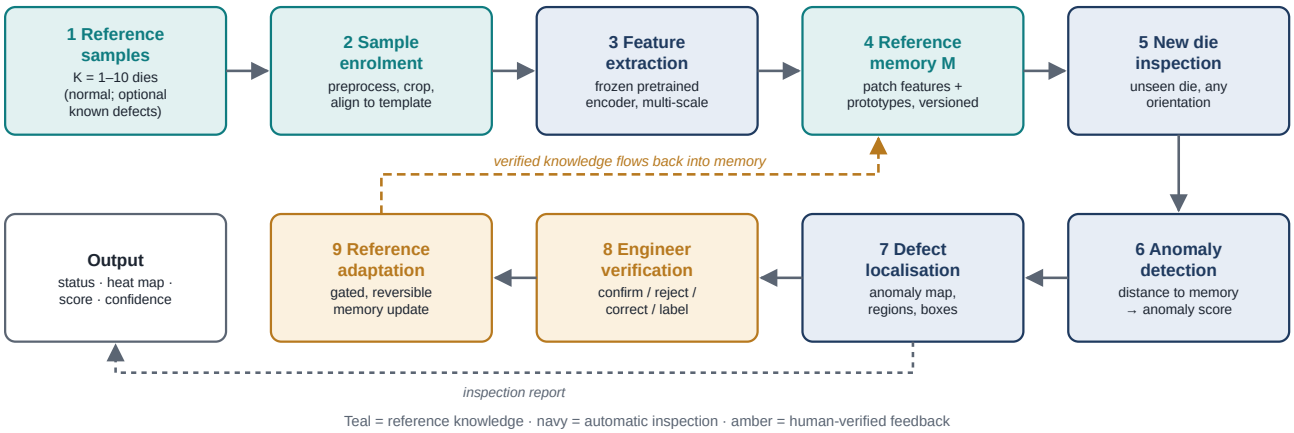


Figure 5. Diagram 1 · Complete research concept. Reference samples are enrolled into a reference memory with a frozen pre-trained encoder (steps 1–4). New dies are inspected against the memory to produce anomaly scores and localised regions (5–7). Engineer verification (8) feeds a gated memory update (9) that improves later inspections. Teal: reference knowledge; navy: automatic inspection; amber: human-verified feedback.

The system has three paths that share one encoder and one memory:

1. **Enrolment path**, run once per die type or recipe. K reference dies are checked for quality, brought into a canonical pose, encoded into multi-scale patch features and stored in the reference memory M after de-duplication and coreset subsampling. Optional: a few verified defect examples per known class are stored as prototypes.
2. **Inspection path**, run for every die. The die is preprocessed and extracted, its pose relative to the references is estimated, and the *references* are warped into the die's frame. Each patch of the die is compared with the memory to give an anomaly map S and an image score A. Thresholding gives regions, which are triaged as known or unknown and given a confidence.
3. **Feedback path**, run for reviewed cases. Engineer decisions pass through a verification gate (quarantine, novelty gate, second review), create a new memory version and are kept only if a regression check on held-out dies passes. Otherwise the memory rolls back.

> **DESIGN PRINCIPLE: LEARN BY REMEMBERING, NOT BY RETRAINING**
>
> The encoder is pretrained by others and stays frozen in the core system. "Learning from samples" and "learning from feedback" both mean *editing the memory*. This keeps enrolment fast, works with K = 1, makes every change auditable and reversible, and avoids catastrophic forgetting of network weights. Fine-tuning the encoder is an optional, later extension (Diagram 8).

9 Detailed System Architecture

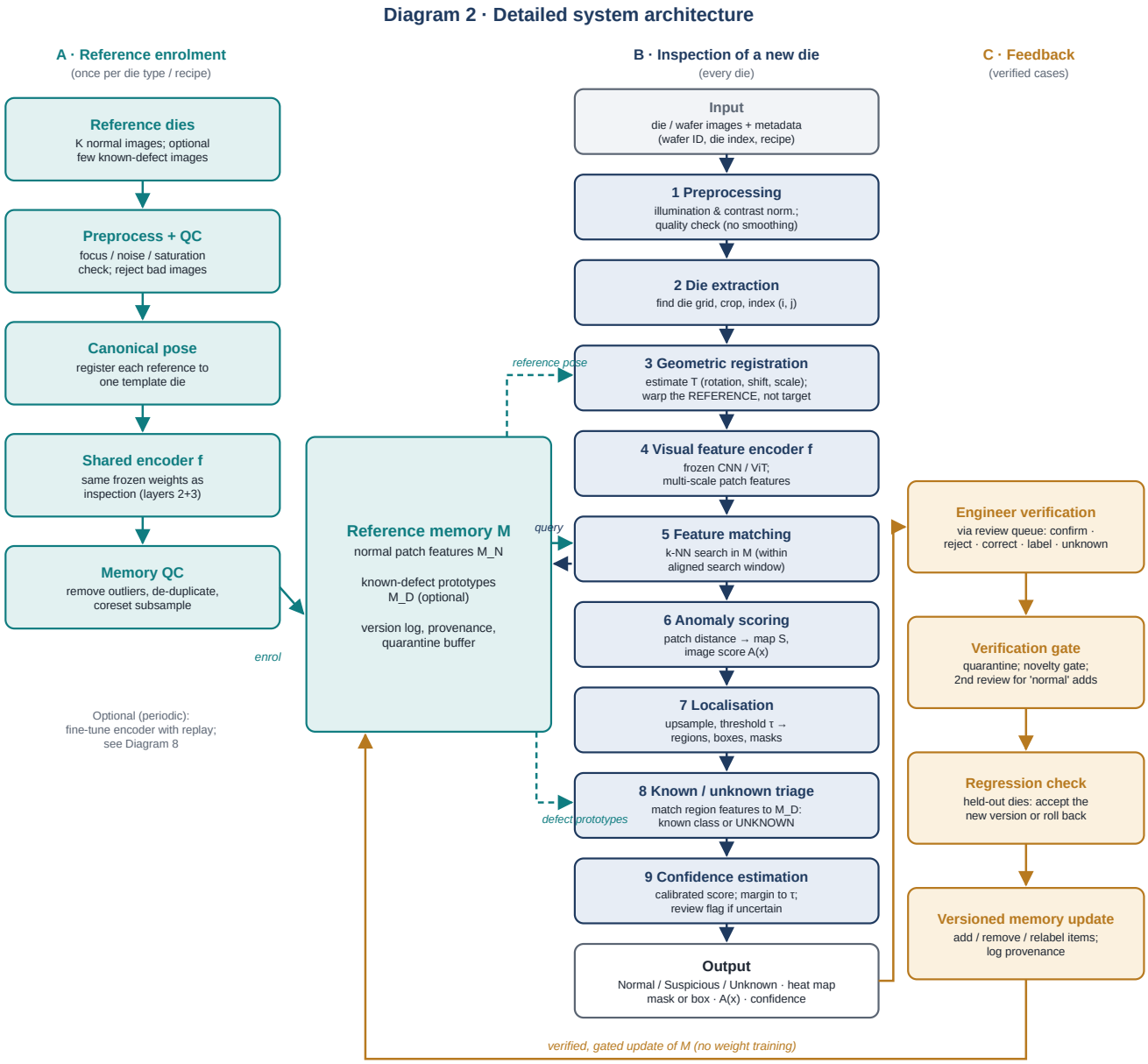


Figure 6. Diagram 2 · Detailed system architecture. Column A enrols reference dies into the reference memory *M*. Column B inspects a new die in ten steps, querying *M* for feature matching and using the references' pose for registration and the defect prototypes for triage. Column C turns engineer decisions into a gated, versioned memory update, routed back into *M* without training any weights.

Table 3. Purpose of every component in Diagram 2.

Component	Purpose	Input → output
Input image collection	Collect die or wafer images with metadata (wafer ID, die index, recipe, tool, magnification).	Files + manifest → image tensors + metadata
Preprocessing	Normalise illumination and contrast; check image quality (focus, noise, saturation). Deliberately no smoothing before scoring.	Raw image → normalised image + quality flag
Die extraction	Locate the die grid in wafer-level images, crop each die, assign (row, column).	Wafer image → die crops + indices
Geometric registration	Estimate rotation, translation and scale relative to the references; warp the *references* to the target frame; mask invalid borders.	Target + references → transform $\hat{T}$, aligned references, validity mask

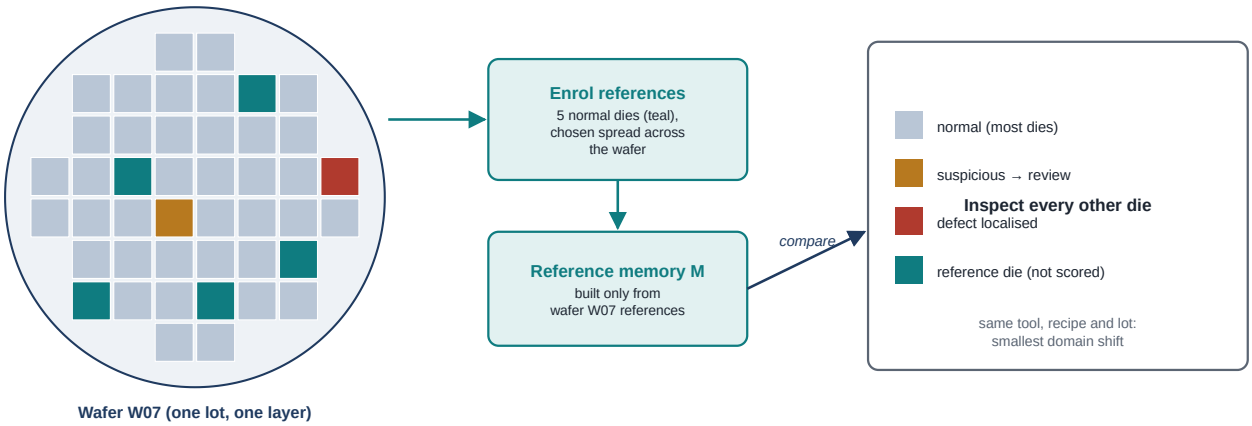
Component	Purpose	Input → output
Visual feature encoder	Turn each image into multi-scale patch features with a frozen pretrained CNN or ViT.	Image → feature map $h \times w \times C$
Reference feature bank (memory)	Store normal patch features (and known-defect prototypes) with provenance and version.	Reference features → index M
Feature matching	Find the nearest memory features for each target patch, optionally within a spatial window around the aligned position.	Target features + M → nearest distances
Anomaly scoring	Convert distances into a patch score map and an image score; normalise with reference statistics.	Distances → S, A
Localisation module	Upsample and threshold the map; extract connected regions, boxes, areas and peak scores.	S → regions, masks, boxes
Optional defect classifier	Compare region features with known-defect prototypes.	Region features → nearest class + distance
Unknown anomaly detection	Assign "unknown" when a region is anomalous but far from every known class.	Distances → {known:c, ambiguous, unknown}
Confidence estimation	Report a calibrated confidence; send low-margin cases to review.	Scores → confidence u, review flag
Feedback mechanism	Record engineer actions (confirm, reject, correct, label, unknown) with reviewer ID and memory version.	Review → feedback records
Reference-memory update	Apply gated, versioned edits to M; regression-check and roll back if needed.	M_t + feedback → M_{t+1}

10 Complete Visual Workflow

This section shows each scenario and mechanism of the system as a diagram. Diagrams 5 and 6 are computed by the toy prototype in `prototype/toy_reference_memory.py` on synthetic dies. They show how the pipeline behaves, not how well it performs.

10.1 Scenario 1: same-wafer inspection

Diagram 3 · Same-wafer inspection (Scenario 1)



Evaluation must keep reference dies out of the test set and report results per wafer (Section 20).

Figure 7. Diagram 3 · Same-wafer inspection. Five normal dies spread across wafer W07 are enrolled as references; every other die on the same wafer is inspected against the resulting memory. This scenario has the smallest domain shift (same lot, tool and recipe) and is the first one to validate.

10.2 Scenario 2: cross-wafer and cross-type inspection

Diagram 4 · Cross-wafer and cross-type generalisation (Scenario 2)

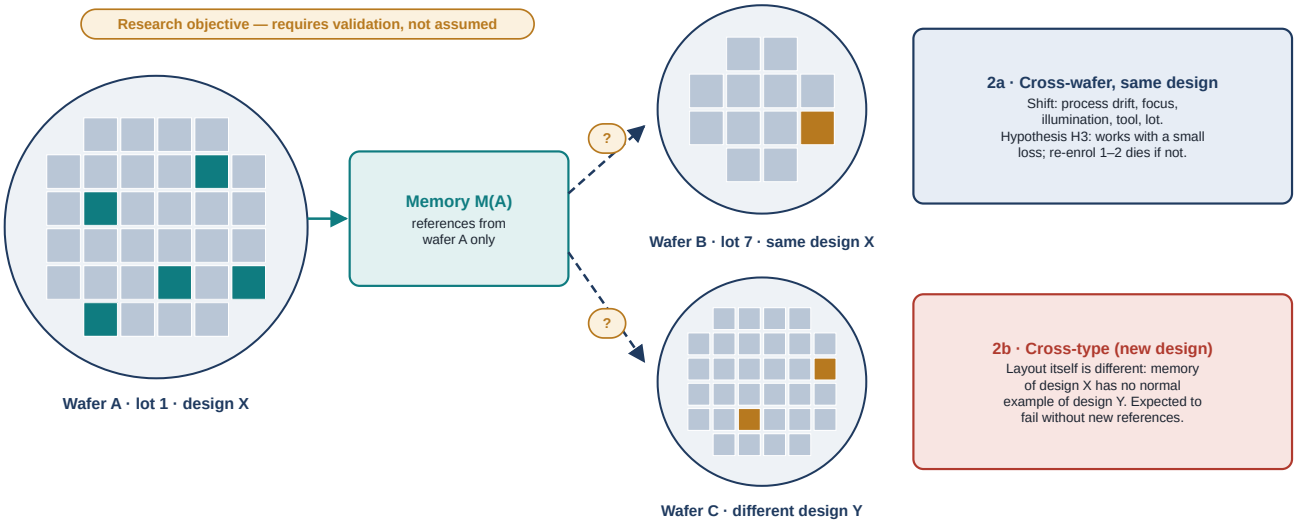


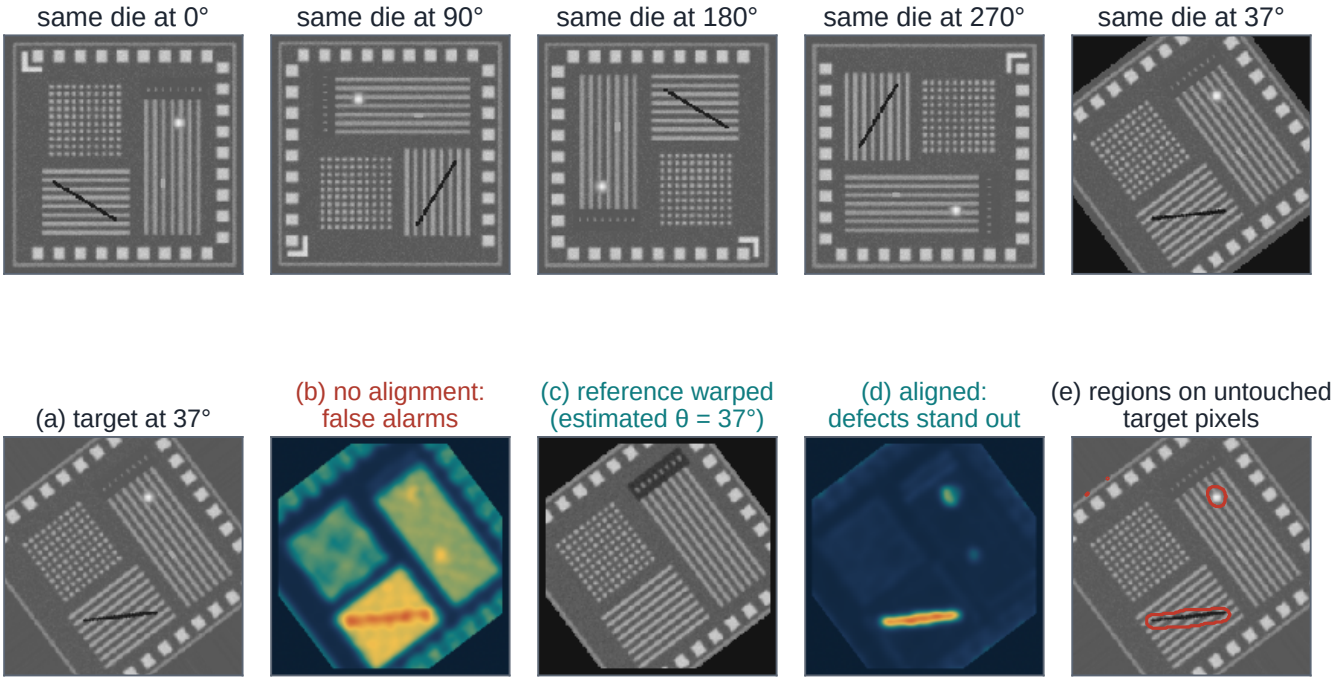
Figure 8. Diagram 4 · Cross-wafer generalisation. References from wafer A are used to inspect wafer B (same design, different lot: Scenario 2a) and wafer C (different design: Scenario 2b). Both are research objectives that require validation. We expect 2a to work with some loss, and 2b to fail without re-enrolling references of the new design.

SAME MECHANISM, DIFFERENT EXPECTATIONS

The same algorithm runs in all scenarios. What changes is how far the target lies from the reference distribution. In Scenario 2b the memory contains no normal example of the new layout, so every structure of design Y looks "abnormal". A reference memory cannot generalise to a design it has never seen; it needs new references. The research question for 2b is therefore not "does it transfer?" but "how few new references are needed, and does anything learned from design X (thresholds, defect prototypes, encoder adaptation) carry over?" (RQ3).

10.3 Rotation-robust inspection

A die can reach the inspector in many poses (see the gallery, Figure 19). Diagram 5 shows, on the toy pipeline, what happens when a rotated die is compared with an unrotated memory, and what alignment changes.



Synthetic conceptual illustration — not real SEM / optical data

Figure 9. Diagram 5 · Rotation-robust inspection. Top: the same die at 0°, 90°, 180°, 270° and 37°. Bottom: (a) a target at 37°; (b) comparing it with an unrotated memory makes every structure look anomalous; (c) the rotation is estimated and the references are warped into the target frame; (d) the aligned comparison leaves only the defects; (e) regions are drawn on the untouched target pixels. SYNTHETIC Toy pipeline on synthetic dies; not an experimental result.

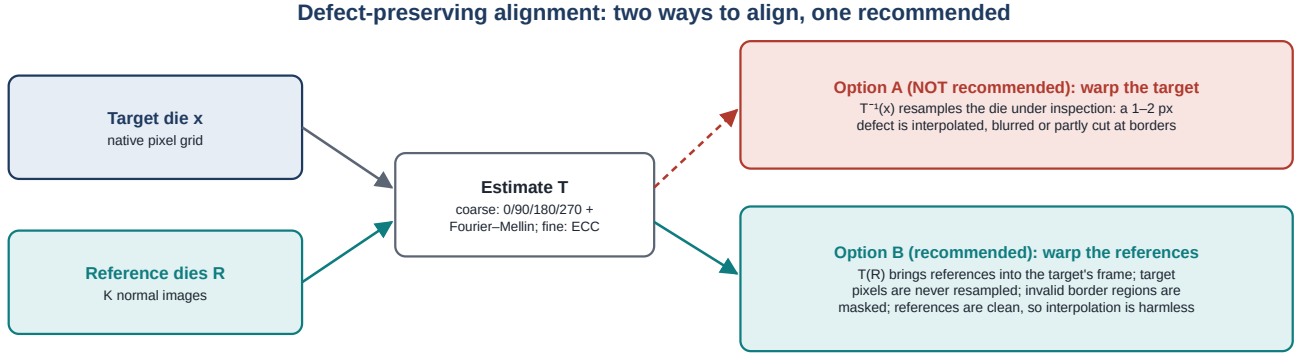
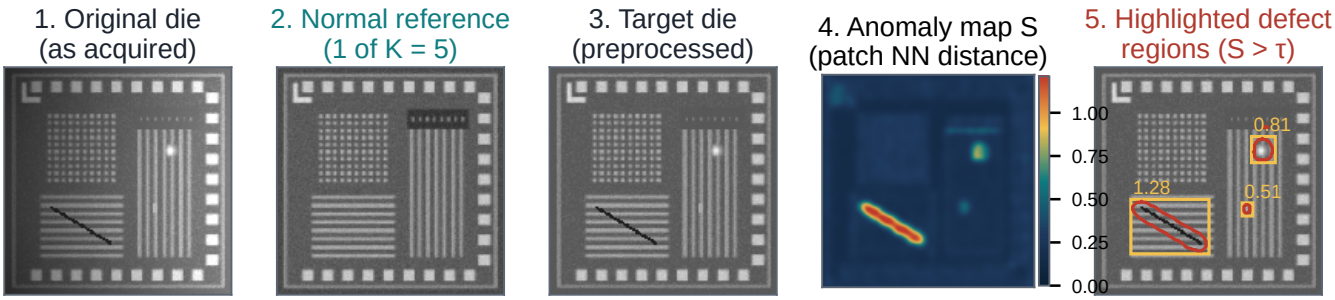


Figure 10. Two ways to apply an estimated transform. Warping the target resamples the image under inspection and can blur or cut small defects. Warping the references keeps the target's pixels intact. This proposal adopts option B and measures the difference with the defect preservation ratio (Section 19.6).

10.4 Defect localisation



Synthetic conceptual illustration — not real SEM / optical data

Figure 11. Diagram 6 · Defect localisation. (1) Original die as acquired, with an illumination gradient; (2) a normal reference; (3) target after preprocessing; (4) anomaly map S = distance of each patch to its nearest reference patch; (5) final highlighted regions where $S > \tau$, with bounding boxes and peak scores. SYNTHETIC Computed by the toy prototype; τ is the 99.9th percentile of scores on held-out synthetic normal dies.

10.5 Feedback-driven adaptation

Diagram 7 · Feedback-driven adaptation cycle

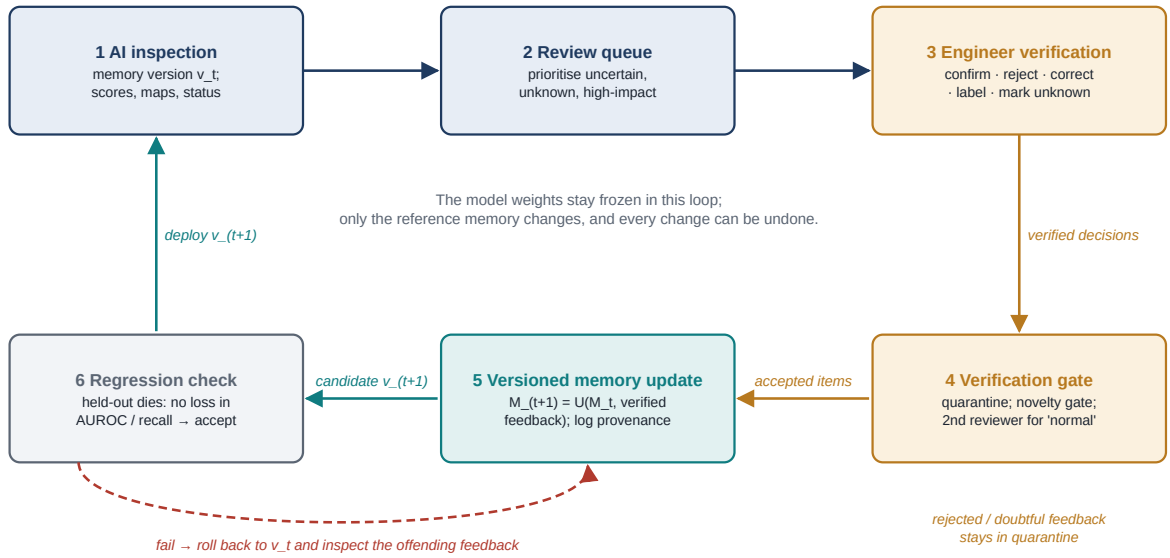
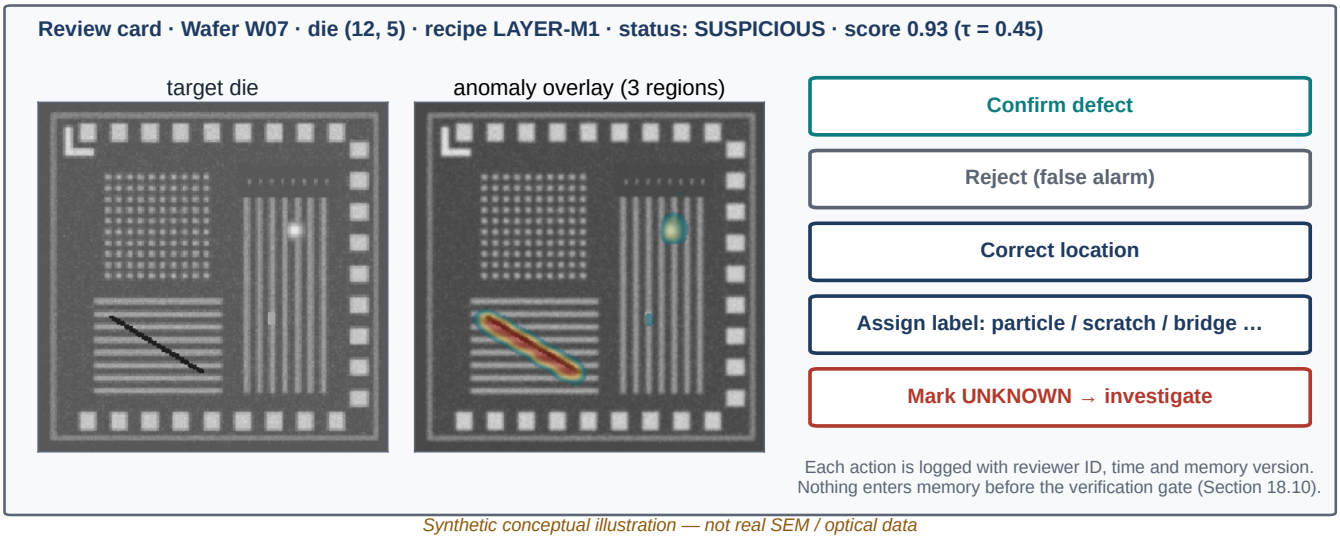


Figure 12. Diagram 7 · Feedback-driven adaptation. Inspection results go to a prioritised review queue. Engineer decisions pass a verification gate, produce a candidate memory version and are deployed only if a regression check on held-out dies passes; otherwise the system rolls back.



Synthetic conceptual illustration — not real SEM / optical data

Figure 13. What the engineer sees: a review card with the die, the anomaly overlay and five actions (confirm, reject, correct location, assign label, mark unknown). It illustrates the information the feedback study needs; building such an interface is not part of this research. SYNTHETIC

10.6 Training vs enrolment vs inference

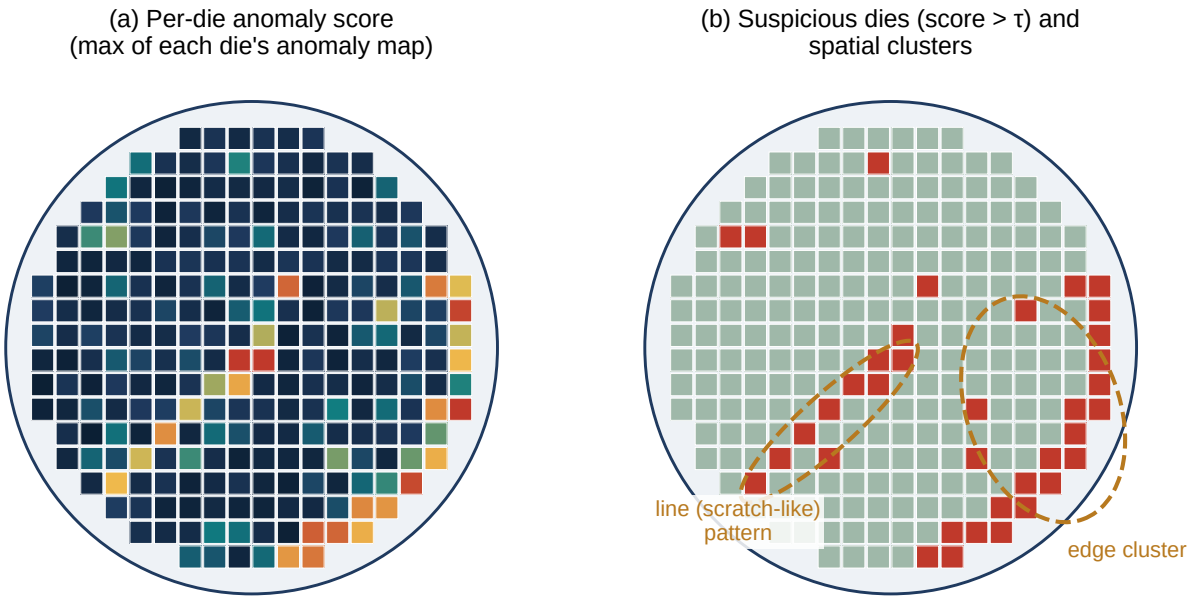
Diagram 8 · Training vs enrolment vs inference vs update

Stage	What happens	How often	What changes	Compute
Offline pre-training WEIGHTS FROZEN	Encoder (e.g. WRN-50, DINOv2) trained on large public data by its authors; we download the weights	once (not by us)	nothing in our project	none
Reference enrolment	K reference dies → preprocess → align → features → memory M (with QC)	per die type / recipe	memory M	seconds–minutes, GPU/CPU
Inference (inspection)	new die → align → features → compare with M → map S, score A, status	every die	nothing (read-only)	ms–s per die
Feedback update	verified feedback → gate → versioned edit of M → regression check	per verified batch	memory M (versioned)	seconds
Optional fine-tuning ADVANCED / OPTIONAL	periodic adaptation of encoder with replay of old data; only after validation	rare (e.g. per quarter)	encoder weights	hours, GPU

Core project = rows 2–4 (no gradient training). Row 5 is an optional extension that must pass the same regression checks.

Figure 14. Diagram 8 · Training vs inference. Offline pre-training is done once by the encoder's authors. In the core system, only enrolment and feedback change anything, and they change only the memory. Optional fine-tuning (bottom row) changes weights and must pass the same regression checks.

10.7 Wafer-level view



Synthetic conceptual illustration — not real SEM / optical data

Figure 15. Aggregating die-level results into a wafer map. (a) Each die coloured by its image anomaly score (the maximum of its anomaly map). (b) Dies above the threshold, with two spatial clusters: an edge cluster and a scratch-like line. Image-level defects (inside a die) and wafer-level patterns (across dies) are different findings and are reported separately. SYNTHETIC

11 Research Objectives

1. **O1 · Few-shot enrolment.** Build and evaluate a reference-memory inspector that works from K = 1, 2, 5, 10 reference dies without retraining, on SEM and optical die data.
2. **O2 · Defect-preserving geometric robustness.** Make decisions stable under 0–360° rotation, translation, ±10% scale and illumination change, while measurably preserving small defects; compare alignment, rotation-augmented memory and invariant features.
3. **O3 · Localisation.** Produce pixel- or region-level anomaly maps and evaluate them against expert masks or boxes, broken down by defect size.
4. **O4 · Unknown-defect handling.** Separate normal, known-class and unknown regions; never force unfamiliar anomalies into known classes with high confidence.
5. **O5 · Transfer scenarios.** Quantify same-wafer, cross-wafer and cross-type performance as separate, explicit evaluation splits.
6. **O6 · Safe feedback adaptation.** Show whether verified feedback improves later inspections on held-out data, and how quarantine, gating and rollback limit damage from wrong feedback.
7. **O7 · Reproducibility.** Release code, configurations, split manifests and the synthetic benchmark, and report negative results.

12 Research Questions and Hypotheses

Each question is paired with a falsifiable hypothesis, its measurement and the experiment that tests it (Artifact 12). Numerical targets are HYPOTHESIS values to be revised after the pilot study; they are not predictions of results.

Table 4. Research questions, hypotheses, measurements and experiments (Artifact 12).

ID	Research question	Hypothesis (falsifiable)	Measured by	Exp.
RQ1	How does reference-based anomaly detection performance change with K = 1, 2, 5, 10 normal reference dies?	H1: Image-level AUROC and pixel AUPRO increase monotonically with K, with diminishing gains after K = 5. Rejected if the K = 10 result is within the 95% confidence interval of K = 1 for most classes.	Image AUROC, AUPRC, pixel AUROC, AUPRO; 5 random reference draws per K	A
RQ2	Does explicit alignment of reference memory to the target die reduce rotation-induced false alarms compared with (a) no alignment and (b) rotation-augmented memory, without weakening true defects?	H2: Alignment-based inspection keeps AUROC within 2 points of the 0° result across a full 0–360° sweep, while the unaligned baseline degrades; defect preservation ratio (DPR) stays ≥ 0.9 for alignment and is lower for target-warping pipelines.	AUROC vs angle curve; DPR; false alarms on normal dies per angle	B
RQ3	Do references enrolled from some dies of a wafer transfer to the remaining dies of the same wafer, and to dies of other wafers or other die types?	H3: Same-wafer transfer is within 3 AUROC points of in-sample performance; cross-wafer transfer loses more; cross-type transfer without new references fails (near chance), and is recovered by enrolling K ≥ 2 references of the new type.	AUROC / AUPRO per split type; performance gap	C, D
RQ4	Can a reference memory plus a small known-defect memory separate normal, known-class and unknown anomalies at pixel / region level?	H4: In leave-one-class-out tests on Carinthia-S, held-out classes are flagged as anomalous (detection AUROC ≥ 0.85, target to be calibrated in the pilot) and are labelled 'unknown' more often than they are forced into a known class.	Unknown-detection AUROC; open-set F1; known-class accuracy; forced-assignment rate	E
RQ5	How accurately do reference-based anomaly maps localise defects compared with expert masks?	H5: Multi-scale patch features localise better than single-scale features, especially for defects smaller than one patch.	Pixel AUROC, AUPRO, Dice/IoU at the validation threshold, by defect-size bin	F, H
RQ6	Do verified feedback updates of the reference memory improve later inspection on held-out dies, without contamination or forgetting?	H6: Gated memory updates improve held-out AUROC and reduce false alarms; with 10% wrong feedback, quarantine + gate limits degradation to below 1 AUROC point; ungated updates degrade more.	ΔAUROC on held-out set; false-alarm rate; forgetting rate; contamination rate	G
RQ7 (optional)	Does combining per-die anomaly scores with wafer-level spatial context improve the ranking of truly defective dies?	H7: Adding a spatial-neighbourhood term improves die-level AUPRC on synthetic wafers with clustered defects, and does not harm it on wafers with random defects.	Die-level AUPRC; per-pattern breakdown	H (wafer-context ablation)

13 Existing Research and Related Papers

13.1 How the review was conducted

Cut-off date: 2 November 2026. **Sources searched:** IEEE Xplore, SpringerLink, ScienceDirect, CVF Open Access, arXiv, ACL Anthology, SPIE, Zenodo and IEEE DataPort records, through publisher-indexed web search. **Search directions:** the 15 directions of the research brief, including few-shot semiconductor defect detection, reference-based and normal-only anomaly detection, SEM defect segmentation, rotation invariance, registration, open-set and unknown defects, continual and incremental learning, human-in-the-loop inspection, cross-domain transfer, wafer-level spatial analysis and self-supervised semiconductor inspection.

> **LIMITATION OF THIS REVIEW**
>
> Direct access to arXiv, Zenodo and Crossref was blocked in the environment where this dossier was produced, so paper details were verified through search results quoting the official records (publisher pages, CVF, ACL Anthology, arXiv listings), supplemented by an earlier verification log in this repository. Every reference carries a verification code (Section 29). Numbers are quoted only where the abstract or official record states them. Before submitting any derived paper, open every link and check every quoted number against the full text. This is not yet a PRISMA-style systematic review; milestone 1 of the roadmap turns it into one.

13.2 The methods you need to know, explained

The research brief lists 18 methods. They fall into six families. For each family: how it works in simple terms, the technical core, and the verdict for this project. Selection: CORE = part of the proposed system; BASELINE = compared against; OPTIONAL = extension; no tag = background only.

Family 1: Memory and nearest-neighbour anomaly detection

Feature-based nearest-neighbour matching CORE. *Simply:* describe every small region of the normal images with a vector of numbers; for a new image, ask whether each region has a close twin among the normal regions. *Technically:* anomaly score = distance to the k-th nearest neighbour in a bank of pretrained features. No training; robust to small data; cost grows with the bank, which is controlled by subsampling.

SPADE [2] BASELINE. Retrieves the K most similar normal images using global features, then does pixel-wise nearest-neighbour matching in their multi-scale feature pyramids. It introduced the "correspondence with similar normal images" idea that this project uses for alignment-aware matching.

PatchCore [1] CORE. Builds a memory bank of locally aggregated mid-level patch features (Wide-ResNet-50 layers 2 and 3), reduces it by greedy coreset subsampling (keeping a subset that covers the feature space), and scores each test patch by its nearest-neighbour distance; the image score is the highest patch score, re-weighted by local density. Reported up to 99.6% image AUROC on MVTec AD. It is the template for the proposed reference memory.

AnomalyDINO [3] BASELINE. The same patch-matching principle with DINOv2 ViT features [46], designed for the few-shot case; 96.6% 1-shot image AUROC on MVTec AD. The strongest simple few-shot baseline.

Family 2: Distribution, flow and reconstruction models

PaDiM [18] BASELINE (aligned data only). Fits a Gaussian per patch position and scores by Mahalanobis distance. It needs many normal images to estimate covariances and assumes alignment, so it serves as a "position-aware" comparison after registration, not as a few-shot method.

FastFlow [19]. Learns a normalizing flow that maps normal features to a simple Gaussian, giving an exact likelihood per location. Needs many normal images; background only.

DRAEM [20] OPTIONAL. Trains a reconstruction network and a segmentation network on normal images with pasted synthetic anomalies. Not few-shot, but its anomaly-synthesis recipe is reused for the defect-injection benchmark.

Family 3: Student–teacher methods

Reverse Distillation [21] and **EfficientAD** [22] BASELINE (full-data upper bound). A student network learns to reproduce a teacher's features on normal images; disagreement marks anomalies. Both need per-category training on many normals. They show what is achievable when plenty of normal data exist, and EfficientAD sets a latency reference.

Family 4: Metric-based few-shot learning

Siamese networks [16]: two weight-sharing encoders embed two images; a learned distance says whether they match. **Matching networks** [15]: classify a query by attention-weighted similarity to a labelled support set. **Prototypical networks** [14] CORE (for known-defect classes): each class is represented by the mean embedding (prototype) of its few examples, and a query takes the class of the nearest prototype. These are classification methods: they need examples of every class and are closed-set by design. In this project prototypes are used *only* for the optional known-defect memory, and always after the normal-reference test (Section 18.8).

Family 5: Encoders: CNNs, Vision Transformers and foundation features

CNN feature extraction (ResNet, Wide-ResNet, EfficientNet, ConvNeXt [47, 48, 49, 50]) CORE: convolution layers produce feature maps whose mid layers combine local detail with some context. **Vision Transformers** [51] split images into patches (e.g. 14 × 14 px for DINOv2) and relate them with self-attention; self-supervised DINOv2 features transfer well without fine-tuning [46] CORE. **CLIP-based** methods (WinCLIP [5], InCTRL [17], SEM-CLIP [52]) add language priors BASELINE. The encoder choice is an experimental variable (Section 18.4).

Family 6: Geometry, adaptation, continual and open-set learning

Image registration methods [31, 24, 26, 25, 27, 28, 29] CORE: estimate the transform between two images from intensities (phase correlation, Fourier–Mellin, ECC) or from matched keypoints (SIFT/ORB + RANSAC). **Spatial Transformer Networks** [30]: a differentiable module that predicts and applies a warp inside a network; used by RegAD [6] for feature-level registration BASELINE. **Rotation-invariant features** (GraphCore [7]) BASELINE.

Test-time adaptation (e.g. Tent [53], which updates normalisation parameters on unlabelled test batches by entropy minimisation) OPTIONAL: attractive for cross-wafer drift, but it adapts to whatever the test batch contains, including defects, so for anomaly detection it needs care. **Continual learning and memory replay** (EWC [35], replay buffers [54], prompt memories [36]) OPTIONAL for encoder fine-tuning; the core system uses memory edits instead. **Open-set anomaly detection** (open-set recognition [32, 33], open-set supervised AD [34]) CORE idea for Stage 8.

13.3 Method comparison and selection

Table 5. Comparison of the investigated methods for the requirements of this project. "K" = number of normal samples needed in practice.

Method	Training needed	Works at K ≤ 10	Localises	Rotation	Update by memory edit	Role here
kNN feature matching / SPADE	None	Yes	Yes	No (needs alignment)	Yes	Core idea / baseline
PatchCore	None	Yes (low-shot study)	Yes	No	Yes	**Core baseline**
AnomalyDINO	None	Yes (1-shot)	Yes	Not studied	Yes	**Core baseline**
PaDiM	Statistics only	Weak (covariance)	Yes	Needs alignment	Partial	Aligned-data comparison
FastFlow	Flow per category	No	Yes	No	No	Background
DRAEM	Per category	No	Yes	No	No	Synthetic-defect recipe
Reverse Distillation	Per category	No	Yes	No	No	Full-data upper bound
EfficientAD	Per category	No	Yes	No	No	Latency reference
RegAD (STN registration)	Meta-training on other categories	Yes	Yes	Learned alignment	Support set swap	Baseline (C1, C2)

Method	Training needed	Works at K ≤ 10	Localises	Rotation	Update by memory edit	Role here
GraphCore	Graph module	Yes	Yes	Invariant	Yes	Baseline (C2)
WinCLIP / InCTRL	None / generalist training	Yes	Yes / No	No	Support set swap	Baseline (cross-type)
Prototypical / Matching / Siamese	Episodic training	Yes (per class)	No	No	Prototype update	Known-defect memory only
Classical registration	None	n/a	n/a	Yes (estimation)	n/a	**Core (Stage 3)**
Test-time adaptation	Online	n/a	n/a	No	No (changes weights)	Optional (cross-wafer)
Continual learning / replay	Retraining	n/a	Depends	No	No	Optional (fine-tuning)
Open-set recognition / DRA	Depends	Partial	Mostly no	No	Partial	Core idea (Stage 8)

SELECTION

Core pipeline: classical registration (coarse rotation search + Fourier–Mellin + ECC refinement) → frozen multi-scale encoder (WRN-50 and DINOv2 ViT-S/14 compared) → PatchCore-style memory with alignment-aware nearest-neighbour search → prototype-based known-defect memory with open-set rejection → gated memory updates. **Baselines:** PatchCore, AnomalyDINO, RegAD, GraphCore (or its "augmentation + PatchCore" variant, which is simpler to reproduce), WinCLIP, and a supervised segmentation upper bound where masks exist. **Not implemented:** FastFlow, Matching and Siamese networks, Tent (unless time allows).

13.4 Paper comparison matrix

The matrix summarises the 34 analysed papers against the five capabilities of the proposed system. Values are assigned only where the paper (abstract, official record or well-documented method) supports them; "Unclear" means we could not confirm it from the sources available. Column definitions: **Few-shot** Yes = main evaluation with ≤ 10 normal or labelled samples per category; **Rotation** Yes = rotation explicitly modelled *and* evaluated; **Localisation** Yes = pixel or region output evaluated; **Unknown** Yes = separates unknown anomalies from known classes, Partial = flags any deviation from normal but has no known-class concept; **Feedback** Yes = learns from human or verified feedback, Partial = continual learning without human verification.

Table 6. Paper comparison matrix (34 papers). Values: Yes / Partial / No / Not evaluated / Unclear.

Paper	Year	Dataset	Method	Few-shot	Rotation	Localisation	Unknown defects	Feedback learning	Limitation
SPADE [2]	2020	MVTec AD; STC	kNN on pretrained feature pyramids	Not evaluated	No	Yes	Partial	No	Memory and search cost; no geometric handling
PaDiM [18]	2021	MVTec AD; STC	Per-position Gaussian, Mahalanobis	No	Partial	Yes	Partial	No	Needs alignment and many normals per position
PatchCore [1]	2022	MVTec AD (+2)	Coreset patch memory + NN distance	Partial	No	Yes	Partial	No	Uncalibrated; no rotation handling
DRAEM [20]	2021	MVTec AD	Synthetic anomalies; recon. + segm. nets	No	No	Yes	Partial	No	Per-category training; synthetic realism
Reverse Distillation [21]	2022	MVTec AD	Teacher encoder, student decoder	No	No	Yes	Partial	No	Needs per-category training data
FastFlow [19]	2021	MVTec AD	2D normalizing flow on features	No	No	Yes	Partial	No	Density fitting needs many normals
EfficientAD [22]	2024	MVTec AD, VisA, LOCO	Light student–teacher + autoencoder	No	No	Yes	Partial	No	Per-category training

Paper	Year	Dataset	Method	Few-shot	Rotation	Localisation	Unknown defects	Feedback learning	Limitation
RegAD [6]	2022	MVTec AD, MPDD	Learned feature registration (STN)	Yes	Partial	Yes	Partial	No	Learned alignment, not die-specific; no defect-preservation test
GraphCore [7]	2023	MVTec AD, MPDD	Rotation-invariant graph features	Yes	Yes	Yes	Partial	No	Invariance may hide orientation-specific defects
WinCLIP [5]	2023	MVTec AD, VisA	CLIP windows + text prompts	Yes	No	Yes	Partial	No	Natural-image language priors
InCTRL [17]	2024	9 datasets	In-context residuals vs few-shot prompts	Yes	No	No	Partial	No	Image-level only
ResAD [55]	2024	MVTec AD, VisA	Residual features vs nearest reference	Partial	No	Yes	Partial	No	Few-shot count unclear; no semiconductor test
AnomalyDINO [3]	2025	MVTec AD, VisA	DINOv2 patch NN matching	Yes	Unclear	Yes	Partial	No	No semiconductor or rotation evaluation
UniVAD [4]	2025	9 datasets	Component clustering + patch/graph matching	Yes	No	Yes	Partial	No	Heavy; not tested on dies
UCAD [36]	2024	MVTec AD, VisA	Continual prompts + key memory	No	No	Yes	Partial	Partial	No human-in-the-loop; category-level tasks
Abbas et al. (HITL memory correction) [10]	2026	MVTec AD	Expert edits PatchCore memory	Yes	No	Unclear	No	Yes	Simulated clean feedback; MVTec only
Prasad et al. (imec) [9]	2024	Private SEM (ADI/AEI)	Continual-learning benchmark	No	No	Partial	No	Partial	Proprietary data; supervised
Zhao et al. (PIRB) [54]	2024	Wafer defects (unverified)	Weight protection + replay buffer	Not evaluated	No	Unclear	Yes	Partial	Classification; dataset details unverified
DRA (open-set supervised AD) [34]	2022	9 datasets	Disentangled seen/unseen anomaly heads	Yes	No	No	Yes	No	Image-level; needs labelled anomalies
Dehaerne et al. (review) [11]	2025	103 papers	Systematic review	Not evaluated	Not evaluated	Not evaluated	Not evaluated	Not evaluated	Review only
Zontak & Cohen [12]	2010	Private SEM	Anisotropic-kernel reference reconstruction	Yes	Partial	Yes	Partial	No	Single clean reference assumed
Ofir et al. (Applied Materials) [56]	2022	Private SEM	U-Net on clean images + defect implants	No	No	Yes	Partial	No	Proprietary; reference-free
SEM-CLIP [52]	2024	Private SEM	CLIP adapted to SEM, few-shot	Yes	No	Yes	No	No	Private data; closed set
Zhao et al. (image-intrinsic normality) [45]	2025	Private IC SEM	Image-intrinsic normality, no references	Not evaluated	Unclear	Yes	Partial	No	Single-image prior; preprint
Zhao et al. (IC-DefectNCD) [57]	2025	Private IC SEM	Intrinsic priors + novel class discovery	Not evaluated	Unclear	Partial	Yes	No	Preprint; private data
Poniatowski et al. (domain adaptation) [43]	2025	SEM defects	Semi-/unsupervised domain adaptation	Partial	No	No	No	No	Classification only

Paper	Year	Dataset	Method	Few-shot	Rotation	Localisation	Unknown defects	Feedback learning	Limitation
LogicQA [58]	2025	MVTec LOCO; private SEM	VLM-generated question checklist	Yes	No	No	Partial	No	No pixel localisation
Dey et al. (ADCDS, imec) [44]	2024	Private SEM (ADI)	Deformable DETR + BoxSnake	No	No	Yes	No	No	Supervised; closed set
Dolgener Cantú et al. (InP golden die) [59]	2025	InP wafers + CAD	Simulated golden die from CAD	Partial	Unclear	Yes	Partial	No	Requires CAD; preprint
Chen (die detection in wafer images) [60]	2020	SAT wafer images	Die-grid clustering + CNN	No	Partial	Partial	No	No	Die-level only
Cai et al. (active learning) [38]	2022	X-ray microscopy	Active learning for segmentation	Not evaluated	No	Yes	No	Yes	Different modality
Wu et al. (WM-811K) [41]	2015	WM-811K	Radon/geometry features + SVM	No	Partial	No	No	No	Maps only; no images
Kim, Yoon, Kim [42]	2024	Sim. + real WBMs	Transfer + ensemble fine-tuning	Yes	No	No	No	No	Wafer maps; closed set
Shin et al. (open-set WBM) [8]	2025	Real WBMs	Open-set: Radon features + OC-SVM	Not evaluated	No	No	Yes	No	Wafer maps only

14 Paper-by-Paper Analysis

Each card begins with a plain-language summary, followed by the 15 fields of the research brief. Papers are grouped by family; within a group, they are ordered by year. "Closeness" (1–5 ★) rates how closely a paper overlaps with the proposed research; the five most relevant papers are discussed further in Section 15. The same data, with links, is in `artifacts/03_literature_review.xlsx`.

A. Reference-based anomaly detection (foundations)

P1. SPADE [2]

N. Cohen, Y. Hoshen · arXiv preprint 2005.02357 · closeness ★★★☆☆

In simple words. SPADE compares a new image with the most similar normal images it has seen, then checks every small region for a close match in those normal images. Regions that find no good match are marked as anomalous.

1. Title	Sub-Image Anomaly Detection with Deep Pyramid Correspondences
2. Authors	N. Cohen, Y. Hoshen
3. Year	2020
4. Venue	arXiv preprint 2005.02357
5. DOI / URL	https://arxiv.org/abs/2005.02357
6. Research problem	Locate anomalies at sub-image (pixel) level using only normal training images.
7. Dataset	MVTec AD; ShanghaiTech Campus (video frames).
8. Input modality	RGB images (industrial objects and textures).
9. Proposed method	Two-stage k-nearest-neighbour (kNN) search: (1) retrieve the K most similar normal images using global ImageNet-pretrained features; (2) for every pixel, find the closest feature in a multi-scale feature pyramid of those K images; the distance is the anomaly score.
10. Architecture	Frozen ImageNet-pretrained ResNet (Wide-ResNet) feature pyramid; no training.
11. What it achieved	Reported a mean PRO of 91.7 on MVTec AD, well above the student and autoencoder baselines in the paper, with virtually no training time.
12. Limitations	Stores all normal features (memory grows with the dataset); kNN search is slow; no explicit handling of rotation; evaluated mainly in the full-data setting.
13. What to learn	The core reference-matching idea of this project: "is there a similar normal patch somewhere in the references?" It needs no defect labels.
14. Relation to this research	Direct ancestor of the proposed reference memory. The image-retrieval stage is similar to choosing the most relevant reference dies.
15. Unexplored	Few-shot behaviour with 1–10 references; semiconductor imagery; rotated targets.

Matrix: few-shot Not evaluated · rotation No · localisation Yes · unknown Partial · feedback No

P2. PaDiM [18]

T. Defard, A. Setkov, A. Loesch, R. Audigier · ICPR 2020 Workshops (LNCS 12664, 2021) · closeness ★★☆☆☆

In simple words. PaDiM learns, for every position in the image, what "normal" looks like as a statistical cloud of features. A new image is anomalous at a position if its feature lies far outside that cloud.

1. Title	PaDiM: A Patch Distribution Modeling Framework for Anomaly Detection and Localization
2. Authors	T. Defard, A. Setkov, A. Loesch, R. Audigier
3. Year	2021
4. Venue	ICPR 2020 Workshops (LNCS 12664, 2021)
5. DOI / URL	https://arxiv.org/abs/2011.08785
6. Research problem	Anomaly detection and localisation with a model that is cheaper than storing all normal features.
7. Dataset	MVTec AD; ShanghaiTech Campus.
8. Input modality	RGB images.
9. Proposed method	For each patch position, fit a multivariate Gaussian (mean and covariance) to pretrained CNN embeddings from normal images; score with Mahalanobis distance.
10. Architecture	Frozen ResNet-18 or Wide-ResNet-50 features from several layers, randomly reduced in dimension.
11. What it achieved	Reported strong localisation on MVTec AD with low inference cost (see paper tables).
12. Limitations	Position-wise statistics assume the objects are aligned; covariance estimation needs many normal images per position, which is a poor fit for 1–5 references. The paper discusses a non-aligned test variant; check its section 4 before citing that result.
13. What to learn	If dies are well registered, position-specific statistics are a strong prior: a via should be at the same place in every die.
14. Relation to this research	Shows why alignment matters. Its per-position modelling becomes an option after geometric registration.
15. Unexplored	Few-shot covariance estimation; rotated dies.

Matrix: few-shot No · rotation Partial · localisation Yes · unknown Partial · feedback No

P3. PatchCore [1]

K. Roth, L. Pemula, J. Zepeda, B. Schölkopf, T. Brox, P. Gehler · CVPR 2022 · closeness ★★★★☆

In simple words. PatchCore keeps a compact "memory bank" of small normal image patches. For a new image, every patch looks up its closest normal patch; if even the closest one is far away, that patch is suspicious.

1. Title	Towards Total Recall in Industrial Anomaly Detection
2. Authors	K. Roth, L. Pemula, J. Zepeda, B. Schölkopf, T. Brox, P. Gehler
3. Year	2022
4. Venue	CVPR 2022
5. DOI / URL	https://doi.org/10.1109/CVPR52688.2022.01392
6. Research problem	Cold-start industrial anomaly detection with normal-only training data.
7. Dataset	MVTec AD (plus two further benchmarks).
8. Input modality	RGB images.
9. Proposed method	Extract locally aggregated mid-level patch features; reduce the memory bank by greedy coreset subsampling; anomaly score = distance to the nearest memory feature; image score from the most anomalous patch.
10. Architecture	Frozen ImageNet-pretrained Wide-ResNet-50 (layers 2 and 3); no gradient training.
11. What it achieved	Up to 99.6% image-level AUROC on MVTec AD reported in the abstract, with strong localisation. The paper also reports a low-shot study.
12. Limitations	Scores are uncalibrated distances; features are pretrained on natural images; no orientation handling; memory grows with data before subsampling.

13. What to learn	The baseline for this project. Enrolment is 'add features to memory', not 'train a network'. That makes few-shot enrolment and feedback updates cheap and reversible.
14. Relation to this research	Proposed reference memory = PatchCore memory built from K reference dies, extended with alignment, open-set triage and verified updates.
15. Unexplored	Semiconductor imagery; controlled rotation; memory updates from feedback; unknown-vs-known separation.

Matrix: few-shot Partial · rotation No · localisation Yes · unknown Partial · feedback No

P4. DRAEM [20]

V. Zavrtanik, M. Kristan, D. Skočaj · ICCV 2021 · closeness ★☆☆☆☆

In simple words. DRAEM paints fake defects onto normal images and trains a network to repair the image and point at the fake defects. After training, it can point at real defects it has never seen.

1. Title	DRAEM – A Discriminatively Trained Reconstruction Embedding for Surface Anomaly Detection
2. Authors	V. Zavrtanik, M. Kristan, D. Skočaj
3. Year	2021
4. Venue	ICCV 2021
5. DOI / URL	https://arxiv.org/abs/2108.07610
6. Research problem	Surface anomaly detection and segmentation without real defect labels.
7. Dataset	MVTec AD.
8. Input modality	RGB images.
9. Proposed method	A reconstructive sub-network learns to remove synthetic anomalies; a discriminative sub-network segments anomalies from the original and reconstructed image together.
10. Architecture	Two U-Net-like encoder–decoders trained end-to-end on synthetic anomalies.
11. What it achieved	Reported strong pixel-level segmentation on MVTec AD (see paper tables).
12. Limitations	Needs many normal images to train; performance depends on how realistic the synthetic anomalies are; trained per category.
13. What to learn	Synthetic defect injection gives pixel-perfect masks for evaluation. This project uses it to test defect preservation under rotation.
14. Relation to this research	Not a few-shot method. Its synthetic-anomaly recipe is reused for the controlled benchmark (Section 17.7).
15. Unexplored	Few-shot training; SEM textures; rotated dies.

Matrix: few-shot No · rotation No · localisation Yes · unknown Partial · feedback No

P5. Reverse Distillation [21]

H. Deng, X. Li · CVPR 2022 · closeness ★☆☆☆☆

In simple words. A 'student' network learns to copy a 'teacher' network's description of normal images. On defects, the student fails to copy, and the disagreement shows where the defect is.

1. Title	Anomaly Detection via Reverse Distillation from One-Class Embedding
2. Authors	H. Deng, X. Li
3. Year	2022
4. Venue	CVPR 2022
5. DOI / URL	https://openaccess.thecvf.com/content/CVPR2022/html/Deng_Anomaly_Detection_via_Reverse_Distillation_From_One-Class_Embedding_CVPR_2022_paper.html
6. Research problem	Unsupervised anomaly detection and localisation.
7. Dataset	MVTec AD (and one-class novelty benchmarks).
8. Input modality	RGB images.

9. Proposed method	Teacher encoder (pretrained, frozen) → one-class bottleneck embedding → student decoder that reconstructs teacher features in reverse order; multi-scale feature discrepancy = anomaly map.
10. Architecture	Wide-ResNet-50 teacher; trainable bottleneck and decoder.
11. What it achieved	Reported state-of-the-art localisation on MVTec AD at publication (see paper tables).
12. Limitations	Requires training on many normal images per category; not designed for few-shot enrolment.
13. What to learn	Multi-scale feature discrepancy is an effective localiser. A useful full-data upper bound.
14. Relation to this research	Optional full-data baseline (upper bound) for experiments where many normal dies exist.
15. Unexplored	Few-shot; rotation; semiconductor data.

Matrix: few-shot No · rotation No · localisation Yes · unknown Partial · feedback No

P6. FastFlow [19]

J. Yu, Y. Zheng, X. Wang, W. Li, Y. Wu, R. Zhao, L. Wu · arXiv preprint 2111.07677 · closeness ★☆☆☆☆

In simple words. FastFlow learns an exact 'probability of being normal' for each image location. Unusual locations get low probability.

1. Title	FastFlow: Unsupervised Anomaly Detection and Localization via 2D Normalizing Flows
2. Authors	J. Yu, Y. Zheng, X. Wang, W. Li, Y. Wu, R. Zhao, L. Wu
3. Year	2021
4. Venue	arXiv preprint 2111.07677
5. DOI / URL	https://arxiv.org/abs/2111.07677
6. Research problem	Density estimation of normal features for anomaly localisation.
7. Dataset	MVTec AD (further datasets in the paper; check the full text).
8. Input modality	RGB images.
9. Proposed method	2D normalizing flow applied to pretrained feature maps; negative log-likelihood per location = anomaly map.
10. Architecture	ResNet / CaiT / DeiT backbones + trainable 2D flow blocks.
11. What it achieved	Reported 99.4% image-level AUC on MVTec AD (abstract), with efficient inference across backbones.
12. Limitations	Needs enough normal data to fit a density model; flow training can be unstable; not few-shot.
13. What to learn	A probabilistic score is easier to threshold than a raw distance, but only when enough data exists to fit it.
14. Relation to this research	Full-data comparison method; not a core component.
15. Unexplored	Few-shot density estimation; semiconductor data.

Matrix: few-shot No · rotation No · localisation Yes · unknown Partial · feedback No

P7. EfficientAD [22]

K. Batzner, L. Heckler, R. König · WACV 2024 · closeness ★☆☆☆☆

In simple words. EfficientAD is designed to be fast. A small student network learns to imitate a teacher on normal images, and an autoencoder checks global consistency.

1. Title	EfficientAD: Accurate Visual Anomaly Detection at Millisecond-Level Latencies
2. Authors	K. Batzner, L. Heckler, R. König
3. Year	2024
4. Venue	WACV 2024
5. DOI / URL	https://openaccess.thecvf.com/content/WACV2024/papers/Batzner_EfficientAD_Accurate_Visual_Anomaly_Detection_at_Millisecond-Level_Latencies_WACV_2024_paper.pdf
6. Research problem	Accurate anomaly detection under strict latency limits.
7. Dataset	MVTec AD, VisA, MVTec LOCO AD.
8. Input modality	RGB images.
9. Proposed method	Patch description network (PDN) teacher distilled from a pretrained backbone; student–teacher discrepancy for local anomalies; autoencoder for logical/global anomalies.
10. Architecture	Lightweight 4-layer convolutional PDN; trained per category.
11. What it achieved	Reported about 2 ms latency and 600 images per second, evaluated on 32 datasets from three industrial anomaly-detection collections.
12. Limitations	Requires per-category training on many normal images; not few-shot.
13. What to learn	Latency matters for in-line inspection. Use it as a speed reference.
14. Relation to this research	Optional full-data, fast baseline for the inference-time comparison.
15. Unexplored	Few-shot and rotated settings.

Matrix: few-shot No · rotation No · localisation Yes · unknown Partial · feedback No

B. Few-shot and generalist anomaly detection

P8. RegAD [6]

C. Huang, H. Guan, A. Jiang, Y. Zhang, M. Spratling, Y.-F. Wang · ECCV 2022 (oral) · closeness ★★★★★

In simple words. RegAD learns how to align images in general. Given a new product with only a few normal samples, it aligns the new image's features to those samples and flags parts that still do not match.

1. Title	Registration Based Few-Shot Anomaly Detection
2. Authors	C. Huang, H. Guan, A. Jiang, Y. Zhang, M. Spratling, Y.-F. Wang
3. Year	2022
4. Venue	ECCV 2022 (oral)
5. DOI / URL	https://doi.org/10.1007/978-3-031-20053-3_18
6. Research problem	Few-shot anomaly detection for new categories without retraining or fine-tuning.
7. Dataset	MVTec AD, MPDD.
8. Input modality	RGB images (MPDD: metal parts in varied positions).
9. Proposed method	Train a category-agnostic feature-registration network (Siamese + spatial transformer modules) on many categories; at test time register test features to the K support images, estimate a normal distribution from the registered support features, and score deviations.
10. Architecture	ResNet-based Siamese network with feature-level spatial transformer networks (STN).
11. What it achieved	Reported AUROC gains over earlier few-shot methods of 5.1%, 6.9%, 8.0% (MVTec AD) and 3.2%, 5.0%, 3.4% (MPDD) for 2-, 4- and 8-shot settings (abstract).

12. Limitations	Registration is learned on natural-object categories; rotation is handled by augmentation and learned alignment, not evaluated as a controlled angle sweep; no semiconductor data; no unknown-vs-known separation.
13. What to learn	Alignment and few-shot detection should be solved together. A learned registration step can generalise to unseen categories.
14. Relation to this research	Closest work for Contribution 1 + 2. This project replaces learned feature registration with explicit, auditable die registration, and measures whether defects survive alignment.
15. Unexplored	Defect preservation under alignment; SEM / die imagery; same-wafer vs cross-wafer splits.

Matrix: few-shot Yes · rotation Partial · localisation Yes · unknown Partial · feedback No

P9. GraphCore [7]

G. Xie, J. Wang, J. Liu, F. Zheng, Y. Jin · ICLR 2023 · closeness ★★★★★

In simple words. When an object is rotated, its patches look different to a normal CNN, so a few-shot memory needs more samples. GraphCore makes the patch description rotation-insensitive, so fewer samples cover more orientations.

1. Title	Pushing the Limits of Fewshot Anomaly Detection in Industry Vision: GraphCore
2. Authors	G. Xie, J. Wang, J. Liu, F. Zheng, Y. Jin
3. Year	2023
4. Venue	ICLR 2023
5. DOI / URL	https://arxiv.org/abs/2301.12082
6. Research problem	Few-shot anomaly detection when objects appear in different orientations.
7. Dataset	MVTec AD, MPDD.
8. Input modality	RGB images.
9. Proposed method	Visual isometric invariant features (VIIF) built with a graph representation of patch features; memory bank and nearest-neighbour scoring as in PatchCore. The paper also describes an 'Augmentation + PatchCore' baseline.
10. Architecture	Pretrained CNN features + graph neural network producing rotation-invariant patch features.
11. What it achieved	Reported average AUC increases of 5.8%, 4.1%, 3.4%, 1.6% (MVTec AD) and 25.5%, 22.0%, 16.9%, 14.1% (MPDD) for 1-, 2-, 4-, 8-shot cases (abstract).
12. Limitations	Rotation invariance can also hide defects whose meaning depends on orientation (for example, a line that should be horizontal). Evaluated on objects, not periodic die layouts.
13. What to learn	Rotation is a major cause of few-shot failure. Invariance and alignment are two competing solutions, and this project should compare both.
14. Relation to this research	Closest work for Contribution 2 (rotation). This project tests invariance against explicit alignment on die imagery, and checks for orientation-dependent defects.
15. Unexplored	Orientation-dependent defects; periodic semiconductor patterns; arbitrary-angle sweeps.

Matrix: few-shot Yes · rotation Yes · localisation Yes · unknown Partial · feedback No

P10. WinCLIP [5]

J. Jeong, Y. Zou, T. Kim, D. Zhang, A. Ravichandran, O. Dabeer · CVPR 2023 · closeness ★★★☆☆

In simple words. WinCLIP uses a vision–language model (CLIP) and text prompts such as 'a photo of a damaged object' to find anomalies with zero or a few normal examples.

1. Title	WinCLIP: Zero-/Few-Shot Anomaly Classification and Segmentation
2. Authors	J. Jeong, Y. Zou, T. Kim, D. Zhang, A. Ravichandran, O. Dabeer
3. Year	2023
4. Venue	CVPR 2023
5. DOI / URL	https://openaccess.thecvf.com/content/CVPR2023/html/Jeong_WinCLIP_Zero-Few-Shot_Anomaly_Classification_and_Segmentation_CVPR_2023_paper.html
6. Research problem	Zero- and few-shot anomaly classification and segmentation.
7. Dataset	MVTec AD, VisA.
8. Input modality	RGB images + text prompts.
9. Proposed method	Compositional state prompts plus window-based CLIP features for zero-shot; WinCLIP+ adds few-shot normal reference matching.
10. Architecture	Frozen CLIP ViT image and text encoders.
11. What it achieved	Zero-shot 91.8% / 85.1% AUROC (classification / segmentation) on MVTec AD and 78.1% / 79.6% on VisA (abstract).
12. Limitations	Text prompts describe natural-object concepts. Semiconductor SEM vocabulary is outside CLIP's training data; coarse windows limit small-defect localisation.
13. What to learn	Language priors help zero-shot detection, but this project's references (normal dies) are more reliable than text for unseen die designs.
14. Relation to this research	Baseline for a 'no references' and few-shot comparison; SEM-CLIP adapts this idea to SEM data.
15. Unexplored	SEM domain prompts; rotation; feedback.

Matrix: few-shot Yes · rotation No · localisation Yes · unknown Partial · feedback No

P11. InCTRL [17]

J. Zhu, G. Pang · CVPR 2024 · closeness ★★★☆☆

In simple words. InCTRL is trained once and then used for new kinds of images. It compares a new image with a few normal examples and learns that 'large differences mean anomaly', whatever the image shows.

1. Title	Toward Generalist Anomaly Detection via In-context Residual Learning with Few-shot Sample Prompts
2. Authors	J. Zhu, G. Pang
3. Year	2024
4. Venue	CVPR 2024
5. DOI / URL	https://openaccess.thecvf.com/content/CVPR2024/html/Zhu_Toward_Generalist_Anomaly_Detection_via_In-context_Residual_Learning_with_Few-shot_CVPR_2024_paper.html
6. Research problem	One generalist anomaly detector that works on new datasets from a few normal samples without retraining.
7. Dataset	Nine datasets: industrial defects, medical images, semantic anomalies.
8. Input modality	RGB images (+ CLIP text).
9. Proposed method	In-context residual learning: residuals between query and few-shot normal prompts at patch and image level, combined with CLIP text priors, to train a generalist scorer.
10. Architecture	CLIP-based backbone with a residual learning head; trained on an auxiliary dataset.
11. What it achieved	Reported to significantly outperform competing methods across the nine datasets (see paper tables).
12. Limitations	Focuses on image-level detection scores; generalisation to SEM / die imagery not tested.

13. What to learn	Training on residuals (query minus reference) can transfer across domains. This matches the cross-wafer question.
14. Relation to this research	Strong candidate for cross-type generalisation (Experiment D) and an argument for residual-based scoring.
15. Unexplored	Pixel-level localisation; semiconductor data; rotation.

Matrix: few-shot Yes · rotation No · localisation No · unknown Partial · feedback No

P12. ResAD [55]

X. Yao et al. · NeurIPS 2024 (spotlight) · closeness ★★★☆☆

In simple words. Instead of learning what each product looks like, ResAD learns what the difference between a sample and its closest normal reference looks like. The differences behave similarly across products.

1. Title	ResAD: A Simple Framework for Class Generalizable Anomaly Detection
2. Authors	X. Yao et al.
3. Year	2024
4. Venue	NeurIPS 2024 (spotlight)
5. DOI / URL	https://proceedings.neurips.cc/paper_files/paper/2024/hash/e2ba11d128034f0c544693374c59e49a-Abstract.html
6. Research problem	Anomaly detection on new classes without retraining.
7. Dataset	Industrial benchmarks incl. MVTec AD and VisA (see paper).
8. Input modality	RGB images.
9. Proposed method	Feature converter computes residual features relative to the nearest normal reference feature; a shallow constraint pulls normal residuals to a hypersphere; a distribution estimator flags out-of-distribution residuals.
10. Architecture	Pretrained backbone + feature converter + shallow constraintor + normalizing-flow-style estimator.
11. What it achieved	Reported strong cross-class generalisation (see paper tables).
12. Limitations	Number of normal references per new class must be checked in the paper; not evaluated on semiconductor data or rotation.
13. What to learn	Residual features are less class-specific than raw features. This idea fits cross-wafer and cross-type transfer.
14. Relation to this research	Candidate component for Contribution 1 (cross-die/cross-type transfer).
15. Unexplored	SEM/die data; geometric variation; feedback.

Matrix: few-shot Partial · rotation No · localisation Yes · unknown Partial · feedback No

P13. AnomalyDINO [3]

S. Damm, M. Laszkiewicz, J. Lederer, A. Fischer · WACV 2025 · closeness ★★★★★

In simple words. AnomalyDINO is PatchCore-style nearest-neighbour matching, but with features from DINOv2, a self-supervised vision transformer. With even one normal image as reference, it finds anomalies very well on standard benchmarks.

1. Title	AnomalyDINO: Boosting Patch-Based Few-Shot Anomaly Detection with DINOv2
2. Authors	S. Damm, M. Laszkiewicz, J. Lederer, A. Fischer
3. Year	2025
4. Venue	WACV 2025
5. DOI / URL	https://openaccess.thecvf.com/content/WACV2025/html/Damm_AnomalyDINO_Boosting_Patch-Based_Few-Shot_Anomaly_Detection_with_DINOv2_WACV_2025_paper.html
6. Research problem	Training-free few-shot anomaly detection and segmentation.
7. Dataset	MVTec AD, VisA.
8. Input modality	RGB images.

9. Proposed method	Patch-level deep nearest-neighbour matching on DINOv2 patch features; masking of background patches; image score from top anomalous patches.
10. Architecture	Frozen DINOv2 ViT (e.g. ViT-S/14); no training.
11. What it achieved	1-shot image AUROC on MVTec AD increased from 93.1% to 96.6% (abstract).
12. Limitations	Natural-image pretrained features; ViT patch size (14 px) limits the smallest detectable defect unless images are upsampled; no rotation study; no semiconductor data.
13. What to learn	A strong, simple, training-free few-shot baseline. Exactly the 'give it a few samples, inspect new ones' setting.
14. Relation to this research	Closest work for Contribution 1. Primary baseline. This project adds die alignment, open-set triage and verified updates, and tests on semiconductor data.
15. Unexplored	SEM/die imagery; small defects vs patch size; rotation; same- vs cross-wafer.

Matrix: few-shot Yes · rotation Unclear · localisation Yes · unknown Partial · feedback No

P14. UniVAD [4]

Z. Gu et al. · CVPR 2025 · closeness ★★★☆☆

In simple words. UniVAD first splits an image into its parts (components), then compares each part with the matching part of a few normal images. It also checks whether the parts relate to each other correctly.

1. Title	UniVAD: A Training-free Unified Model for Few-shot Visual Anomaly Detection
2. Authors	Z. Gu et al.
3. Year	2025
4. Venue	CVPR 2025
5. DOI / URL	https://openaccess.thecvf.com/content/CVPR2025/html/Gu_UniVAD_A_Training-free_Unified_Model_for_Few-shot_Visual_Anomaly_Detection_CVPR_2025_paper.html
6. Research problem	One training-free few-shot detector across industrial, logical and medical domains.
7. Dataset	Nine datasets incl. MVTec AD, VisA, MVTec LOCO, Brain MRI, Liver CT.
8. Input modality	RGB and medical images.
9. Proposed method	Contextual component clustering (with vision foundation models), component-aware patch matching for structural anomalies, graph-based component aggregation for logical anomalies.
10. Architecture	Frozen foundation models (segmentation + feature encoders); no training on the target domain.
11. What it achieved	Reported state-of-the-art few-shot performance across domains (see paper tables).
12. Limitations	Heavy foundation-model pipeline; component segmentation may not transfer to dense periodic SEM layouts; no rotation study.
13. What to learn	Component-aware matching restricts comparison to the 'same kind of part'. This is close to region-aware matching in die layouts.
14. Relation to this research	Baseline for few-shot detection; idea for region-restricted matching.
15. Unexplored	Semiconductor data; compute-limited deployment.

Matrix: few-shot Yes · rotation No · localisation Yes · unknown Partial · feedback No

C. Continual learning, feedback and open-set

P15. UCAD [36]

J. Liu et al. · AAAI 2024 · closeness ★★★☆☆

In simple words. UCAD learns new product types one after another without forgetting the old ones, by storing small 'prompts' and a key–knowledge memory for each task.

1. Title	Unsupervised Continual Anomaly Detection with Contrastively-learned Prompt
2. Authors	J. Liu et al.
3. Year	2024
4. Venue	AAAI 2024
5. DOI / URL	https://doi.org/10.1609/aaai.v38i4.28153
6. Research problem	Task-agnostic continual learning for unsupervised anomaly detection and segmentation.
7. Dataset	MVTec AD, VisA (as sequential tasks).
8. Input modality	RGB images.
9. Proposed method	Continual prompting module with key–prompt–knowledge memory; structure-based contrastive learning using SAM.
10. Architecture	Frozen ViT backbone + learnable prompts + SAM-guided contrastive learning.
11. What it achieved	Reported improvements of 15.6% (detection) and 26.6% (segmentation) over previous continual AD methods (arXiv abstract).
12. Limitations	No human verification of what is learned; tasks are product categories, not drifting inspection streams; not semiconductor.
13. What to learn	Memory-based continual learning can avoid catastrophic forgetting without retraining the backbone.
14. Relation to this research	Shows continual AD is feasible. This project's feedback loop is simpler (memory edits) and adds human verification.
15. Unexplored	Human-verified updates; noisy feedback; semiconductor drift.

Matrix: few-shot No · rotation No · localisation Yes · unknown Partial · feedback Partial

P16. Abbas et al. (HITL memory correction) [10]

A. Abbas, S. Abbas, K. Adhikari · arXiv preprint 2608.17775 (2026) · closeness ★★★★★

In simple words. An expert corrects a PatchCore detector's mistakes by editing its memory of normal patches. No retraining. When the detector raises a false alarm, normal patches from that image are added to memory, but only if they are genuinely new.

1. Title	Training-Free Human-in-the-Loop Anomaly Detection via Memory Bank Correction
2. Authors	A. Abbas, S. Abbas, K. Adhikari
3. Year	2026
4. Venue	arXiv preprint 2608.17775 (2026)
5. DOI / URL	https://arxiv.org/abs/2608.17775
6. Research problem	Correcting a memory-bank anomaly detector with expert feedback, without retraining or original training data.
7. Dataset	MVTec AD (15 categories).
8. Input modality	RGB images.
9. Proposed method	Start from a bank of 10 normal 'golden' samples; for false positives, add patches through a self-calibrating novelty gate (only patches beyond the median pool-normal nearest-neighbour distance); held-out evaluation with 20 splits per category and Holm-corrected Wilcoxon tests. Feedback simulated from ground truth.
10. Architecture	PatchCore (frozen backbone + memory bank) with memory editing operations.

11. What it achieved	Corrections recovered a median 66% of the gap to a fully trained bank and improved 12 of 15 MVTec AD categories without harming any (abstract). Passive and active querying were statistically indistinguishable; a defect-memory extension failed.
12. Limitations	Preprint; simulated, noise-free feedback; natural-object data; no rotation; no unknown-defect triage.
13. What to learn	Crucial evidence and a warning: adding corrected images to memory inflates naive evaluation through memorisation. A held-out protocol is mandatory.
14. Relation to this research	Closest work for Contribution 5. This project extends it to semiconductor dies, noisy/incorrect feedback, quarantine and rollback, and known/unknown defect memories.
15. Unexplored	Noisy or adversarial feedback; reference contamination; semiconductor data; defect memory that works.

Matrix: few-shot Yes · rotation No · localisation Unclear · unknown No · feedback Yes

P17. Prasad et al. (imec) [9]

A. Prasad, B. Dey, V. Blanco, S. Halder · ECML PKDD 2024 Industry Track; arXiv 2407.12724 · closeness ★★★☆☆

In simple words. When a defect detector is taught new defect types from a new process step, it tends to forget old types. This paper measures how well continual-learning methods prevent that on real fab SEM images.

1. Title	An Evaluation of Continual Learning for Advanced Node Semiconductor Defect Inspection
2. Authors	A. Prasad, B. Dey, V. Blanco, S. Halder
3. Year	2024
4. Venue	ECML PKDD 2024 Industry Track; arXiv 2407.12724
5. DOI / URL	https://arxiv.org/abs/2407.12724
6. Research problem	Incremental learning of new SEM defect classes across process steps (ADI → AEI) without forgetting.
7. Dataset	Resist-wafer SEM images at After-Development Inspection (ADI) and After-Etch Inspection (AEI); not public.
8. Input modality	SEM images.
9. Proposed method	Task-incremental continual learning benchmark against conventional fine-tuning.
10. Architecture	Supervised defect detection network (see paper).
11. What it achieved	Reported better results than conventional supervised training and showed that plain fine-tuning quickly forgets earlier defect classes.
12. Limitations	Proprietary data; supervised detection needs box labels; no few-shot normal-reference setting.
13. What to learn	Forgetting is real on semiconductor data. Feedback updates must be checked for forgetting, not only for gains.
14. Relation to this research	Motivates the forgetting-rate metric and replay-based safeguards in Stage 10.
15. Unexplored	Normal-reference memories; open-set; human verification.

Matrix: few-shot No · rotation No · localisation Partial · unknown No · feedback Partial

P18. Zhao et al. (PIRB) [54]

Z. Zhao, J. Wang, Q. Tao, A. Li, Y. Chen · Reliability Engineering & System Safety 244 (2024) · closeness ★★☆☆☆

In simple words. A wafer-defect model that keeps learning online as new, previously unknown defect types appear, while protecting what it learned before.

1. Title	An Unknown Wafer Surface Defect Detection Approach Based on Incremental Learning for Reliability Analysis
2. Authors	Z. Zhao, J. Wang, Q. Tao, A. Li, Y. Chen
3. Year	2024
4. Venue	Reliability Engineering & System Safety 244 (2024)
5. DOI / URL	https://ideas.repec.org/a/eee/reensy/v244y2024ics0951832024000413.html
6. Research problem	Online detection of unknown wafer defects with incremental learning.

7. Dataset	Wafer defect data (dataset name not confirmed in this session; check the paper).
8. Input modality	Wafer defect images/maps (check paper).
9. Proposed method	PIRB: restricts updates of important weights to retain old knowledge and keeps a tiny reference buffer of previously seen defect patterns.
10. Architecture	Deep classifier with weight-importance regularisation + replay buffer.
11. What it achieved	Reported 60% shorter training time than offline learning and 10% higher total accuracy than compared methods (abstract).
12. Limitations	Classifier-based; no pixel localisation reported in the abstract; dataset details unverified here.
13. What to learn	A small replay buffer plus weight protection is a practical anti-forgetting recipe.
14. Relation to this research	Supports the 'small verified buffer' design of the feedback stage.
15. Unexplored	Normal-reference memory; human verification; localisation.

Matrix: few-shot Not evaluated · rotation No · localisation Unclear · unknown Yes · feedback Partial

P19. DRA (open-set supervised AD) [34]

C. Ding, G. Pang, C. Shen · CVPR 2022 · closeness ★★★☆☆

In simple words. DRA uses a few labelled defect examples to learn 'seen' anomalies, but keeps a separate path to catch completely new anomalies so they are not ignored.

1. Title	Catching Both Gray and Black Swans: Open-set Supervised Anomaly Detection
2. Authors	C. Ding, G. Pang, C. Shen
3. Year	2022
4. Venue	CVPR 2022
5. DOI / URL	https://openaccess.thecvf.com/content/CVPR2022/html/Ding_Catching_Both_Gray_and_Black_Swans_Open-Set_Supervised_Anomaly_Detection_CVPR_2022_paper.html
6. Research problem	Using a few labelled anomalies without overfitting to them, while still detecting unseen anomaly types.
7. Dataset	Nine real-world datasets incl. MVTec AD, AITEX, SDD, ELPV, Mastcam, BrainMRI, HeadCT, Hyper-Kvasir.
8. Input modality	Images.
9. Proposed method	Disentangled representations for seen anomalies, pseudo (augmented) anomalies and latent residual anomalies.
10. Architecture	CNN backbone with multiple heads.
11. What it achieved	Reported better detection of both seen and unseen anomalies than prior methods (see paper tables).
12. Limitations	Image-level focus; needs some labelled anomalies; not semiconductor.
13. What to learn	Few known defects should help, not narrow, detection. Keep a 'deviation from normal' path that is independent of known classes.
14. Relation to this research	Design rule for Stage 8: known-defect memory is added next to, not instead of, the normal-reference score.
15. Unexplored	Pixel localisation; semiconductor; rotation.

Matrix: few-shot Yes · rotation No · localisation No · unknown Yes · feedback No

D. Semiconductor inspection

P20. Dehaerne et al. (review) [11]

E. Dehaerne, B. Dey, V. Blanco, J. Davis · SPIE journal (2025); arXiv 2409.06833 · closeness ★★★☆☆

In simple words. A careful survey of 103 papers on automatically finding defects in SEM images of chips. It explains which approaches were popular and what is still hard.

1. Title	Scanning Electron Microscopy-Based Automatic Defect Inspection for Semiconductor Manufacturing: A Systematic Review
2. Authors	E. Dehaerne, B. Dey, V. Blanco, J. Davis
3. Year	2025
4. Venue	SPIE journal (2025); arXiv 2409.06833
5. DOI / URL	https://arxiv.org/abs/2409.06833
6. Research problem	Systematic review of automatic SEM defect inspection algorithms.
7. Dataset	103 selected publications.
8. Input modality	SEM images (review).
9. Proposed method	Systematic literature review (database search, screening, taxonomy).
10. Architecture	—
11. What it achieved	Reports that reference-based defect detection was the most common algorithm type until 2020, after which deep learning became dominant, especially for classification.
12. Limitations	Review; does not benchmark methods; literature cut-off before 2025 developments.
13. What to learn	Reference-based comparison is the historical backbone of SEM inspection. A learned reference memory is a natural evolution, not a break.
14. Relation to this research	Positions the project: it connects the classical reference paradigm with modern few-shot deep features.
15. Unexplored	Few-shot reference memories; feedback loops.

Matrix: few-shot Not evaluated · rotation Not evaluated · localisation Not evaluated · unknown Not evaluated · feedback Not evaluated

P21. Zontak & Cohen [12]

M. Zontak, I. Cohen · Machine Vision and Applications 21 (2010) · closeness ★★★☆☆

In simple words. Instead of perfectly aligning a die image to a reference die, this method tries to rebuild the inspected image from pieces of the reference. Whatever cannot be rebuilt is a defect.

1. Title	Defect Detection in Patterned Wafers Using Anisotropic Kernels
2. Authors	M. Zontak, I. Cohen
3. Year	2010
4. Venue	Machine Vision and Applications 21 (2010)
5. DOI / URL	https://doi.org/10.1007/s00138-008-0146-y
6. Research problem	Defect detection in patterned wafers without explicit registration.
7. Dataset	Patterned-wafer SEM images (not public).
8. Input modality	SEM images.
9. Proposed method	Anisotropic-kernel reconstruction of the inspected image from a reference image; unreconstructable regions flagged.
10. Architecture	Classical non-parametric reconstruction; no learning.
11. What it achieved	Demonstrated registration-free reference-based detection on patterned wafers.
12. Limitations	Assumes the reference is defect-free; single reference; hand-designed.

13. What to learn	Classical precedent for 'compare with a reference, without pixel-perfect alignment'. Patch matching with a search window does the same in feature space.
14. Relation to this research	Historical anchor for the alignment-tolerant matching design (Stage 5).
15. Unexplored	Learned features; few-shot memories; open-set.

Matrix: few-shot Yes · rotation Partial · localisation Yes · unknown Partial · feedback No

P22. Ofir et al. (Applied Materials) [56]

N. Ofir, R. Yacobi, O. Granoviter, B. Levant, O. Shtalrid · ICIP 2022 · closeness ★★☆☆☆

In simple words. An industry team trains a segmentation network on clean SEM backgrounds with pasted-in fake defects, so it learns to find defects in a single SEM image without needing a reference image.

1. Title	Automatic Defect Segmentation by Unsupervised Anomaly Learning
2. Authors	N. Ofir, R. Yacobi, O. Granoviter, B. Levant, O. Shtalrid
3. Year	2022
4. Venue	ICIP 2022
5. DOI / URL	https://arxiv.org/abs/2202.02998
6. Research problem	SEM defect segmentation without manual labels and without a reference image.
7. Dataset	Industrial SEM defect-review images (not public).
8. Input modality	SEM images.
9. Proposed method	U-Net-style network trained on automatically generated samples: clean backgrounds with copy-paste 'defect implants', plus contrastive-style feature losses.
10. Architecture	U-Net-style encoder–decoder.
11. What it achieved	Demonstrated label-free defect segmentation from the defect image alone.
12. Limitations	Proprietary data; synthetic-implant realism; no few-shot reference enrolment.
13. What to learn	Industry already uses reference-free and implant-based training. Synthetic implants are a valid evaluation tool if reported honestly.
14. Relation to this research	Supports the synthetic defect-injection benchmark; offers a reference-free comparison point.
15. Unexplored	Reference-guided few-shot; rotation; feedback.

Matrix: few-shot No · rotation No · localisation Yes · unknown Partial · feedback No

P23. SEM-CLIP [52]

Q. Jin, Y. Jiang, X. Lu, Y. Liu, Y. Chen, D. Gao, Q. Sun, C. Zhuo · ICCAD 2024 · closeness ★★★★☆

In simple words. SEM-CLIP adapts a vision–language model to SEM images of chips, so it can classify and outline nanoscale defects after seeing only a few examples.

1. Title	SEM-CLIP: Precise Few-Shot Learning for Nanoscale Defect Detection in Scanning Electron Microscope Image
2. Authors	Q. Jin, Y. Jiang, X. Lu, Y. Liu, Y. Chen, D. Gao, Q. Sun, C. Zhuo
3. Year	2024
4. Venue	ICCAD 2024
5. DOI / URL	https://arxiv.org/abs/2502.14884
6. Research problem	Few-shot defect classification and segmentation in SEM images.
7. Dataset	In-house SEM defect dataset (not public).
8. Input modality	SEM images + text prompts.

9. Proposed method	CLIP adapted with SEM-specific prompts and mechanisms that focus on defect regions and suppress background.
10. Architecture	CLIP ViT encoders + adaptation modules.
11. What it achieved	Reported to surpass PromptAD in the 10-shot setting on the in-house dataset.
12. Limitations	Private data; no public reproduction; rotation and unknown defects not addressed.
13. What to learn	Few-shot learning on real SEM defects is feasible, and domain adaptation of foundation models matters.
14. Relation to this research	Closest semiconductor few-shot work; a comparison point if a public implementation exists.
15. Unexplored	Public benchmark; normal-reference memory; open-set; feedback.

Matrix: few-shot Yes · rotation No · localisation Yes · unknown No · feedback No

P24. Zhao et al. (image-intrinsic normality) [45]

B. Zhao, Q. Shi, S. Lyu, Y. Lu · arXiv preprint 2509.09375 (2025) · closeness ★★★★★

In simple words. Because chip layouts change between products and are hard to align, this method stops comparing with external reference images. It finds what is normal inside the test image itself (repeated structures) and segments what does not fit.

1. Title	Unsupervised Integrated-Circuit Defect Segmentation via Image-Intrinsic Normality
2. Authors	B. Zhao, Q. Shi, S. Lyu, Y. Lu
3. Year	2025
4. Venue	arXiv preprint 2509.09375 (2025)
5. DOI / URL	https://arxiv.org/abs/2509.09375
6. Research problem	Unsupervised IC defect segmentation without external normal references.
7. Dataset	IC SEM images (see paper).
8. Input modality	SEM images.
9. Proposed method	Learnable normal-information extractor aggregates representative normal features from the test image; a coherence loss ties them to normal regions; defects segmented as non-coherent regions.
10. Architecture	Deep encoder with a normal-feature extractor head.
11. What it achieved	Reported effective segmentation on IC data (numbers not retrieved in this session).
12. Limitations	Preprint; relies on repetition within each image; may fail when a defect repeats or a structure is unique within the field of view.
13. What to learn	An important counter-argument: external-reference matching is brittle for IC images when layouts vary and alignment is hard. The proposal must test this risk directly.
14. Relation to this research	Closest semiconductor competitor to the reference-guided idea. A combined score (external reference + intrinsic normality) is a natural extension.
15. Unexplored	Combining intrinsic and reference cues; feedback; wafer context.

Matrix: few-shot Not evaluated · rotation Unclear · localisation Yes · unknown Partial · feedback No

P25. Zhao et al. (IC-DefectNCD) [57]

B. Zhao, X. Wang, S. Lyu, Y. Lu · arXiv preprint 2511.03120 (2025) · closeness ★★★★☆

In simple words. Extends the previous paper: it finds defects without a support set and also groups new defect types that nobody has named yet.

1. Title	Image-Intrinsic Priors for Integrated Circuit Defect Detection and Novel Class Discovery via Self-Supervised Learning
2. Authors	B. Zhao, X. Wang, S. Lyu, Y. Lu
3. Year	2025
4. Venue	arXiv preprint 2511.03120 (2025)
5. DOI / URL	https://arxiv.org/abs/2511.03120
6. Research problem	Support-set-free IC defect detection plus novel class discovery (NCD) of unseen defect types.
7. Dataset	Real-world IC SEM dataset (paper describes several fabrication stages; details not verified here).
8. Input modality	SEM images.
9. Proposed method	Image-intrinsic priors for detection; self-supervised classifier with IC defect priors; semi-supervised k-means to estimate the number of unknown classes.
10. Architecture	Self-supervised encoder + clustering head.
11. What it achieved	Reported validation on a real-world IC dataset (numbers not retrieved in this session).
12. Limitations	Preprint; private data; NCD needs a batch of unknown samples to cluster.
13. What to learn	'Unknown defect' handling can go further than flagging: unknowns can be clustered into candidate new classes for engineer review.
14. Relation to this research	Informs Stage 8 (unknown triage) and the engineer workflow 'mark for investigation'.
15. Unexplored	Reference-guided NCD; human verification of discovered classes.

Matrix: few-shot Not evaluated · rotation Unclear · localisation Partial · unknown Yes · feedback No

P26. Poniatowski et al. (domain adaptation) [43]

A. Poniatowski, N. Gentner, M. Barusco, D. Dalle Pezze, S. Salti, G. A. Susto · IEEE T-ASE 23 (2026); arXiv 2506.15260 · closeness ★★★☆☆

In simple words. A defect classifier trained on SEM images from one setting often fails in another. This Infineon / university study tests domain-adaptation methods that adapt the classifier with few or no labels from the new setting.

1. Title	Domain Adaptation for Image Classification of Defects in Semiconductor Manufacturing
2. Authors	A. Poniatowski, N. Gentner, M. Barusco, D. Dalle Pezze, S. Salti, G. A. Susto
3. Year	2025
4. Venue	IEEE T-ASE 23 (2026); arXiv 2506.15260
5. DOI / URL	https://arxiv.org/abs/2506.15260
6. Research problem	Cross-domain SEM defect classification (source → target domain shift).
7. Dataset	SEM defect images (see paper for exact datasets).
8. Input modality	SEM images.
9. Proposed method	Semi-supervised (SSDA) and unsupervised (UDA) domain adaptation methods applied to SEM defect classification.
10. Architecture	CNN classifiers with domain-adaptation objectives.
11. What it achieved	Reported as the first application of SSDA/UDA to this kind of semiconductor image classification (authors' claim).
12. Limitations	Classification only; closed set; no localisation.

13. What to learn	Cross-domain shift is real in semiconductor SEM data. Cross-wafer and cross-type inspection cannot be assumed to work.
14. Relation to this research	Evidence for treating cross-wafer generalisation (Contribution 1, scenario 2) as a hypothesis to test.
15. Unexplored	Normal-reference memories under domain shift; localisation.

Matrix: few-shot Partial · rotation No · localisation No · unknown No · feedback No

P27. LogicQA [58]

Y. Kwon, D. Moon, Y. Oh, H. Yoon · ACL 2025 (Industry Track) · closeness ★★☆☆☆

In simple words. A vision–language model writes a checklist of questions about what a normal image should look like, then answers them for each new image. Wrong answers point to logical anomalies, and the answers explain the decision in words.

1. Title	LogicQA: Logical Anomaly Detection with Vision Language Model Generated Questions
2. Authors	Y. Kwon, D. Moon, Y. Oh, H. Yoon
3. Year	2025
4. Venue	ACL 2025 (Industry Track)
5. DOI / URL	https://doi.org/10.18653/v1/2025.acl-industry.29
6. Research problem	Training-free, few-shot logical anomaly detection with explanations.
7. Dataset	MVTec LOCO AD; proprietary semiconductor SEM dataset.
8. Input modality	Images + language.
9. Proposed method	VLM-generated question checklist from few normal images; answers aggregated into anomaly scores and explanations.
10. Architecture	Large vision–language model (prompting only).
11. What it achieved	Reported 87.6% AUROC and 87.0% F1-max for logical anomalies on MVTec LOCO AD; SEM results described as strong without public numbers.
12. Limitations	Logical (global) anomalies, not pixel localisation; proprietary SEM data; VLM cost.
13. What to learn	Explanations help engineers verify decisions, which is relevant to the feedback interface.
14. Relation to this research	Possible extension for engineer-facing explanations; not a core component.
15. Unexplored	Pixel-level localisation; rotation.

Matrix: few-shot Yes · rotation No · localisation No · unknown Partial · feedback No

P28. Dey et al. (ADCDS, imec) [44]

B. Dey, M. Monden, V. Blanco, S. Halder, S. De Gendt · ECCV 2024 Workshops · closeness ★★☆☆☆

In simple words. A pipeline that classifies, boxes and outlines nanoscale defects in SEM images. To avoid drawing pixel masks by hand, it learns masks from bounding boxes only.

1. Title	Advancing SEM Based Nano-Scale Defect Analysis in Semiconductor Manufacturing for Advanced IC Nodes
2. Authors	B. Dey, M. Monden, V. Blanco, S. Halder, S. De Gendt
3. Year	2024
4. Venue	ECCV 2024 Workshops
5. DOI / URL	https://doi.org/10.1007/978-3-031-92805-5_15
6. Research problem	Unified classification, detection and segmentation of SEM defects with cheap (box-level) labels.
7. Dataset	ADI SEM dataset (imec; not public).
8. Input modality	SEM images.
9. Proposed method	Deformable DETR detection feeds BoxSnake box-supervised instance segmentation.

10. Architecture	Deformable DETR + BoxSnake.
11. What it achieved	Demonstrated pixel-precise masks from box supervision on ADI SEM data (see paper for numbers).
12. Limitations	Supervised; closed set; private data.
13. What to learn	Pixel masks are expensive in fabs. Box-level annotation is a realistic compromise, and evaluation must match the annotation level.
14. Relation to this research	Informs the annotation strategy (Stage 7) and the supervised upper-bound baseline.
15. Unexplored	Few-shot normal references; unknown defects.

Matrix: few-shot No · rotation No · localisation Yes · unknown No · feedback No

P29. Dolgener Cantú et al. (InP golden die) [59]

E. Dolgener Cantú, R. K. Wittmann, O. Abdeen, P. Wagner, W. Samek, M. Baier, S. Lapuschkin · arXiv preprint 2506.10713 (2025) · closeness ★★★★☆

In simple words. For small-volume InP wafers there are too few identical dies to compare against. This work generates a realistic 'golden die' image from the CAD design and compares the real die with it.

1. Title	Deep Learning-based Multi Project InP Wafer Simulation for Unsupervised Surface Defect Detection
2. Authors	E. Dolgener Cantú, R. K. Wittmann, O. Abdeen, P. Wagner, W. Samek, M. Baier, S. Lapuschkin
3. Year	2025
4. Venue	arXiv preprint 2506.10713 (2025)
5. DOI / URL	https://arxiv.org/abs/2506.10713
6. Research problem	Reference-based defect detection when few repeated dies exist (multi-project wafers).
7. Dataset	InP multi-project wafer images with CAD layouts (see paper).
8. Input modality	Optical wafer images + CAD.
9. Proposed method	Deep network trained to render photo-realistic wafer images from CAD; pixel-wise comparison of simulated and real images.
10. Architecture	Image-to-image generative network (see paper).
11. What it achieved	Demonstrated simulated golden references for unsupervised detection (numbers not retrieved here).
12. Limitations	Preprint; needs CAD data; simulation errors become false alarms.
13. What to learn	When repeated dies are scarce, the reference can be synthesised. This is a fallback for cross-type inspection with very few samples.
14. Relation to this research	Alternative reference source for Contribution 1; relevant to cross-type experiments.
15. Unexplored	Combination with few real references; rotation.

Matrix: few-shot Partial · rotation Unclear · localisation Yes · unknown Partial · feedback No

P30. Chen (die detection in wafer images) [60]

H.-C. Chen · Applied Sciences 10(10):3423 (2020) · closeness ★★☆☆☆

In simple words. From a picture of a whole wafer, the method finds each die's row and column, builds a wafer map, and classifies defective and abnormal dies.

1. Title	Automated Detection and Classification of Defective and Abnormal Dies in Wafer Images
2. Authors	H.-C. Chen
3. Year	2020
4. Venue	Applied Sciences 10(10):3423 (2020)
5. DOI / URL	https://doi.org/10.3390/app10103423
6. Research problem	Die extraction and indexing from whole-wafer images, then defective-die classification.
7. Dataset	Scanning acoustic tomography (SAT) wafer images.
8. Input modality	SAT wafer images.
9. Proposed method	Clustering to estimate die rows and columns (instead of template matching), wafer-map construction, CNN classification.
10. Architecture	Clustering + CNN classifier.
11. What it achieved	Reported good identification of defective-die patterns and a wafer map for diagnosis.
12. Limitations	Die-level classification, no pixel localisation; single imaging modality.
13. What to learn	Die extraction and indexing (row, column) is the bridge between die images and wafer maps.
14. Relation to this research	Informs Stage 1 (die extraction) and Stage 9 (wafer-level indexing).
15. Unexplored	Few-shot anomaly detection inside each die.

Matrix: few-shot No · rotation Partial · localisation Partial · unknown No · feedback No

P31. Cai et al. (active learning) [38]

L. Cai, R. S. Pahwa, X. Xu, J. Wang, R. Chang, L. Zhang, C.-S. Foo · ICIP 2022 (arXiv 2507.17359) · closeness ★★☆☆☆

In simple words. Instead of asking an expert to label everything, the model picks the images it would learn most from. This reduces annotation effort for semiconductor defect segmentation.

1. Title	Exploring Active Learning for Semiconductor Defect Segmentation
2. Authors	L. Cai, R. S. Pahwa, X. Xu, J. Wang, R. Chang, L. Zhang, C.-S. Foo
3. Year	2022
4. Venue	ICIP 2022 (arXiv 2507.17359)
5. DOI / URL	https://arxiv.org/abs/2507.17359
6. Research problem	Reduce annotation cost for semiconductor defect segmentation.
7. Dataset	X-ray microscopy scans of semiconductor samples (see paper).
8. Input modality	X-ray microscopy.
9. Proposed method	Active learning sample selection for segmentation, addressing domain shift and class imbalance.
10. Architecture	Segmentation network + acquisition function.
11. What it achieved	Reported reduced annotation effort; identified domain shift and class imbalance as the main obstacles.
12. Limitations	X-ray, not SEM/optical; supervised segmentation.
13. What to learn	Engineer time is the scarce resource, so feedback requests should be prioritised.
14. Relation to this research	Supports an active-query option in the feedback loop.
15. Unexplored	Normal-reference memories; open-set.

Matrix: few-shot Not evaluated · rotation No · localisation Yes · unknown No · feedback Yes

E. Wafer-map (wafer-level) analysis

P32. Wu et al. (WM-811K) [41]

M.-J. Wu, J.-S. R. Jang, J.-L. Chen · IEEE Trans. Semiconductor Manufacturing 28(1) (2015) · closeness ★★☆☆☆

In simple words. This paper introduced WM-811K, a very large collection of wafer maps: pictures that show which dies on a wafer passed or failed. It also classifies the spatial failure patterns such as 'edge ring' or 'scratch'.

1. Title	Wafer Map Failure Pattern Recognition and Similarity Ranking for Large-Scale Data Sets
2. Authors	M.-J. Wu, J.-S. R. Jang, J.-L. Chen
3. Year	2015
4. Venue	IEEE Trans. Semiconductor Manufacturing 28(1) (2015)
5. DOI / URL	https://doi.org/10.1109/TSM.2014.2364237
6. Research problem	Recognise and retrieve wafer-level failure patterns at scale.
7. Dataset	WM-811K: 811,457 wafer maps; 172,950 labelled into 9 classes (incl. 'none').
8. Input modality	Wafer maps (die pass/fail grids), not images of die surfaces.
9. Proposed method	Radon-transform and geometry features; SVM classification; similarity ranking.
10. Architecture	Hand-crafted features + SVM.
11. What it achieved	Released the dataset that became the standard wafer-map benchmark.
12. Limitations	No die-surface appearance; no pixel masks; label imbalance (most labelled maps are 'none').
13. What to learn	Wafer-level spatial patterns are a different problem from die-surface defects. Do not mix the two.
14. Relation to this research	Data source for the optional wafer-level context study (Stage 9, Contribution 4) only.
15. Unexplored	Linking die-image anomalies to wafer-map patterns.

Matrix: few-shot No · rotation Partial · localisation No · unknown No · feedback No

P33. Kim, Yoon, Kim [42]

H. Kim, H. Yoon, H. Kim · J. Manufacturing Science and Engineering (ASME), 2024 · closeness ★★☆☆☆

In simple words. To recognise a new wafer-map failure pattern from only a few examples, the model is first trained on simulated wafer maps and then fine-tuned on the few real ones.

1. Title	Few-Shot Classification of Wafer Bin Maps Using Transfer Learning and Ensemble Learning
2. Authors	H. Kim, H. Yoon, H. Kim
3. Year	2024
4. Venue	J. Manufacturing Science and Engineering (ASME), 2024
5. DOI / URL	https://doi.org/10.1115/1.4065255
6. Research problem	Few-shot classification of wafer bin maps.
7. Dataset	Simulated wafer bin maps for pre-training; a real WBM dataset for evaluation (see paper).
8. Input modality	Wafer bin maps.
9. Proposed method	Pre-train on simulated maps; fine-tune several classification heads (ensemble) on few real maps.
10. Architecture	CNN with ensemble of classification layers.
11. What it achieved	Reported to outperform meta-learning methods commonly used for few-shot WBM classification on a real dataset.
12. Limitations	Wafer-map level; closed set.

13. What to learn	Simulation-based pre-training is a practical answer to label scarcity in fabs.
14. Relation to this research	Supports synthetic pre-training and the few-shot wafer-level extension.
15. Unexplored	Die-image level; open-set.

Matrix: few-shot Yes · rotation No · localisation No · unknown No · feedback No

P34. Shin et al. (open-set WBM) [8]

J.-S. Shin, M.-J. Kim, B.-S. Kim, D.-H. Lee · Computers in Industry 164:104208 (2025) · closeness ★★★☆☆

In simple words. In fabs, engineers usually spot new wafer-map patterns by experience. This method automatically flags maps whose pattern does not belong to any known class.

1. Title	Enhanced Detection of Unknown Defect Patterns on Wafer Bin Maps Based on an Open-Set Recognition Approach
2. Authors	J.-S. Shin, M.-J. Kim, B.-S. Kim, D.-H. Lee
3. Year	2025
4. Venue	Computers in Industry 164:104208 (2025)
5. DOI / URL	https://doi.org/10.1016/j.compind.2024.104208
6. Research problem	Detect unknown defect patterns on wafer bin maps.
7. Dataset	Wafer bin maps incl. new patterns from real manufacturing (see paper).
8. Input modality	Wafer bin maps.
9. Proposed method	Open-set recognition with constrained mean filtering and Radon-transform features; one-class SVM.
10. Architecture	Feature extraction + one-class SVM.
11. What it achieved	Reported over 98% detection performance for various defect classes on new patterns from real manufacturing.
12. Limitations	Wafer-map level; no die-image localisation.
13. What to learn	Open-set recognition works at wafer level. The same idea is needed inside each die image.
14. Relation to this research	Evidence for Stage 8 design at wafer level; motivates die-level open-set triage.
15. Unexplored	Die-image open-set localisation.

Matrix: few-shot Not evaluated · rotation No · localisation No · unknown Yes · feedback No

15 Research Gap

15.1 The closest existing research

Five papers overlap most with the proposed work. For each we state the overlap, the difference, what is already done, what remains open and how this project could extend it.

AnomalyDINO (WACV 2025) [3]

Overlap	Same core mechanism: a few normal images, frozen self-supervised patch features, nearest-neighbour patch scores, pixel-level maps, no training.
Difference	Natural-object benchmarks; no alignment step; no known/unknown triage; no feedback; no semiconductor data; no transfer splits.
Already accomplished	Showed that 1-shot patch matching with DINOv2 reaches 96.6% image AUROC on MVTec AD.
Remaining gap	Behaviour on SEM and optical die images, on defects smaller than the 14-px ViT patch, and under arbitrary rotation.
How this work extends it	Use as the primary few-shot baseline and encoder option; add alignment, DPR evaluation, triage and gated updates; report by defect size.

RegAD (ECCV 2022) [6]

Overlap	Few-shot anomaly detection that aligns test features to a few normal support images; category-agnostic, no re-training for new categories.
Difference	Alignment is learned (feature-level STN) on natural-object categories; defect preservation is not measured; no unknown triage or feedback.
Already accomplished	3–8% AUROC gains over earlier few-shot methods on MVTec AD and MPDD at 2–8 shots.
Remaining gap	Whether alignment (learned or classical) weakens real defects; explicit, auditable die registration; semiconductor evaluation.
How this work extends it	Compare learned registration with classical 'warp the reference' registration under a controlled rotation sweep, with DPR on injected defects.

GraphCore (ICLR 2023) [7]

Overlap	Targets the same failure mode: few-shot memories break under rotation; proposes rotation-invariant patch features.
Difference	Solves rotation by invariance, not alignment; evaluated on objects (MVTec AD, MPDD), not periodic die layouts.
Already accomplished	Large few-shot gains, especially on MPDD (e.g. +25.5% AUC at 1-shot).
Remaining gap	Invariance cannot detect orientation-dependent defects (a correct structure in the wrong orientation); not tested on dies.
How this work extends it	Include orientation-dependent defects in the synthetic benchmark and compare invariance with alignment on DPR and false alarms.

Abbas et al., HITL memory bank correction (arXiv 2026) [10]

Overlap	Same feedback mechanism: experts correct a PatchCore memory directly, without retraining, through a novelty gate; starts from 10 golden samples.
Difference	Simulated, noise-free feedback; MVTec AD only; no unknown triage; no rollback or quarantine; their defect-memory extension failed.
Already accomplished	Recovered a median 66% of the gap to a full bank; improved 12 of 15 categories, harmed none; showed naive evaluation is inflated by memorisation.
Remaining gap	Robustness to wrong feedback; contamination and drift monitoring; semiconductor data; a defect memory that helps triage rather than detection.

How this work extends it	Adopt their held-out protocol and gate as the baseline; add noisy-feedback arms, quarantine, second review, rollback and a canary set.

Zhao et al., image-intrinsic normality for IC defects (arXiv 2025) and IC-DefectNCD (arXiv 2025) [45]

Overlap	Semiconductor (IC SEM) defect localisation without labels, and (in the follow-up) discovery of novel defect classes, which is very close to the unknown-defect goal.
Difference	Deliberately avoids external reference images, arguing they are brittle for IC images; private data; no feedback loop.
Already accomplished	Reference-free unsupervised segmentation and support-set-free novel class discovery on real IC SEM data (numbers not verified here).
Remaining gap	A direct comparison of reference-memory vs intrinsic-normality cues on public data, and their combination.
How this work extends it	Use an intrinsic self-similarity score as the 'without reference memory' ablation; test whether combining both cues helps on unpatterned SEM.

15.2 Gap statement

DEFENSIBLE RESEARCH GAP

Few-shot, reference-based anomaly detection with frozen pretrained features is established on natural-object benchmarks [1, 3, 4]. Learned registration [6] and rotation-invariant features [7] have improved few-shot detection under pose variation. Open-set recognition is studied for wafer maps [8]. Continual learning has been benchmarked on fab SEM data [9]. Training-free memory correction from simulated expert feedback has been demonstrated on MVTec AD [10]. Semiconductor-specific work uses private data [52, 45, 57, 44] and, in one line of work, argues *against* external references for IC images [45].

Within the literature we reviewed, **we found no study that, on semiconductor die imagery with public, reproducible splits:**

1. measures whether geometric alignment or rotation invariance **preserves or weakens genuine defects** (not only whether detection accuracy holds), including orientation-dependent defects;
2. performs **pixel-level three-way triage** (normal / known class / unknown) driven by normal references plus a small known-defect memory, evaluated by leave-one-class-out on expert masks;
3. evaluates **memory updates from imperfect engineer feedback** (wrong labels, wrong locations) with quarantine, gating and rollback, under a held-out protocol that rules out memorisation;
4. separates **same-wafer, cross-wafer and cross-type** transfer as distinct, explicitly defined evaluation conditions.

This is a gap in **evaluated combinations and evaluation protocols**, not a claim that the individual components are new.

WHAT WOULD CLOSE THE GAP (AND HOW THE PROJECT WOULD RESPOND)

If a paper appears that already performs items 1–3 on semiconductor data, the project becomes a reproduction-and-extension study on public data, focusing on whichever item remains open. A monthly literature alert is part of the risk plan (Section 26).

16 Novelty and Original Contributions

16.1 Novelty matrix (Artifact 2)

Table 7. Novelty matrix: proposed contributions vs closest existing research. The bar shows how strong and testable the remaining gap is (■ = 1 point of 5), as judged from the reviewed literature.

ID	Contribution	Closest prior work	Already covered	Still open	Type of claim	Status / gap
C1	**Sample-guided cross-die knowledge transfer**	PatchCore, SPADE (memory matching); AnomalyDINO, UniVAD, WinCLIP (few-shot); RegAD, InCTRL, ResAD (cross-category transfer)	Few-shot reference matching on natural-object benchmarks is **established**. Cross-category generalisation exists for natural objects.	Systematic same-wafer vs cross-wafer vs cross-type evaluation on die imagery; explicit evidence of when transfer fails.	Combination of existing methods + new evaluation protocol	Partially explored ■■■□□
C2	**Defect-preserving geometric alignment**	RegAD (learned registration); GraphCore (rotation invariance); PaDiM (alignment-dependent); classical registration (ECC, Fourier–Mellin); Zontak & Cohen (registration-free)	Alignment and invariance for few-shot AD exist. Classical die registration is standard industrial practice.	Measuring whether alignment or invariance *removes or weakens* real defects (defect preservation), including orientation-dependent defects; 'warp the reference, not the target' design.	Potentially novel evaluation + design principle	Gap found in reviewed literature ■■■■□
C3	**Reference-guided unknown-anomaly localisation**	One-class AD (PatchCore etc.) flags any deviation; DRA (open-set supervised AD); Shin et al. (open-set at wafer-map level); IC-DefectNCD (novel class discovery in IC SEM)	Detecting deviations is established; open-set recognition exists at image or wafer-map level; NCD exists on private IC data.	Pixel-level three-way triage (normal / known class / unknown) driven by normal references plus a small known-defect memory, evaluated with leave-one-class-out on public SEM masks.	Combination with potentially novel evaluation	Partially explored ■■■■□
C4	**Cross-die and wafer-level contextual analysis**	WM-811K line of work (wafer-map pattern recognition); Chen 2020 (die indexing); industrial die-to-die practice	Wafer-map pattern recognition is mature; die-image inspection is mature; they are usually studied separately.	Linking per-die anomaly scores to wafer-level spatial statistics. **No public dataset links die images to wafer coordinates**, so this can only be tested on synthetic or partner data.	Proposed combination; unverified hypothesis	Weakly testable with public data ■■□□□
C5	**Verified feedback-driven reference adaptation**	Abbas et al. 2026 (HITL memory correction, simulated feedback); UCAD (continual AD); Prasad et al., PIRB (continual learning in semiconductors); Cai et al. (active learning)	Training-free memory correction from feedback was shown on MVTec AD with clean simulated feedback (preprint).	Noisy or wrong feedback, quarantine and rollback, contamination and false-positive drift, forgetting on semiconductor data.	Extension of very recent work	Recently explored (2026 preprint) ■■■□□

16.2 Classification of claims

Table 8. What kind of claim each part of the proposal makes.

Category	Items
Existing established methods	Pretrained-feature nearest-neighbour anomaly detection; coreset memory; classical registration (ECC, Fourier–Mellin, feature + RANSAC); prototypical classification; open-set rejection by distance threshold; wafer-map pattern recognition; replay and regularisation for continual learning.
Proposed combinations of existing methods	Registration that warps references instead of the target, combined with alignment-aware patch matching; normal-memory test followed by prototype triage; memory edits gated by a novelty test plus a held-out regression check with rollback; per-die scores aggregated to wafer maps.
Potentially novel contributions	Defect preservation ratio (DPR) as an evaluation of alignment and preprocessing; the joint public protocol separating same-wafer / cross-wafer / cross-type transfer on die imagery; pixel-level leave-one-class-out open-set triage on Carinthia-S; noisy-feedback robustness study of memory editing in a semiconductor setting.
Unverified hypotheses	H1–H7 (Section 12): for example, that warping references preserves defects better than warping targets; that held-out classes are flagged as unknown rather than misclassified; that gated updates resist 10% wrong feedback; that wafer context improves die ranking.

16.3 Recommended contributions for a student project

PRIORITISE THESE

1. **C2 · Defect-preserving alignment (highest priority).** Clear gap, cheap to test (rotations and injected defects are fully controlled), and it produces a new, reusable evaluation tool (DPR). Feasible on public and synthetic data.
2. **C3 · Reference-guided unknown-anomaly localisation.** Testable on public data with real expert masks (Carinthia-S, leave-one-class-out). Results are meaningful even if the answer is negative.
3. **C5 · Verified feedback adaptation under noisy feedback** (if time allows). Builds directly on a 2026 preprint [10], so its novelty is the noisy-feedback robustness and the semiconductor setting. Simulation is cheap; the main risk is evaluation leakage, which the protocol addresses.

C1 is not a separate contribution but the shared experimental backbone: every result is reported for K = 1–10 and for the transfer splits. **C4** (wafer-level context) is an optional extension because no public dataset links die images to wafer coordinates.

17 Dataset Investigation

The core experiment needs, ideally: (i) many images of **patterned dies of one design**, so that some can be references and others targets; (ii) **pixel-level defect masks**; (iii) **wafer and die coordinates**, to form same-wafer and cross-wafer splits; (iv) **several defect classes**, to hold some out as unknown; and (v) **several designs**, for cross-type tests. This section checks what public data actually offers. All counts were verified against the official record (through search results quoting it) unless the table says otherwise.

17.1 Carinthia SEM defect dataset

What it is. Zenodo record 10715190 (version v2, published 27 February 2024), credited to KAI GmbH, Infineon Technologies Dresden and Infineon Technologies Austria [39]. It contains **4,591 SEM images of defects on one production layer of unstructured (unpatterned) semiconductor wafers**, unevenly distributed over **six defect classes**, packaged as a single `data.zip` with a separate HTML description. Labels are image-level (a CSV mapping image path to label).

- **SEM availability:** yes, openly downloadable from Zenodo.
- **Defect categories:** six classes. The class names should be taken from the record's description file; they were not confirmed in this session.
- **Resolution and annotation format:** image-level labels in CSV. Resolution not verified (a third-party listing mentions 480 × 480 px).
- **Class imbalance:** the record states that the distribution is uneven; exact counts per class must be computed in the dataset audit (milestone 2).
- **Access and permissions:** open download. The **licence was not verified**. Note also that Carinthia-S cites the concept DOI 10.5281/zenodo.10696644, which differs from the v2 record; cite the version actually used.

17.2 Carinthia-S

What it is. Zenodo record 16895427 (v1, 20 August 2025), credited to KAI GmbH and the University of Sarajevo, Faculty of Electrical Engineering [61]. It is described as an enhanced version of Carinthia with **expert-validated binary segmentation masks for each of the 4,591 defect images**.

- **Suitability for pixel-level localisation:** high. It is the best public source of real SEM defect masks found in this review.
- **Annotation quality:** masks are described as expert-validated. Inter-annotator agreement is not reported in the sources we could access. The audit should inspect a random sample, checking anti-aliased edges, empty masks and class consistency.
- **Key limitation:** the wafers are **unpatterned**, so there is no repeated die layout and no die-to-die correspondence. Rotation tests on these images test texture robustness, not layout alignment. Whether a defect-free subset exists is not confirmed; if not, normal references must be built from defect-free regions of defect images (masked by the expert masks).
- **Access and DOI:** open download via Zenodo; DOI via the record; **licence not verified**.

17.3 WM-811K

What it is. The wafer-map dataset of Wu, Jang and Chen [41]: **811,457 wafer maps**, of which **172,950 are labelled** into **nine classes**: Center, Donut, Edge-Loc, Edge-Ring, Loc, Near-full, Random, Scratch and None. Each map is a grid of per-die values (no die, pass, fail).

- **Representation:** categorical die-level grids, not images of die surfaces.

- **Limitations for microscopic die-image inspection:** no die appearance, no pixel masks, no possibility of testing rotation of die images, alignment, localisation inside a die or reference matching of die structures. The labelled part is dominated by "None".
- **Use here:** only for the optional wafer-level study (Stage 9, RQ7), such as few-shot or open-set recognition of spatial patterns, and as a realistic source of spatial cluster shapes for the synthetic wafer benchmark.

17.4 MVTec AD

What it is. The standard industrial anomaly-detection benchmark [62, 23]: **5,354 high-resolution colour images in 15 object and texture categories**, with a defect-free training set, a test set with more than 70 defect types, and **pixel-precise masks**. Licence: CC BY-NC-SA 4.0 (non-commercial).

- **Suitability:** excellent for initial method development, and for checking the implementation by reproducing published PatchCore and AnomalyDINO numbers. The few-shot protocol (sampling K normal training images) is well established.
- **Domain difference:** natural colour images of objects and textures, not SEM or optical die images; most categories are roughly aligned; defects are often larger relative to the image than typical die defects. MVTec AD 2 [63] adds lighting changes and very small defects and is useful for illumination robustness.

17.5 Additional datasets

Wafer Surface Defect (IEEE DataPort) [40]: **500 optical images, 680 × 680 px, each showing one individual die**; it includes defect-free dies and four defect types (particle, scratch, stain, liquid residual), annotated with YOLO-format bounding boxes. It is the closest public match to the "normal reference dies → inspect other dies" setting, but it is small, has boxes rather than masks, and has no wafer coordinates. Per-class counts and the licence were not verified, and access may require an IEEE DataPort account.

MixedWM38 [64]: about 38,015 wafer maps (52 × 52) with 38 patterns (1 normal, 8 single, 29 mixed). Wafer-level only; the count comes from one secondary source and should be confirmed.

VisA [65]: 10,821 images (9,621 normal, 1,200 anomalous) in 12 subsets with pixel masks; four subsets are printed circuit boards. It is the best public proxy for **patterned, repeated designs** and is useful for testing registration on real repeated layouts. The licence statements conflict between sources.

MPDD [66]: metal parts in varied positions and orientations, used by RegAD and GraphCore to evaluate pose variation. Counts not verified here.

Private semiconductor datasets. Most recent semiconductor papers (SEM-CLIP, IC-DefectNCD, imec ADI/AEI, LogicQA's SEM set) use private data [52, 57, 9, 58]. Their results cannot be reproduced and serve only as context.

17.6 Dataset comparison table (Artifact 4)

Table 9. Dataset comparison (Artifact 4). Figures as stated in official records; unverified items are marked.

Dataset	Modality	Samples	Labels	Segmentation masks	Official link	Intended use here	Limitations
Carinthia [39]	Grayscale SEM images of defects on one production layer of **unstructured (unpatterned)** wafers	4,591 images	6 defect classes, unevenly distributed; image-level class label (CSV: image path, label)	No (image-level only)	zenodo.org/records/10715190	Real SEM defect appearance; known-class prototypes; leave-one-class-out unknown tests	No repeated die layout, no die-to-die references, no wafer coordinates; resolution and licence not verified in this session; record DOI (v2) differs from the concept DOI cited by Carinthia-S

Dataset	Modality	Samples	Labels	Segmentation masks	Official link	Intended use here	Limitations
Carinthia-S [61]	Same SEM images as Carinthia	4,591 images + 4,591 masks	6 classes (as Carinthia)	Yes: expert-validated binary segmentation masks, one per image	zenodo.org/records/16895427	**Core public SEM set** for pixel-level localisation and leave-one-class-out unknown-defect experiments	Unpatterned surfaces, so no die-to-die correspondence; whether a defect-free subset exists is not confirmed (third-party listings mention a 'no visible defect' class and 480×480 px images, not verified); licence not verified
Wafer Surface Defect [40]	Optical images; **each image is one individual die**	500 images, 680×680 px	Defect-free dies + 4 defect types: particle, scratch, stain, liquid residual	No: YOLO-format bounding boxes (class, x, y, w, h)	doi.org/10.21227/7x9f-zt52	Closest public match to the 'few normal reference dies → inspect other dies' setting; box-level localisation	Small; per-class counts and licence not verified; access may require an IEEE DataPort account; no wafer coordinates; boxes only
WM-811K [41]	**Wafer maps** (die-level pass/fail grids), not die-surface images	811,457 wafer maps; 172,950 labelled	9 classes: Center, Donut, Edge-Loc, Edge-Ring, Loc, Near-full, Random, Scratch, None	No (map-level labels)	doi.org/10.1109/TSM.2014.2364237	Wafer-level spatial-pattern context (Stage 9, Contribution 4); few-shot and open-set at wafer level	No die appearance; cannot test pixel localisation, rotation of die images or die-to-die matching; heavy class imbalance; licence unclear
MixedWM38 [64]	**Wafer maps** (52×52 grids)	38,015 maps (single secondary source)	38 patterns: 1 normal, 8 single-defect, 29 mixed (~1,000 per class)	No	github.com/Junliangwangdhu/WaferMap	Mixed wafer-level patterns; wafer-level unknown-pattern tests	Maps only; count from one secondary source; repository notes corrected labels (C7, C9)
MVTec AD [62]	High-resolution RGB images of industrial objects and textures	5,354 images, 15 categories	Normal training set; test set with >70 defect types	Yes: pixel-precise masks	www.mvtec.com/research-teaching/datasets/mvtec-ad	Method development; reproducing PatchCore / AnomalyDINO numbers to validate the implementation	Not semiconductor; CC BY-NC-SA 4.0 (non-commercial); objects are roughly aligned in most categories
MVTec AD 2 [63]	High-resolution images, 8 scenarios incl. lighting changes and very small defects	>8,000 images	Public test set with masks + private test set (evaluation server)	Yes (public test)	www.mvtec.com/research-teaching/datasets/mvtec-ad-2	Illumination robustness; small-defect behaviour	Not semiconductor; CC BY-NC-SA 4.0
VisA (PCB subsets) [65]	RGB images; 12 subsets, 4 of them printed circuit boards	10,821 images (9,621 normal, 1,200 anomalous)	Normal / anomalous per image	Yes	github.com/amazon-science/spot-diff	Best public **patterned, repeated-design proxy**: registration and alignment tests on real repeated layouts	PCBs, not semiconductor dies; licence statements conflict between sources
MPDD [66]	RGB images of metal parts in varied positions and orientations	Not verified in this session	Normal / defective	Yes (per repository description; not verified here)	github.com/stepanje/MPDD	Orientation / position variation benchmark used by RegAD and GraphCore	Not semiconductor; counts and licence not verified here
KolektorSDD2 [67]	Surface images of production items	3,335 images (356 defective, 2,979 defect-free)	Defective / defect-free	Yes: pixel annotations	www.vicos.si/resources/kolektorsdd2/	Optional surface-defect sanity check with many normals	Not semiconductor; CC BY-NC-SA 4.0 per ViCoS page

17.7 Can the core experiment be done with public data?

HONEST ANSWER: ONLY PARTLY

No public dataset found in this review combines patterned dies of one design, pixel masks, wafer/die coordinates and multiple designs. Specifically:

- **Few-shot detection and localisation** can be tested on real data: MVTec AD (method check), Carinthia-S (SEM with masks; references from normal regions) and Wafer Surface Defect (die-level references; box-level localisation).
- **Unknown-defect detection** can be tested on real data: leave-one-class-out on Carinthia-S (6 classes) and on Wafer Surface Defect (4 classes).
- **Rotation robustness** can be tested on real images by applying controlled rotations, but layout alignment of real *patterned* dies can only be tested on VisA PCBs (proxy) and on synthetic dies.
- **Same-wafer vs cross-wafer** splits **cannot** be formed on any public semiconductor image dataset, because none provides wafer IDs and die coordinates for die images. They are tested on synthetic wafers and, if possible, on partner data.
- **Cross-type** transfer can be *approximated* across datasets (e.g. enrol on Wafer Surface Defect, test on Carinthia-S), with the caveat that modality also changes.

Realistic experimental alternative: a four-tier data plan

Table 10. Tiered data plan. Results from each tier are reported separately and never pooled.

Tier	Data	What it can show	Claim strength
T0 · Method check	MVTec AD, MVTec AD 2, VisA	Implementation correctness; few-shot behaviour; illumination robustness; registration on repeated layouts (VisA PCB)	Not semiconductor evidence
T1 · Real SEM	Carinthia / Carinthia-S	Pixel localisation on real SEM defects; known/unknown triage; texture-level rotation robustness	Real data, unpatterned surfaces
T2 · Real die images	Wafer Surface Defect	Die-level few-shot references; box-level localisation; leave-one-class-out with 4 classes	Real dies; small set; boxes only
T3 · Synthetic patterned wafers	Procedurally generated dies on synthetic wafers, with defects injected from real Carinthia-S masks and procedural shapes	Same-wafer vs cross-wafer vs cross-type; exact rotation ground truth; DPR; wafer-level clusters (shapes inspired by WM-811K)	Controlled, exact ground truth; external validity limited
T4 · Partner data (if obtained)	Real patterned die images with wafer coordinates from a university lab or industry partner	External validity of T3 findings	Strongest; not guaranteed

Synthetic patterned-wafer benchmark (T3) design

1. **Layouts:** several die designs built from parametric primitives (pad rings, via arrays, line gratings, logic blocks, orientation marks), rendered with an SEM-like or optical-like appearance model (blur, edge brightening, shot-like noise). The toy generator in `prototype/` is the starting point.
2. **Wafers:** each design is placed on a die grid inside a wafer outline. Each wafer gets its own nuisance parameters (global brightness, contrast, focus blur, noise level, small layout bias) to create controlled wafer-to-wafer shift.
3. **Geometry:** each die image gets a recorded transform (rotation in {0, 90, 180, 270} plus a residual of up to ±15°, or any angle in 0–360° for the sweep; translation; scale within ±10%).
4. **Defects:** injected with recorded masks. Shapes are cut from Carinthia-S masks (real shape statistics) or drawn procedurally (bridge, open, missing via, particle, scratch, stain), with sizes from 1 px upwards so small-defect behaviour can be studied. Orientation-dependent defects (e.g. a correctly shaped line at a wrong angle) are included on purpose to test invariance.

5. **Spatial patterns:** defective dies are placed randomly or in clusters (edge ring, line, local blob) inspired by WM-811K classes [41].
6. **Realism checks:** compare intensity histograms, noise power spectra and edge profiles with real SEM (Carinthia) and optical (Wafer Surface Defect) images. Report synthetic results only as synthetic.

17.8 Annotation needed for each output

Table 11. Annotation level required by each output and metric.

Output	Minimum annotation	Metrics enabled	Available in
Normal / anomalous decision	Image-level label	Image AUROC, AUPRC, F1	All image datasets
Known defect class	Image- or region-level class	Accuracy, macro-F1, open-set F1	Carinthia(-S), Wafer Surface Defect, MVTec AD
Bounding boxes	Boxes	Box IoU, detection AP	Wafer Surface Defect; derivable from masks
Pixel anomaly map / mask	Pixel masks	Pixel AUROC, AUPRO, Dice, IoU	Carinthia-S, MVTec AD, VisA, synthetic
Rotation robustness, DPR	Known transform + mask	AUROC vs angle, DPR	Any masked set with controlled rotation; synthetic
Same- vs cross-wafer	Wafer ID + die index	Per-split AUROC/AUPRO gap	Synthetic; partner data

18 Proposed Technical Methodology

The methodology follows the ten stages of the research brief. For each stage: what it does, which options were compared, what is recommended for the first implementation, and what can go wrong.

18.1 Stage 1: Data input

Table 12. Required and optional inputs.

Input	Required?	Details
Reference die collection	Yes (K ≥ 1)	K images of normal dies of one design and layer, same modality as the targets. Prefer dies spread across the wafer (centre and edge) to capture natural variation. For SEM sets without normal images, normal regions of defect images, masked by the expert mask, serve as references.
Normal samples for calibration	Strongly recommended	A few extra normal dies, not used as references, to set the threshold (Section 18.6). With K = 1 and none available, use augmentation-based self-scoring.
Known defective samples	Optional	A few verified defect examples per known class (with masks or boxes) to build defect prototypes for triage.
Target die images	Yes	Dies to inspect; any orientation; same design (Scenarios 1, 2a) or new design (2b).
Metadata	Optional	Wafer ID, lot, die index (i, j), layer, tool, magnification, timestamp. Required for wafer-level analysis and for leak-free splits.

Minimum viable input: one normal reference die and a set of target dies. **Recommended:** K = 5 references, 3–5 calibration normals, metadata, and 2–5 examples per known defect class.

18.2 Stage 2: Image preprocessing

Table 13. Preprocessing options and their risk to small defects.

Operation	Purpose	Risk to defects	Decision
Resizing	Match the encoder's input size	**High.** Downsampling by a factor of 4 can shrink a 3-pixel particle below one pixel. Prefer tiling at native resolution.	Tile, do not shrink
Noise reduction (Gaussian, median, non-local means)	Suppress sensor or shot noise	**High.** Removes exactly the small, high-frequency structures that small defects are made of; a median filter erases isolated bright pixels.	Avoid before scoring
Contrast normalisation (per-image z-score, percentile stretch)	Remove gain/offset differences	Low, if computed globally. Local equalisation (CLAHE) can amplify noise or flatten low-contrast stains.	Global only
Illumination correction	Remove slow brightness gradients	Medium. A large-scale background estimate (e.g. very wide Gaussian) is safe for small defects but can absorb large, faint stains.	Wide kernel; ablate
Image quality assessment	Reject blurred, saturated or charged images	None to defects; prevents false alarms from bad images.	Yes
Die cropping and extraction	Isolate each die, assign (i, j)	Low; cropping too tight can cut defects at the die edge.	Crop with margin

> **RULE**
>
> Any operation that is applied to the target must also be applied, identically, to the references, and every preprocessing choice is checked with the defect preservation ratio (Section 19.6) on injected defects before it is adopted.

18.3 Stage 3: Geometric alignment

Table 14. Alignment options.

Approach	How it works	Strengths	Weaknesses on dies
Feature-based (SIFT/ORB + RANSAC) [27, 28, 29]	Match keypoints, robustly fit a transform	Handles large rotation/scale	Periodic layouts give many identical keypoints, so matches are ambiguous; few keypoints on smooth SEM regions
Template matching (normalised cross-correlation)	Slide a template, take the correlation peak	Simple, robust for translation	Rotation needs a search over angles; peaks repeat with the layout pitch
Classical intensity registration: phase correlation [25], Fourier–Mellin [26], ECC [24]	Estimate shift (and rotation/scale via log-polar FFT); refine by maximising a correlation coefficient	Global, sub-pixel; ECC is invariant to brightness and contrast	Local optima; Fourier–Mellin is sensitive to large non-overlapping content
Learned registration (STN, RegAD) [30, 6]	Network predicts the warp	Learns from data; differentiable	Needs training data; opaque; may also "align away" defects
Rotation augmentation of the memory	Store references at several angles	No estimation step; robust to 90° steps	Memory grows; fine angles need many copies; can make the memory accept wrongly oriented structures
Invariant features (GraphCore) [7]	Features that do not change with rotation	No alignment needed	Blind to orientation-dependent defects

PRACTICAL BASELINE (RECOMMENDED FIRST IMPLEMENTATION)

1. **Coarse orientation:** evaluate the four 90° rotations of the reference template against the target with normalised cross-correlation on downsampled images; keep the best.
2. **Residual rotation and scale:** Fourier–Mellin (log-polar FFT) on the coarse-aligned pair; or a fine angle search in ±15°.
3. **Refinement:** ECC with a Euclidean (rotation + translation) or similarity model on a Gaussian pyramid, using die-edge and seal-ring regions to avoid pitch ambiguity.
4. **Apply to the references, not the target:** warp each reference into the target frame (bilinear), compute a validity mask for regions the references do not cover, and exclude them from scoring.
5. **Quality check:** if the post-alignment correlation is below a threshold, flag the die as "alignment uncertain" and fall back to a rotation-augmented memory.

Limitations: large non-rigid distortions (e.g. SEM charging drift) are not modelled; periodic structures can still cause pitch-sized errors; very small dies with little structure may not align. All are measured by target registration error on synthetic data.

18.4 Stage 4: Feature extraction

What a feature vector is. A list of numbers (for example 1,024 of them) that describes a region of an image. A good encoder places similar-looking regions close together in this space, so "how different is this region?" becomes "how far apart are these vectors?".

What a feature map is. Encoders keep the spatial layout: a 256 × 256 image becomes, say, a 32 × 32 grid of feature vectors. Each grid cell describes a small patch of the image, its *receptive field*. Comparing feature maps cell by cell gives a spatial anomaly map.

Why multiple scales matter. Early layers describe fine detail (edges, small dots) at high resolution; deeper layers describe larger context (a via in an array, the end of a line) at low resolution. A 2-pixel missing via is visible in early layers; a large stain is better captured in deep layers. PatchCore uses mid layers 2 and 3 and averages each feature with its neighbours, a compromise between detail and context [1].

Why tiny defects need spatially informative features. If one feature cell covers 16 × 16 pixels and a defect occupies 2 × 2 of them, the defect changes only a small part of that cell's description and can be lost in natural variation. Remedies: features from earlier layers, smaller stride, larger input (tiling at native resolution), or upsampling the image so that the defect spans more of a patch. Experiment F reports results by defect size so this effect becomes visible.

Table 15. Encoder candidates.

Encoder	Type	Patch / stride	Pros	Cons	Plan
Wide-ResNet-50 (layers 2+3) [48]	CNN, ImageNet	8–16 px	PatchCore default; reproducible; good locality	Natural-image bias	**Primary**
ResNet-18 [47]	CNN	8–16 px	Fast; low memory	Weaker features	Low-GPU fallback
EfficientNet [49]	CNN	varies	Efficient	Less used in AD; tuning needed	Optional
ConvNeXt [50]	Modern CNN	varies	Strong features	Heavier	Optional
DINOv2 ViT-S/14 [46]	Self-supervised ViT	14 px	Strong few-shot results [3]	Fixed 14-px patches; small defects need upsampling	**Primary**
CLIP ViT [68]	Vision–language	14–32 px	Text prompts (WinCLIP)	Coarse; natural-image vocabulary	Baseline only
Self-supervised on SEM	DINO-style fine-tuning on unlabelled SEM	—	Domain-specific features	Compute; risk of collapse	Advanced extension

18.5 Stage 5: Reference memory

How the model "remembers" without retraining. Enrolment passes each reference through the frozen encoder and appends its patch features to a list, the memory M. That is the whole learning step. Inspection asks nearest-neighbour questions of M. Adding, removing or relabelling items changes the system's behaviour immediately and reversibly; no gradient descent is involved.

- **Feature bank (normal memory M_N).** All patch features of the aligned references, with metadata (reference ID, position, scale, memory version).
- **Coreset subsampling.** Greedy selection of a subset that covers the bank (e.g. 10–25%), keeping search fast [1]. With $K \leq 10$ dies the bank is small, so subsampling may be unnecessary.
- **Nearest-neighbour matching with an alignment-aware window.** Because references are aligned, a target patch at position p is first compared with reference patches near p (window of $\pm w$ cells). This is faster and stricter: a via-array patch cannot be "explained" by a similar via array elsewhere on the die. A global search is kept as a second score; disagreement between the two flags misalignment.
- **Prototype representations.** Mean features of groups (e.g. of each defect class) for compact storage and classification [14].
- **Known-defect memory M_D.** Region features of verified defects, grouped by class, kept separate from M_N so that defects never contaminate the normal memory.
- **Memory quality control.** Before admission: remove features from low-quality images; detect outlier references (a reference whose patches are far from the other references' patches may contain a defect: leave-one-out scoring); deduplicate; record provenance and version.

18.6 Stage 6: Anomaly detection

Feature distance and similarity. For each target patch feature, compute its Euclidean distance to the nearest memory feature (or 1 − cosine similarity). Distance 0 means "an identical patch exists in the references".

Normality and anomaly score. The anomaly score of a patch is that nearest distance; a normality score is its negative or a monotone transform. The image score $A(x)$ is the maximum patch score, or the mean of the top-q% of patch scores, which is more robust to single noisy pixels.

Score normalisation. Distances depend on the encoder and the die type. To compare across die types, standardise with statistics from leave-one-out scoring of the references (each reference scored against the others) and the calibration normals: $z = (s - \mu_{ref}) / \sigma_{ref}$.

Threshold selection. Choose τ so that a target false-alarm rate α on normal dies is met: τ = the (1 − α) quantile of image scores of held-out normals (for image decisions) or of pixel scores (for maps). With K = 1 and no held-out normals, build pseudo-normals by small random shifts, rotations and noise of the reference and score them. This underestimates natural variation, so α is then reported as optimistic. Thresholds are always selected on validation data, never on test data.

Confidence and uncertainty. Report (i) the margin $(A - \tau)/\sigma_{ref}$; (ii) agreement between the windowed and global nearest-neighbour scores; (iii) optionally, variation of the decision across the K leave-one-out sub-memories (a cheap ensemble). Low margin or disagreement sends the die to review. Scores can be mapped to calibrated probabilities with a validation set (isotonic regression or temperature-style scaling [69]) when enough labelled validation data exist.

18.7 Stage 7: Defect localisation

- **Pixel-level anomaly maps:** the patch score grid is bilinearly upsampled to image size and lightly smoothed (Gaussian $\sigma \approx 2$–4 px). This smoothing affects the *map*, not the image, and is applied identically to all methods.
- **Patch-level localisation:** the raw grid itself; useful when masks are coarse.
- **Segmentation:** threshold the map at τ_{pix}; remove regions below a minimum area; report each connected region with area, peak score and mean score.
- **Bounding boxes:** the tight box of each region, for comparison with box annotations (Wafer Surface Defect).
- **Multi-scale localisation:** compute maps from two feature levels and combine them (maximum or mean), so that small and large defects are both visible; ablated in Experiment H.

Annotation needed for each output is listed in Section 17.8. Localisation is reported in the target's own coordinates, which is guaranteed by the "warp the reference" design.

18.8 Stage 8: Unknown defect detection

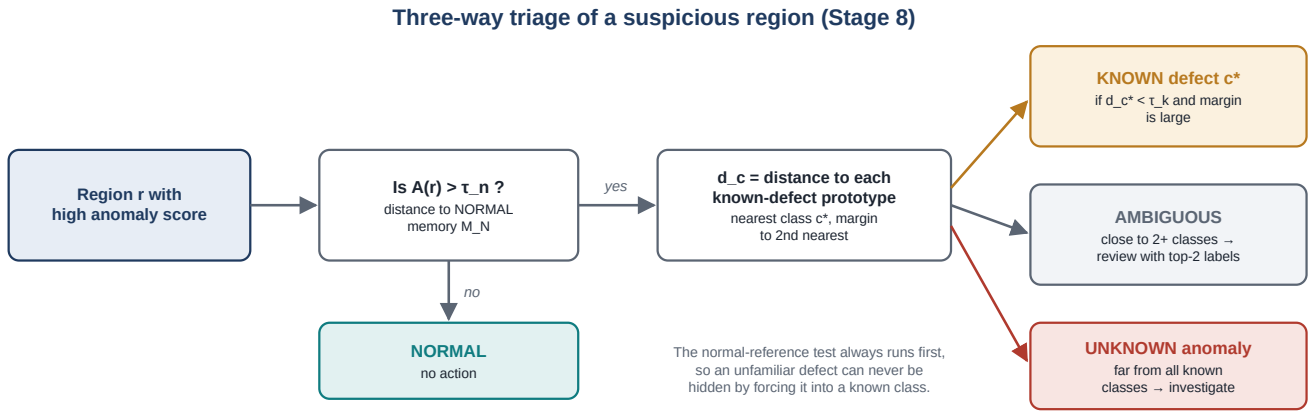


Figure 16. Three-way triage. The normal-memory test always runs first. Only regions that are anomalous are compared with known-defect prototypes; a region far from all known classes is reported as an unknown anomaly, and a region close to two classes is sent to review as ambiguous.

The decision for each anomalous region *r* uses two distances: to the normal memory (is it abnormal?) and to known-defect prototypes (is it something we know?).

1. **Known normal patterns:** $A(r) \le \tau_n \rightarrow$ normal.
2. **Known defect types:** $A(r) > \tau_n$ and the nearest prototype distance $d_{c*} < \tau_k$ with a clear margin to the second-nearest class $\rightarrow$ known class $c*$.
3. **Unknown suspicious patterns:** $A(r) > \tau_n$ and $d_{c*} \ge \tau_k \rightarrow$ unknown anomaly; low margin $\rightarrow$ ambiguous (review with top-2 candidates).

τ_k is set per class from the spread of that class's own examples (e.g. the maximum leave-one-out distance times a safety factor), so classes with few, tight examples get tight boundaries, as in open-set recognition [32]. Because step 1 depends only on normal references, an unknown defect is always flagged as anomalous even if triage is wrong, which follows the design rule of [34]. Unknown regions can later be clustered for engineer review, similar to novel class discovery [57], but that is an extension.

18.9 Stage 9: Wafer-level analysis

Image-level defects vs wafer-map failure patterns. An image-level defect is a localised abnormality inside one die (a particle at pixel 412, 87). A wafer-map failure pattern is a spatial arrangement of abnormal *dies* across the wafer (an edge ring). They are different findings, computed from different data, and reported separately.

- **Die indexing:** (row, column) from die extraction or metadata; wafer coordinates in mm from the die pitch.
- **Wafer-map visualisation:** per-die image score A_{ij}, per-die status, and per-die defect count/area (Figure 15).
- **Defect clustering:** DBSCAN on coordinates of suspicious dies; spatial autocorrelation (e.g. join-count statistics) to test whether failures are clustered or random.
- **Repeated spatial patterns:** compare the binary suspicious-die map with WM-811K-style pattern classes using a wafer-map classifier trained on WM-811K, as an optional context label.
- **Context score (RQ7, optional):** $A'_{ij} = A_{ij} + \lambda \cdot \text{mean}(A \text{ of neighbouring dies})$, to test whether spatial context improves die ranking. Risk: context can raise false alarms for good dies inside a cluster, so it is evaluated, not assumed.
- **Repeater detection:** an anomaly at the same *within-die* position on many dies suggests a systematic (reticle- or design-related) cause, and also warns that such a defect may have contaminated the references.

18.10 Stage 10: Engineer feedback and adaptation

Table 16. Adaptation options compared.

Option	What changes	Speed	Reversible	Forgetting risk	Contamination risk	Verdict
Reference-memory update	Normal memory items	Instant	Yes (versioned)	None for weights; memory can drift	High if wrong "normal" labels are admitted; mitigated by gates	**Implement first**
Prototype update	Known-defect prototypes	Instant	Yes	Low	Medium (mislabelled class)	Implement second
Replay-based continual learning	Encoder or head weights, with stored old samples	Minutes–hours	Only by checkpoint	Reduced by replay	Medium	Optional extension
Periodic fine-tuning	Encoder weights	Hours	Checkpoint	High without replay [9]	Medium	Optional, gated
Test-time adaptation	Normalisation statistics	Online	Partly	Medium	High (adapts to defects too)	Not for core

Safe feedback workflow (recommended first):

1. **Record:** every engineer action (confirm, reject, correct location, assign label, mark unknown) is stored with reviewer ID, time, die ID and the memory version that produced the result.
2. **Route by action:**
   - *Reject (false alarm)* → candidate **normal** patches from the flagged region enter a **quarantine buffer**.
   - *Confirm + label* → region features become candidate **known-defect** items for class *c*.
   - *Correct location* → corrected mask defines which patches are defect and which are normal.
   - *Mark unknown* → region stored in an **investigation set**; it never enters the normal memory.
3. **Gate:** a candidate normal patch is admitted only if (a) it is novel (its distance to M_N exceeds the median nearest-neighbour distance among existing normals, a self-calibrating gate following [10]), (b) it is *not* close to any known-defect prototype, and (c) a second reviewer confirmed it, or the same pattern was independently rejected as a false alarm on at least *n* different dies.

4. **Version and regression check:** admitted items create a candidate memory M_{t+1}. It is evaluated on a fixed held-out validation set of normal and defective dies (never used for feedback). It is accepted only if recall at the operating threshold does not drop and AUROC does not decrease beyond a tolerance; otherwise it rolls back to M_t.
5. **Drift monitoring:** track the memory size, the share of feedback-derived items, the false-alarm rate on a fixed set of golden normal dies, and the detection rate on a fixed set of known defects (a "canary" set). Alarm if either drifts.

> **WHY MEMORY UPDATES FIRST**
>
> They need no GPU training, act immediately, and are fully auditable and reversible. They also directly test the question of the research brief, "does feedback improve detection or merely memorise samples?", because a held-out protocol can measure improvement on dies that never generated feedback. Fine-tuning is only worth adding if memory updates plateau.

19 Mathematical Formulation

This section uses only distances, minima and averages. Every symbol is explained in plain English right after it appears.

19.1 Symbols

Table 17. Symbols used in the formulation.

Symbol	Meaning in plain English
$R = \{r_1, \ldots, r_K\}$	The reference set: K images of normal dies that the engineer provides.
x	The target die image we want to inspect (height H, width W pixels).
T, $\hat{T}$	The true and the estimated geometric transform (rotation angle θ, shift t, scale s) between a reference and the target.
f	The frozen feature encoder. f(x) is a grid of h × w feature vectors, each with C numbers. $f_p(x)$ is the vector at grid position p.
$M = M_N \cup M_D$	The reference memory: normal patch features M_N, plus optional known-defect prototypes $M_D = \{\mu_c\}$.
$d(a, b)$	Distance between two feature vectors, here Euclidean: d(a, b) = ‖a − b‖. Small = similar.
$s(p)$	Anomaly score of the patch at position p.
S	The spatial anomaly map: s(p) for every position, upsampled to H × W pixels.
$A(x)$	The image-level anomaly score of the whole die.
τ, τ_{pix}, τ_k	Thresholds: for image decisions, for pixels, and for accepting a known class.
U	The feedback update function, which turns the memory at time t into the memory at time t + 1.

19.2 Conceptual flow

Reference dies R → **feature extraction** $f(r_k)$ → **reference memory** M → **target features** f(x) → **feature comparison** d(·, M) → **anomaly map** S → **defect decision** (A(x) > τ ?)

19.3 Enrolment

$$M_N = \bigcup_{k=1..K} \{\, f_p(\, \hat{T}_k(r_k)\,) : p \text{ valid} \,\} \tag{1}$$

In words: align each reference, run it through the encoder, and put every valid patch feature into the memory. "Valid" excludes border regions with no image data after warping. If the memory is large, a coreset subset is kept.

19.4 Anomaly scoring

$$s(p) = \min_{m \in M_N(p)} \| f_p(x) - m \| \tag{2}$$

In words: for the patch at position p of the target, find the most similar normal patch in the memory and take the distance. $M_N(p)$ is either the whole normal memory (global search) or only the reference patches within a window around p (aligned search). A large distance means "nothing like this exists in the references".

$$S = G_\sigma * \text{Upsample}(\, [\, s(p)\,]_p\,), \quad A(x) = \text{mean of the top-}q\%\text{ values of } s \tag{3}$$

In words: the anomaly map is the grid of patch scores, enlarged to the image size and lightly smoothed (G_σ is a Gaussian blur of width σ applied to the map, not to the image). The die's score summarises its worst patches; q = 1% is a typical choice, and q → 0 gives the maximum.

$$\text{decision}(x) = \text{anomalous if } A(x) > \tau, \quad \tau = \text{Quantile}_{1-\alpha}\{\, A(x') : x' \text{ normal, held-out} \,\} \tag{4}$$

In words: the threshold is set so that only a fraction α (e.g. 1%) of normal validation dies would be flagged. Pixels whose map value exceeds τ_{pix} (chosen the same way on pixel scores) form the predicted defect mask.

19.5 Known / unknown triage

$$c^* = \text{argmin}_c \| g(r) - \mu_c \|, \quad y(r) = \text{known:}c^* \text{ if } \|g(r) - \mu_{c^*}\| < \tau_{k,c^*}, \text{ otherwise unknown} \tag{5}$$

In words: g(r) is the average feature of an anomalous region r; μ_c is the prototype (average feature) of known class c. The region takes the nearest class only if it is close enough to it; otherwise it is "unknown".

19.6 Defect preservation ratio (proposed evaluation)

$$\text{DPR}(P) = s_{\text{defect}}(\text{ pipeline with step } P \text{ }) \,/\, s_{\text{defect}}(\text{ reference pipeline without } P, \text{ ideal alignment }) \tag{6}$$

In words: inject a defect with a known mask, and measure its peak (or mean) anomaly score with and without a processing step P (an alignment variant, a denoising filter, a resize). DPR ≈ 1 means the step keeps the defect as visible as before; DPR < 1 means the step weakens it. Reported by defect size and type. Because the injected defect is known exactly, DPR isolates the effect of P from the rest of the pipeline.

19.7 Feedback update

$$M_{t+1} = U(M_t, F_t) = (M_{N,t} \cup \{ z \in Z_t^{\text{normal}} : \text{gate}(z) \}) \cup (M_{D,t} \oplus Z_t^{\text{defect}}) \tag{7}$$

$$\text{gate}(z) = [\min_{m \in M_{N,t}} \|z - m\| > \delta_t] \wedge [\min_c \|z - \mu_c\| > \tau_k] \wedge [\text{verified}] \tag{8}$$

In words: F_t is the batch of engineer decisions; Z_t^{normal} are patch features from regions the engineers called normal, and Z_t^{defect} features from confirmed, labelled defects. A candidate normal patch z enters the memory only if it is genuinely new (farther than δ_t, the median nearest-neighbour distance among current normal patches), is not similar to a known defect, and has been verified. Defect features update the class prototypes ($\oplus$ = add the example and recompute the class mean). M_{t+1} is kept only if the held-out regression check passes; otherwise $M_{t+1} = M_t$.

19.8 Worked example of feature-based anomaly scoring

Use tiny 2-number features to keep the arithmetic visible. Suppose K = 2 references give four normal patch features at one aligned position window:

$$M_N(p) = \{ (1.0, 2.0), (1.2, 1.9), (0.9, 2.1), (1.1, 2.2) \}$$

Normal target patch $f_p(x) = (1.05, 2.05)$. Distances: to (1.0, 2.0) = √(0.05² + 0.05²) = 0.071; to (1.2, 1.9) = 0.212; to (0.9, 2.1) = 0.158; to (1.1, 2.2) = 0.158. Minimum: **s(p) = 0.071**.

Defective target patch (a bright particle raises both feature values) $f_p(x) = (2.0, 3.0)$. Distances: 1.414, 1.360, 1.421, 1.204. Minimum: **s(p) = 1.204**.

Threshold. Scoring held-out normal patches gives values between 0.03 and 0.25; with α = 1% suppose $\tau_{pix} = 0.30$. The first patch (0.071) is normal; the second (1.204) is anomalous, and it is about 4× above the threshold, which is a confident decision.

Triage. Known prototypes: $\mu_{particle} = (2.1, 3.1)$ with $\tau_k = 0.4$; $\mu_{scratch} = (0.2, 0.5)$ with $\tau_k = 0.4$. Distance of the region (2.0, 3.0) to $\mu_{particle} = 0.141 < 0.4$ → **known: particle**. A region at (3.5, 0.8) would be 2.69 from the particle prototype and 3.31 from the scratch prototype, beyond both τ_k → **unknown anomaly**, even though it is clearly anomalous (its distance to normal memory is large).

RUNNABLE VERSION

The same computation, with real (toy) images, multi-scale patch descriptors, a calibrated threshold and rotation alignment, is in `prototype/toy_reference_memory.py`. It runs in about 10 seconds on a CPU and produced Diagrams 5 and 6. It uses synthetic dies, so its numbers are illustrations, not results.

20 Experimental Design

Eight experiments, A–H, test the research questions. All are designed so that they can be run, and so that a negative outcome is informative. **No results are reported in this dossier; none have been produced.** This section, together with Sections 21–22, forms the experimental protocol (Artifact 7; also in `artifacts/07_experimental_protocol.md`).

20.1 General protocol

- **Repetitions:** every few-shot setting is run with 5 independent random draws of the reference set (fixed, logged seeds). Report mean ± standard deviation and 95% bootstrap confidence intervals over test images.
- **Statistics:** paired comparisons between methods use the same reference draws; significance with a Wilcoxon signed-rank test across categories or draws, Holm-corrected for multiple comparisons (as in [10]).
- **Thresholds and hyper-parameters** (τ, window size, coreset ratio, q) are tuned on validation data only, then frozen before testing.
- **Reporting:** every table states the data tier (T0–T4); synthetic results are never merged with real-data results.

20.2 Data splits without leakage

Table 18. Split rules. A split manifest lists file hashes per split; the evaluation script refuses to run if any test hash appears in a reference, calibration or feedback set.

Dataset	Split rule
MVTec AD / VisA	Official train/test split. References drawn only from the official normal training set; calibration normals from the remaining training normals.
Carinthia-S	Stratified split by class into reference-source / validation / test (e.g. 20 / 20 / 60%), after near-duplicate clustering by perceptual hash so re-captures of one site stay together. Normal references are taken from defect-free regions of reference-source images only.
Wafer Surface Defect	Normal dies split into reference pool / calibration / test; defective dies only in validation and test. Near-duplicate clustering as above.
Synthetic wafers (T3)	Split by **wafer** and by **design**: same-wafer tests use references and targets from one wafer but never the same die; cross-wafer tests use disjoint wafers; cross-type tests use a design never seen in enrolment or tuning.
Partner data (T4)	Split by lot and wafer; never place dies of one wafer in both tuning and test; keep the acquisition date order (train on earlier, test on later) to mimic deployment.
Feedback experiments	Three disjoint pools: **feedback stream** (dies whose review decisions are fed back), **held-out evaluation** (never fed back) and **canary set** (fixed normals + known defects to monitor drift).

20.3 Experiments

Table 19. Experiment plan (Artifact 7).

Exp.	Question	Data	Procedure	Primary metrics
A Few-shot	RQ1: performance vs K	T0 MVTec AD; T1 Carinthia-S; T2 Wafer Surface Defect	$K \in \{1, 2, 5, 10\}$, 5 draws each; encoders WRN-50 and DINOv2; with and without memory QC	Image AUROC, AUPRC; pixel AUROC, AUPRO (where masks exist)
B Rotation robustness	RQ2: alignment vs augmentation vs invariance; defect preservation	T0, T1, T2 with controlled transforms; T3 synthetic	Angle sweep 0–360° in 15° steps plus 90° multiples; translation ±10% of die size; scale ±10%; illumination gradients. Pipelines: none / warp-target / warp-reference / rotation-augmented memory / GraphCore-style invariance. Inject defects (1–20 px) for DPR	AUROC vs angle; false alarms on normals per angle; DPR by size; target registration error
C Cross-die (same wafer)	RQ3: references from some dies transfer to other dies of the same wafer	T3 synthetic; T4 if available	Enrol K dies of a wafer (spread vs clustered selection); test on all other dies of that wafer	AUROC, AUPRO; gap to in-sample upper bound

Exp.	Question	Data	Procedure	Primary metrics
D Cross-wafer and cross-type	RQ3: transfer to other wafers and designs	T3 (wafer shift levels; new designs); T4; cross-dataset proxy (T2 → T1)	Enrol on wafer A, test on wafers B…; then on a new design with 0, 1, 2, 5 new references; optional per-wafer feature normalisation	Per-split AUROC/AUPRO; performance vs shift level; references needed to recover
E Unknown defects	RQ4: unknown vs known triage	T1 Carinthia-S (6 classes); T2 (4 classes); T3	Leave-one-class-out: build prototypes from all classes except one; test regions of the held-out class. Repeat for every class	Unknown-detection AUROC; open-set F1; forced-assignment rate; known-class accuracy
F Localisation	RQ5: map quality	T0, T1 (masks), T2 (boxes), T3	Single- vs multi-scale maps; native-resolution tiling vs resizing; threshold at validation-selected τ_{pix}	Pixel AUROC, AUPRO, Dice, IoU (masks); box IoU / AP (boxes); all by defect-size bin
G Feedback adaptation	RQ6: improvement without contamination or forgetting	T0 MVTec AD (comparison with [10]); T1; T3	Simulated review stream from ground truth with controlled error rates (0%, 5%, 10%, 20% wrong labels; 10% wrong locations). Arms: no update / ungated update / gated update / gated + second review / gated + rollback	ΔAUROC and false-alarm rate on held-out set over time; forgetting rate on canary defects; contamination rate
H Ablations	Contribution of each component	Best tier per component	Remove one component at a time (Section 22.2)	Same as the affected experiment

20.4 Experiment G in more detail

Because the risk of fooling ourselves is highest here, the protocol is spelled out. (1) Start from memory M_0 built from K references. (2) Process the feedback stream in batches of *b* dies; for each flagged or sampled die, generate the "engineer decision" from ground truth, then corrupt a controlled fraction of decisions (label flips, shifted locations, defect called normal). (3) Apply the update arm. (4) After each batch, evaluate on the held-out set and the canary set, never on dies from the stream. (5) Report learning curves, final gain, worst-case dip, and how often rollback fired. A useful negative control is "update with random normal patches": if it helps as much as the feedback, the gain comes from simply having more memory, not from the feedback.

21 Evaluation Metrics

Table 20. Metrics and the tasks they apply to.

Metric	Definition (simple)	Task	Notes
AUROC (image)	Probability that a random defective die scores higher than a random normal die	Detection (A–E, G)	Threshold-free; optimistic when anomalies are rare
AUPRC	Area under precision–recall curve	Detection with rare defects	Preferred headline metric for imbalanced data
Precision, Recall, F1	Share of alarms that are real; share of real defects found; harmonic mean	Operating point at τ	Report recall at fixed false-alarm rate (e.g. 1%)
Pixel AUROC	AUROC computed over all pixels	Localisation (F)	Dominated by large defects and by easy background pixels
AUPRO / PRO	Mean per-region overlap up to 30% false-positive rate; every defect region weighs equally [23]	Localisation (F)	Fairer to small defects than pixel AUROC
Dice, IoU	Overlap between predicted and true masks at τ_{pix}	Segmentation (F)	Need masks; sensitive to threshold
Box IoU / AP	Overlap of predicted and true boxes; detection average precision	Box localisation (T2)	When only boxes exist
Unknown-detection AUROC; open-set F1	How well held-out classes are labelled unknown; F1 over known classes + "unknown"	Unknown defects (E)	Also report forced-assignment rate
DPR	Defect score with a processing step ÷ without it (Eq. 6)	Rotation / preprocessing (B)	Proposed metric; by defect size
Target registration error	Mean pixel distance between true and estimated positions of points after alignment	Alignment (B)	Needs known transforms (T3) or fiducials
Inference time	Wall-clock ms per die (feature extraction + search), median of 100 runs, hardware stated	All	Separate enrolment time
Memory consumption	Memory bank size (MB, number of vectors) and peak GPU/CPU RAM	All	Grows with K and feedback
Adaptation performance	Held-out AUROC/AUPRC gain after n feedback batches	Feedback (G)	Only on dies never fed back
Forgetting rate	Drop in detection of canary known defects (or old categories) after updates	Feedback (G)	Must stay ≈ 0
Contamination rate	Share of admitted normal items that come from true defects	Feedback (G)	Measurable only in simulation

22 Baselines and Ablation Studies

22.1 Baselines

Table 21. Planned baselines. Implementations: official code where available, otherwise Anomalib [70].

Baseline	Why it is included	Experiments
PatchCore [1]	Canonical memory-bank method; the system's direct ancestor	A–H
AnomalyDINO [3]	Strongest simple few-shot baseline	A, B, D, F
RegAD [6]	Learned registration for few-shot AD (alignment alternative)	A, B, D
GraphCore [7] or "augmentation + PatchCore"	Rotation-invariance alternative	B
WinCLIP [5]	Zero/few-shot with language priors; cross-type reference point	A, D
Classical die-to-die difference	Industrial-style pixel subtraction after alignment; sanity check	B, F
Supervised U-Net segmentation	Upper bound when many labelled masks exist (Carinthia-S)	F, E (known classes)
Reverse Distillation / EfficientAD [21, 22]	Full-data upper bound and latency reference (optional)	A (K = all)
Abbas et al. memory correction [10]	Closest feedback method; ungated vs gated comparison	G

22.2 Ablation studies

Table 22. Ablations requested in the research brief and what each tests.

Ablation	Implementation	Expected effect (hypothesis)
Without geometric alignment	Skip Stage 3; global nearest-neighbour search only	Large AUROC drop for rotated targets; little change at 0°
Warp target instead of references	Apply $\hat{T}^{-1}$ to the target	Lower DPR for defects ≤ 3 px
Without reference memory	Replace memory matching by a reference-free score (e.g. intrinsic self-similarity, in the spirit of [45]) or by raw pixel difference	Shows what the memory contributes; may win on unpatterned SEM
Without multi-scale features	Single feature layer	Lower AUPRO for small defects
Without unknown-defect handling	Closed-set nearest prototype for every anomalous region	High forced-assignment rate on held-out classes
Without feedback adaptation	Memory fixed at M_0	Baseline for Experiment G
Without feedback gate / rollback	Admit all feedback	Degradation under noisy feedback
Without wafer-level context	$\lambda = 0$ in the context score	No change on random-defect wafers; lower AUPRC on clustered ones
Without memory QC	Admit all reference patches	Sensitivity to a defective reference

23 Implementation Architecture

The implementation is a **research prototype**: Python modules, configuration files and scripts that reproduce every experiment. There is no front end or production service.

23.1 Tools

Table 23. Software stack and the role of each tool.

Tool	Role in this project
Python 3.11	Language for all modules and experiments.
PyTorch + torchvision / timm	Load frozen pretrained encoders (WRN-50, DINOv2), extract multi-scale feature maps on GPU or CPU.
FAISS [37] (or scikit-learn NearestNeighbors)	Fast nearest-neighbour search in the reference memory.
OpenCV	Image I/O, ECC alignment (`findTransformECC`), warping, ORB features, connected components.
scikit-image	Phase correlation (`phase_cross_correlation`), log-polar transforms for Fourier–Mellin, region properties.
NumPy / SciPy	Array maths, Gaussian filtering, statistics, bootstrap confidence intervals, Wilcoxon tests.
scikit-learn	AUROC, AUPRC, precision–recall, DBSCAN for wafer clusters, isotonic calibration.
Albumentations	Controlled, logged geometric and photometric transforms for rotation and illumination experiments and synthetic data.
Anomalib [70]	Reference implementations of PatchCore, PaDiM, EfficientAD and others for baselines; AUPRO implementation.
Matplotlib	Anomaly maps, overlays, curves, wafer maps.
Hydra or plain YAML + pydantic	Configuration of every experiment; configs are saved with results.
MLflow or CSV/JSON logs	Experiment tracking (parameters, metrics, artefacts, git commit).
pytest	Unit tests for metrics, splits (leakage check) and alignment on synthetic transforms.

23.2 Project directory structure

```
research/
├── configs/                # YAML: datasets, encoders, experiments A–H
├── datasets/               # loaders + split manifests (hash lists) per dataset
│   ├── carinthia_s.py  wafer_surface_defect.py  mvtec.py  wm811k.py
│   └── synthetic/          # patterned-die and wafer generator (T3)
├── preprocessing/          # quality checks, illumination/contrast normalisation, die extraction
├── registration/           # coarse 90° search, Fourier–Mellin, ECC, warp-reference, validity masks
├── feature_extraction/     # frozen encoders, multi-scale patch features, feature cache
├── reference_memory/       # memory bank, coreset, windowed k-NN, QC, versioning
├── anomaly_detection/      # patch scores, image scores, normalisation, thresholds
├── localization/           # map upsampling, regions, boxes, masks
├── unknown_detection/      # prototypes, open-set triage, ambiguity
├── feedback/               # feedback records, simulator, gate, quarantine, rollback
├── wafer_analysis/         # die indexing, wafer maps, clustering, context score
├── evaluation/             # metrics (AUROC, AUPRO, DPR, forgetting), statistics
├── visualization/          # overlays, curves, wafer maps, gallery
├── experiments/            # run_A_fewshot.py … run_H_ablation.py
├── notebooks/              # exploration only; never the source of reported numbers
├── results/                # one folder per run: config.yaml, metrics.json, maps/, log.txt
└── tests/                  # pytest: leakage, metrics, alignment
```

23.3 Modules, inputs and outputs

Table 24. Module interfaces.

Module	Input	Output	Key functions
datasets	Raw files, split manifest	`Sample(image: float32[H,W], mask: bool[H,W]|None, label, meta: dict)`	`load_split()`, `check_leakage()`
preprocessing	Sample	Normalised image, `quality: dict`	`normalize()`, `quality_check()`, `extract_dies()`
registration	target image, reference images	$\hat{T}$ (2×3 matrix), warped references, validity mask, alignment score	`coarse_rot90()`, `fourier_mellin()`, `ecc_refine()`, `warp_reference()`
feature_extraction	Image(s)	`float16[h,w,C]` per scale; cached to disk by image hash	`Encoder.extract()`
reference_memory	Feature maps + validity masks + metadata	`Memory` object (FAISS index, item metadata, version)	`enrol()`, `query(window=)`, `add()`, `remove()`, `snapshot()`, `rollback()`
anomaly_detection	Distances	Patch map `[h,w]`, image score, z-score	`score_patches()`, `image_score()`, `calibrate_threshold()`
localization	Patch map, τ_{pix}	Pixel map `[H,W]`, list of regions (bbox, area, peak, mask)	`upsample()`, `regions()`
unknown_detection	Region features, prototypes	`{normal | known:c | ambiguous | unknown}`, distances	`triage()`
feedback	Results + engineer (or simulated) actions	Feedback records; candidate memory; accept/rollback decision	`simulate()`, `gate()`, `apply_update()`, `regression_check()`
wafer_analysis	Per-die results + (i, j)	Wafer map arrays, clusters, context scores	`wafer_map()`, `cluster()`, `context_score()`
evaluation	Predictions + ground truth	`metrics.json`	`auroc()`, `aupro()`, `dpr()`, `forgetting()`, `bootstrap_ci()`

23.4 Pseudocode: reference enrolment

```
def enrol(reference_images, encoder, cfg):
    template = choose_template(reference_images)          # e.g. the most "central" reference
    items = []
    for r in reference_images:
        q = quality_check(r)
        if not q.ok: log.warn("reference rejected", q); continue
        r = normalize(r)                                  # global contrast only, no smoothing
        T = register(template, r)                         # canonical pose of this reference
        r_c, valid = warp(r, T)                           # references may be warped: they are clean
        for scale, fmap in encoder.extract(r_c).items():
            items += patches(fmap, valid, meta={ref_id, scale, position})
    items = memory_qc(items)       # leave-one-out outlier check, de-duplication, optional coreset
    M = Memory(items, version=0, provenance="enrolment", cfg_hash=hash(cfg))
    M.prototypes = build_prototypes(cfg.known_defects, encoder)   # optional, kept separate
    tau = calibrate_threshold(M, cfg.calibration_normals, alpha=cfg.alpha)
    return M, tau
```

23.5 Pseudocode: inference on a new die

```
def inspect(x, M, tau, encoder, cfg):
    q = quality_check(x)
    if not q.ok: return Result(status="REVIEW: image quality", quality=q)
    x = normalize(x)                                    # same operation as for references
    T, align_score = register(M.template, x)            # coarse 90° + Fourier–Mellin + ECC
    refs_in_x_frame, valid = warp_references(M.reference_images, T)   # target untouched
    feats = encoder.extract(x)                          # multi-scale feature maps of the target
    s_win = M.query(feats, window=cfg.window, aligned_to=T)          # Eq. (2), aligned search
    s_glob = M.query(feats, window=None)                # global search, misalignment check
    S = smooth(upsample(combine_scales(s_win)), cfg.sigma) * valid    # Eq. (3)
    A = top_q_mean(S, cfg.q)
    regions = connected_regions(S > tau.pixel, min_area=cfg.min_area)
    for r in regions:
        r.label, r.distances = triage(region_feature(feats, r), M.prototypes)   # Eq. (5)
    status = ("NORMAL" if A <= tau.image else
              "UNKNOWN" if any(r.label == "unknown" for r in regions) else "SUSPICIOUS")
    conf = confidence(A, tau, agreement(s_win, s_glob), align_score)
    return Result(status, A, S, regions, conf, T, memory_version=M.version)
```

23.6 Pseudocode: feedback update

```
def feedback_round(M, batch, heldout, canary, cfg):
    cand_normal, cand_defect = [], []
    for fb in batch:                                  # fb: action, die, region, label, reviewer
        log.record(fb, memory_version=M.version)
        if fb.action == "reject":                     # false alarm: region is normal
            quarantine.add(features_in(fb.region), fb)
        elif fb.action in ("confirm", "label", "correct"):
            cand_defect += [(features_in(fb.corrected_region), fb.label)]
        elif fb.action == "unknown":
            investigation_set.add(fb)                 # never enters normal memory
    released = quarantine.release(min_votes=cfg.min_votes, second_review=cfg.second_review)
    delta = median_nn_distance(M.normal)              # self-calibrating novelty gate
    for z in released:
        if nn_distance(z, M.normal) > delta and min_proto_distance(z, M.prototypes) > cfg.tau_k:
            cand_normal.append(z)                     # Eq. (8)
    M_new = M.snapshot().add_normal(cand_normal).update_prototypes(cand_defect)   # Eq. (7)
    before, after = evaluate(M, heldout), evaluate(M_new, heldout)
    drift_ok = evaluate(M_new, canary).recall >= evaluate(M, canary).recall - cfg.tol
    if after.recall_at_tau >= before.recall_at_tau and after.auroc >= before.auroc - cfg.tol and
drift_ok:
        return M_new                                  # accepted: version t+1
    log.warn("update rejected, rolling back"); return M
```

23.7 Reproducible experiment configuration

```
# configs/exp_B_rotation.yaml  (full example in artifacts/example_experiment_config.yaml)
experiment: B_rotation
seed: [0, 1, 2, 3, 4]               # 5 reference draws
data: {dataset: carinthia_s, tier: T1, split_manifest: manifests/carinthia_s_v1.json}
references: {K: [1, 2, 5, 10], selection: random_stratified}
encoder: {name: wide_resnet50_2, layers: [layer2, layer3], weights: imagenet1k_v1, input:
native_tiles_448}
memory: {coreset_ratio: 1.0, window: 3, metric: l2}
alignment: {pipeline: [none, warp_target, warp_reference, rot_aug_memory], coarse: rot90_ncc,
            fine: [fourier_mellin, ecc_euclidean], ecc_iters: 200}
transforms: {angles_deg: "0:360:15", translate_frac: 0.1, scale: [0.9, 1.1], illumination_gradient: [0,
0.3]}
defect_injection: {sizes_px: [1, 2, 3, 5, 10, 20], types: [particle, scratch, bridge, missing_via]}
scoring: {image_score: top_q_mean, q: 0.01, map_sigma: 4}
threshold: {alpha: 0.01, source: validation}
metrics: [auroc, auprc, pixel_auroc, aupro, dpr, false_alarm_rate, tre_px, ms_per_die]
logging: {tracker: mlflow, save_maps: 20, git_commit: auto}
```

23.8 Logging and evaluation strategy

- Every run writes `results/<experiment>/<timestamp>_<hash>/` containing the resolved config, the git commit, environment versions, split manifest hashes, `metrics.json`, per-image predictions (CSV) and a sample of anomaly maps.
- Features are cached by image hash and encoder name, so experiments that only change scoring do not recompute features.
- Reported tables are generated by a script from `metrics.json` files, never typed by hand.
- Unit tests: AUPRO and DPR on hand-made cases; alignment recovers known synthetic transforms within a tolerance; the leakage checker fails on a deliberately leaked manifest.
- Compute budget: on one consumer GPU (8–12 GB), WRN-50 features for a 448 × 448 tile take on the order of tens of milliseconds and $K \leq 10$ memories fit easily in memory; everything also runs (slowly) on CPU. This is an estimate to be confirmed in milestone 3, not a measurement.

24 Required Artifacts

Table 25. The twelve required artifacts and where to find them. All files are in the `research-sample-guided-inspection/` folder of the repository.

#	Artifact	In this document	Separate file
1	Research problem definition	Section 6.2	`artifacts/01_problem_definition.md`
2	Novelty matrix	Section 16.1	`artifacts/02_novelty_matrix.csv`
3	Literature review spreadsheet	Sections 13.4, 14	`artifacts/03_literature_review.xlsx` and `.csv`
4	Dataset comparison table	Section 17.6	`artifacts/04_dataset_inventory.csv` (also a sheet in the XLSX)
5	Architecture diagram	Diagram 2 (Section 9)	`figures/dia_02_architecture.svg`, `diagrams/02_architecture.mmd`
6	End-to-end workflow diagram	Figure 17 (Section 25)	`figures/dia_09_workflow.svg`, `diagrams/09_research_workflow.mmd`
7	Experimental protocol	Sections 20–22	`artifacts/07_experimental_protocol.md`, `artifacts/example_experiment_config.yaml`
8	Research risk register	Section 26.1	`artifacts/08_risk_register.csv`
9	Implementation roadmap	Section 25	`artifacts/09_roadmap.csv`, `figures/dia_10_gantt.svg`
10	Glossary	Section 30.1	`artifacts/10_glossary.md`
11	Visual example gallery	Section 30.2	`figures/` (all SVG, editable)
12	Research questions and hypotheses	Section 12	`artifacts/12_research_questions.md`

Mermaid sources of all eight required diagrams are in `diagrams/*.mmd` and can be edited and re-rendered in any Mermaid editor. The SVG versions used in this PDF are generated by `src/figures.py`.

25 Research Roadmap

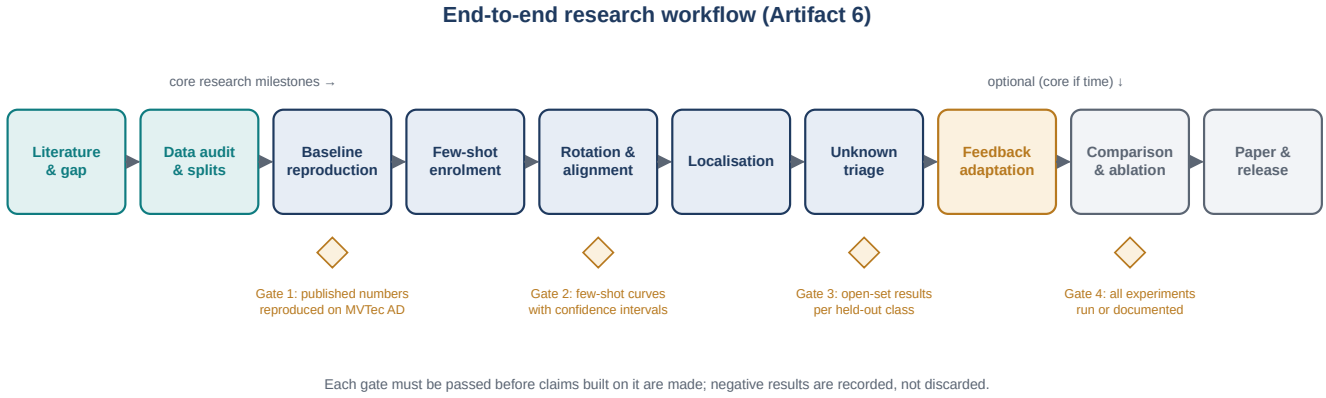


Figure 17. End-to-end research workflow (Artifact 6). Ten phases with four decision gates. A gate must be passed before any claim that depends on it is made.

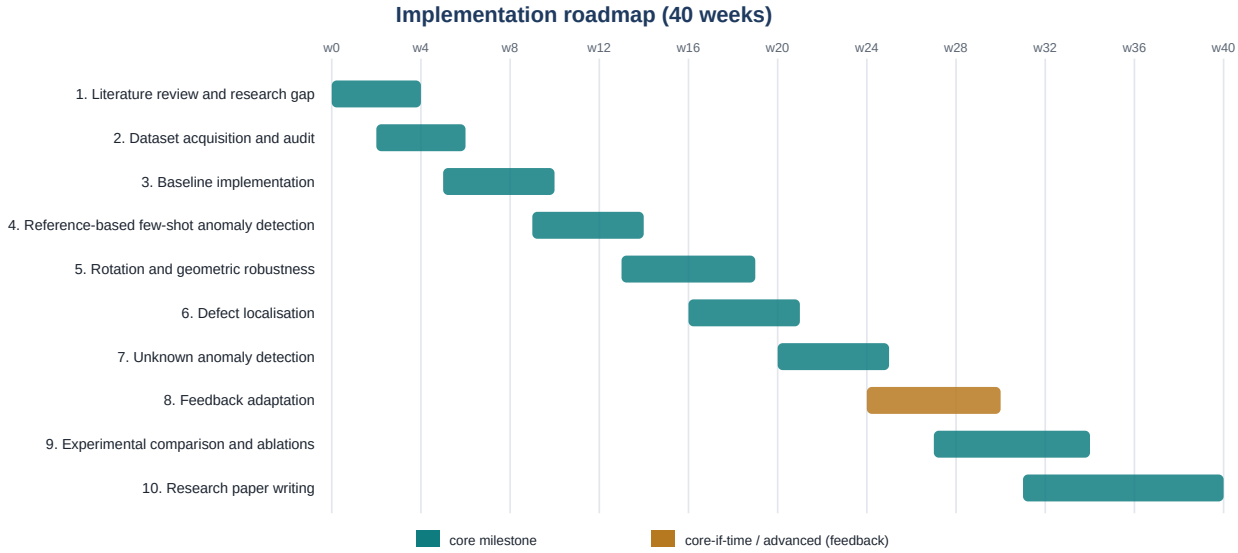


Figure 18. Implementation roadmap over 40 weeks (Artifact 9). Teal bars are core milestones; the amber bar (feedback adaptation) is core if time allows and is otherwise an advanced extension. Phases overlap where they do not depend on each other.

Table 26. Research execution roadmap: tasks, outputs, challenges and completion criteria per phase (Artifact 9).

#	Phase	Main tasks	Expected output	Technical challenges	Completion criteria
1	**Literature review and research gap** core · weeks 1–4 · depends on —	Systematic search (IEEE, Springer, ScienceDirect, CVF, arXiv, SPIE); read the 5 closest papers in full; set up monthly alerts.	Literature spreadsheet (updated), 2-page gap statement, final RQs	Paywalls; fast-moving field	Supervisor agrees the gap statement and RQs
2	**Dataset acquisition and audit** core · weeks 3–6 · depends on 1	Download Carinthia-S, MVTec AD, Wafer Surface Defect, WM-811K; verify counts, classes, masks, licences; build split manifests.	Dataset audit report; versioned manifests with hashes	Access approvals; undocumented annotation details	Every dataset has verified counts, licence and leak-free splits
3	**Baseline implementation** core · weeks 6–10 · depends on 2	Implement PatchCore-style memory with WRN-50 and DINOv2 encoders; reproduce published MVTec AD numbers within tolerance.	Baseline code + reproduction table	Matching reported numbers	Reproduced AUROC within ~1 point of published values for several categories

#	Phase	Main tasks	Expected output	Technical challenges	Completion criteria
4	**Reference-based few-shot anomaly detection** core · weeks 10–14 · depends on 3	Few-shot enrolment (K = 1, 2, 5, 10); threshold calibration; Experiment A on MVTec AD, Carinthia-S, Wafer Surface Defect.	Few-shot curves with confidence intervals	Calibration at K = 1	Experiment A complete with 5 draws per K
5	**Rotation and geometric robustness** core · weeks 14–19 · depends on 4	Implement coarse + fine registration, 'warp the reference', rotation-augmented memory; synthetic defect injection; DPR metric; Experiment B.	Angle-sweep results; DPR analysis	Periodic patterns; interpolation artefacts	Experiment B complete; DPR reported for all pipelines
6	**Defect localisation** core · weeks 17–21 · depends on 4	Multi-scale anomaly maps; size-binned evaluation; Experiment F.	Localisation results and qualitative gallery	Small defects	Pixel AUROC / AUPRO / Dice reported by size bin
7	**Unknown anomaly detection** core · weeks 21–25 · depends on 6	Known-defect memory; three-way triage; leave-one-class-out on Carinthia-S; Experiment E.	Open-set results	Defining known vs unknown fairly	Every class held out once; per-class results
8	**Feedback adaptation** core-if-time · weeks 25–30 · depends on 4, 7	Simulated feedback stream with controlled error rate; gated updates, quarantine, rollback; Experiment G.	Before/after and noisy-feedback results	Avoiding evaluation leakage	Held-out gains and forgetting reported with significance tests
9	**Experimental comparison and ablations** core · weeks 28–34 · depends on 5–8	Full comparison against baselines; Experiments C, D, H; optional wafer-context study (RQ7).	Final result tables; ablation table	Compute time across seeds	All planned experiments run or documented as infeasible
10	**Research paper writing** core · weeks 32–40 · depends on 9	Write paper / thesis; release code and synthetic benchmark; reproducibility checklist.	Paper draft; public repository	Honest reporting of negative results	Supervisor-approved draft; code reproduces tables

CORE VS OPTIONAL

Core research milestones: phases 1–7, 9 and 10, which deliver C2 (defect-preserving alignment), C3 (unknown triage) and the few-shot/transfer protocol (C1). **Core-if-time:** phase 8 (feedback, C5). **Optional advanced extensions:** wafer-level context (C4, RQ7), self-supervised SEM encoder adaptation, test-time adaptation for cross-wafer drift, clustering unknowns into candidate new classes, and partner-data validation (T4) if access is obtained.

26 Risks and Limitations

26.1 Research risk register (Artifact 8)

Table 27. Risk register (Artifact 8). Probability and impact: High / Medium / Low.

#	Risk	Why it matters	Probability	Impact	Mitigation
1	**No public patterned-die dataset with masks and wafer coordinates**	Core cross-die / cross-wafer claims cannot be tested on real data	High	High	Use a tiered data plan: public SEM (Carinthia-S), die-level optical (Wafer Surface Defect), synthetic patterned dies with known ground truth, and seek partner data; report synthetic results as synthetic.
2	**Pretrained features (ImageNet, DINOv2) do not represent SEM textures well**	Few-shot performance may be poor on semiconductor data	Medium	High	Compare several encoders; optional self-supervised adaptation on unlabelled SEM images; report negative results.
3	**Defects smaller than one feature patch are missed**	Tiny defects are often the most critical	High	High	Multi-scale features; higher input resolution or tiling; size-binned evaluation.
4	**Alignment or preprocessing removes or blurs real defects**	Silent false negatives	Medium	High	Warp the reference, not the target; defect preservation ratio (DPR) test with injected defects; no smoothing before scoring.
5	**Periodic layouts make feature-based registration ambiguous**	Wrong alignment by one pitch produces false alarms	Medium	Medium	Coarse alignment from die edges / global correlation; check the residual; fallback to rotation-augmented memory.
6	**K = 1 gives no data to calibrate a threshold**	Threshold is arbitrary	High	Medium	Augmentation-based self-scoring; report threshold-free metrics (AUROC) alongside thresholded ones.
7	**Wrong engineer feedback contaminates the normal memory**	Defects become 'normal' and are missed later	Medium	High	Quarantine buffer, novelty gate, second review for additions, versioned memory with rollback, held-out regression check.
8	**Feedback evaluation inflated by memorisation**	Claimed improvement is not real	High	High	Strict held-out protocol (as in Abbas et al.): updated samples never appear in the evaluation set.
9	**Data leakage between splits (same wafer, rescans)**	Overestimated generalisation	Medium	High	Split by wafer / lot / design; perceptual-hash de-duplication; hash-checked split manifests.
10	**Licence or access restrictions on datasets**	Results cannot be published or reproduced	Medium	Medium	Confirm licences on official records before use; keep a data-access log; prefer openly licensed data.
11	**Limited GPU resources**	Cannot run large backbones or many seeds	Medium	Medium	Frozen backbones, cached features, coreset subsampling, small ViT-S / WRN-50; CPU-feasible baseline.
12	**Novelty overlap with a paper published during the project**	Contribution weakened	Medium	Medium	Monthly literature alert; frame contributions as evaluated hypotheses; position as a semiconductor-specific study.
13	**Unknown-class experiments depend on how classes are defined**	Open-set results may not generalise	Medium	Medium	Leave-one-class-out over every class; report per-class results and variance.
14	**Cross-wafer domain shift larger than expected**	Same strategy fails across wafers	High	Medium	Treat as a finding; test re-enrolment with few new references and feature normalisation.

26.2 Critical feasibility assessment

Is a small reference collection sufficient? For same-wafer inspection of a stable process, probably yes for clear defects: published few-shot results on natural-object benchmarks are strong at K = 1–4 [3, 7]. For subtle defects close to the natural variation, no: K = 1 sees one sample of "normal", so false alarms and thresholds will be poorly controlled. Experiment A measures where the curve flattens; the honest expectation is that K = 5–10 is needed for stable thresholds.

Can normal-only references identify all defect types? No. They cannot identify (i) defects that look normal at the encoder's resolution; (ii) systematic defects present in every reference; (iii) defects whose appearance falls within natural variation (low-contrast stains under strong illumination change); (iv) "logical" defects in which every part looks normal but something is in the wrong place, unless matching is position-aware. The windowed (aligned) search partly addresses (iv); (ii) needs design data or golden references [59].

Are different wafer types different enough to break transfer? Different *designs*, yes: a memory of design X contains no normal example of design Y (Scenario 2b), so it must be re-enrolled. Different *wafers* of the same design differ by process drift and imaging conditions; the size of the effect is an empirical question (Experiment D), and domain-shift studies on SEM classification suggest it is not negligible [43].

Can rotation normalisation remove meaningful defect information? Yes. Warping the target resamples it, which blurs 1–2 px defects and can cut defects near borders; rotation-invariant features cannot distinguish a correctly shaped but wrongly oriented structure. The design (warp references, mask borders) and the DPR metric exist to measure and limit this; it is not assumed to be solved.

Do public datasets support cross-wafer evaluation? No (Section 17.7). Cross-wafer claims rest on synthetic wafers unless partner data is obtained. This is the most important limitation of the study, and it is stated in every result table.

Are pixel-level masks available? Yes for SEM (Carinthia-S) and for natural-object sets; no for the die-level optical set (boxes only). Pixel-level claims on patterned dies come from synthetic data only.

Does feedback improve detection or merely memorise samples? That is exactly what Experiment G tests: improvement is measured only on dies that never generated feedback, with a "random extra normals" control. Abbas et al. showed that naive evaluation inflates gains through memorisation [10]. If held-out gains vanish, the conclusion is that feedback only memorises, which is still a useful negative result.

Can the system run with limited GPU resources? Yes for the core: frozen encoders, cached features and $K \leq 10$ memories fit a consumer GPU, and the toy pipeline runs on a laptop CPU. Large foundation-model baselines (UniVAD, CLIP-based) may be too heavy; they are optional. Self-supervised SEM pre-training is the only component that needs substantial compute, and it is an extension.

26.3 Smallest scientifically meaningful prototype

> **MINIMUM VIABLE STUDY (ABOUT 16–20 WEEKS)**
>
> 1. PatchCore-style memory with WRN-50 and DINOv2 encoders, reproduced on MVTec AD (sanity gate).
> 2. Few-shot curves K = 1, 2, 5, 10 on Carinthia-S and Wafer Surface Defect (Experiment A).
> 3. Rotation sweep with four pipelines and DPR on injected defects, on Carinthia-S and a small synthetic die set (Experiment B, Contribution C2).
> 4. Leave-one-class-out unknown triage on Carinthia-S (Experiment E, Contribution C3).
>
> This already answers RQ1, RQ2 and RQ4 with public data and is publishable as a workshop or conference paper if executed carefully. Everything else builds on it.

27 Expected Contributions

If the plan is carried out, the expected outputs are listed below. These are outputs of the work, not predicted outcomes; each hypothesis may be confirmed or rejected.

1. **An evaluated, open reference-memory inspector for die images** covering K-shot enrolment, alignment, localisation, triage and gated updates, with reproducible configurations.
2. **Evidence on defect-preserving alignment (C2):** a comparison of no alignment, target warping, reference warping, rotation-augmented memories and invariant features under a full rotation sweep, with the defect preservation ratio as a reusable evaluation tool.
3. **Evidence on unknown-defect triage (C3):** leave-one-class-out results on public SEM masks showing how often unfamiliar defects are flagged as unknown versus forced into known classes.
4. **Evidence on safe feedback (C5):** held-out measurements of whether verified memory edits help, and how gating and rollback behave under realistic error rates.
5. **A transfer protocol (C1):** explicit same-wafer / cross-wafer / cross-type splits and a synthetic patterned-wafer benchmark with exact ground truth, released for others to use.
6. **Negative results where they occur**, for example if reference-free intrinsic normality [45] beats reference memories on unpatterned SEM, or if feedback gains vanish on held-out data.

28 Conclusion

The research idea, "give the AI a few sample dies, let it learn their patterns, inspect new dies, show where the defects are, and improve from verified feedback", is technically feasible with existing building blocks. Frozen pretrained encoders and a reference memory provide few-shot learning without retraining; classical registration applied to the references keeps targets untouched; nearest-neighbour distances give interpretable anomaly maps; distance-based open-set triage keeps unknown defects visible; and memory edits make feedback auditable and reversible.

The literature shows that each block exists, so novelty must come from the **combination, the semiconductor setting and the evaluation**: whether alignment preserves defects, whether unknown defects are localised and kept apart from known ones, whether feedback survives realistic mistakes, and how transfer differs between same-wafer, cross-wafer and cross-type inspection. The main limitation is data: no public patterned-die dataset with masks and wafer coordinates exists, so cross-wafer conclusions will rest on synthetic data unless partner data becomes available.

What to do first: reproduce PatchCore on MVTec AD; run few-shot curves on Carinthia-S; build the rotation sweep with "warp the reference" and the DPR metric; then add unknown triage. That sequence produces a meaningful result within about four months and lays the foundation for the feedback study.

29 References

Numbered in order of first citation. Verification codes: **S** = checked in this session through search results quoting the official record (publisher, CVF, ACL Anthology, arXiv, Zenodo, IEEE DataPort); **P** = checked in an earlier verification log in this repository; **K** = standard bibliographic knowledge, not re-fetched (direct access to arXiv, Zenodo and Crossref was blocked). Open every link once before external distribution.

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30 Appendix and Glossary

30.1 Glossary (Artifact 10)

Ablation study
An experiment that removes one component to measure how much that component contributes.

Affine transform
A geometric mapping that combines rotation, scaling, shear and translation; straight lines stay straight.

Anomaly
Anything that differs from the learned notion of 'normal'. Not every anomaly is a defect (it may be noise or a harmless variation).

Anomaly detection
Finding samples or regions that do not look like normal data, usually trained with normal data only.

Anomaly map
An image in which each pixel's value is that location's anomaly score; also called a heatmap.

Anomaly score
A number that says how abnormal a sample (image score) or location (pixel score) is. Higher means more suspicious.

AUPRC
Area under the precision–recall curve. More informative than AUROC when defects are rare.

AUPRO / PRO
Per-region overlap: a localisation metric that gives every defect region equal weight, regardless of its size, integrated up to a false-positive rate of 30%.

AUROC
Area under the ROC curve: the probability that a random anomalous sample gets a higher score than a random normal one. 0.5 = chance, 1.0 = perfect.

Backbone / encoder
The pretrained neural network that converts an image into feature maps.

Bounding box
A rectangle around a defect, given by its position, width and height.

Calibration
Making scores behave like probabilities, so a 'confidence of 0.9' is correct about 90% of the time.

Catastrophic forgetting
When a model updated on new data loses performance on what it had learned before.

Continual learning
Learning from a stream of new data or tasks over time without forgetting earlier knowledge.

Coreset subsampling
Selecting a small, representative subset of memory features that covers the whole feature space.

Cross-die inspection
Inspecting dies that were not used as references, with references enrolled from other dies.

Cross-type inspection
Inspecting a different die design (product) from the one the references came from.

Cross-wafer inspection
Inspecting dies from a wafer different from the reference wafer.

Defect
A physical imperfection on the die (particle, scratch, bridge, open, residue, ...) that may affect yield or reliability.

Defect preservation ratio (DPR)
Proposed in this dossier: anomaly score of an injected defect after a processing step divided by its score without that step. Values below 1 mean the step weakens the defect.

Defect localisation
Saying where a defect is: a heatmap, box or pixel mask.

Die
One rectangular chip on a wafer; after dicing it becomes an individual integrated circuit.

Die-to-die inspection
Classical method that compares a die with neighbouring dies of the same design; differences are defect candidates.

Dice / IoU
Overlap measures between predicted and true masks. IoU = intersection/union; Dice = 2·intersection/(sum of areas).

DINOv2
A self-supervised Vision Transformer whose patch features transfer well to many tasks without fine-tuning.

Domain shift
Change in data distribution between training and use, such as a different tool, layer, product or illumination.

ECC alignment
Enhanced Correlation Coefficient: an iterative, brightness-insensitive method for refining an image alignment.

Enrolment
Adding the features of reference dies to the memory. No network weights change.

Feature map
A grid of feature vectors, one per image location, produced by the encoder.

Feature vector
A list of numbers that describes a patch or an image in a way that makes similar things close together.

Few-shot learning
Learning or adapting from a handful (for example 1–10) of examples.

Fourier–Mellin registration
Estimating rotation and scale with Fourier transforms and log-polar resampling, followed by translation.

Ground truth
The reference annotation (label, box or mask) used to evaluate predictions.

Human-in-the-loop
A workflow in which people verify or correct the model's outputs and their decisions are fed back to the system.

Illumination correction
Removing slow brightness variations caused by the imaging system, not by the sample.

Image registration
Finding the geometric transform that maps one image onto another so that corresponding structures overlap.

Inference
Using a trained or enrolled model on new data.

k-nearest neighbours (kNN)
Finding the k stored items closest to a query; the distance to them is used as a score.

Known-defect memory
A small store of features or prototypes of verified, named defect classes.

Leakage
When information from the test set influences training or tuning, inflating results.

Mahalanobis distance
A distance that accounts for the spread and correlation of the normal data.

Memory bank / reference memory
The stored set of normal (and optionally defect) features that new dies are compared with.

Novelty gate
A rule that only admits new memory items if they are sufficiently different from what is already stored.

Open-set recognition
Classification that can answer 'none of the known classes' instead of forcing a known label.

Optical inspection
Wafer inspection with light microscopy; fast, but with lower resolution than SEM.

Patch
A small square region of an image (or one cell of a feature map).

Pixel-level mask
A binary image in which defect pixels are 1 and normal pixels are 0.

Precision / Recall / F1
Precision = share of alarms that are real defects; recall = share of real defects found; F1 = their harmonic mean.

Prototype
The average feature of a class; used for nearest-prototype classification.

Prototypical network
A few-shot classifier that compares a query with class prototypes.

Quarantine buffer
Temporary store for feedback-derived memory items that are not used until verified.

Reference die
A die image accepted as normal and used for comparison.

Rotation robustness
Giving the same correct decision whatever the die's orientation.

SEM
Scanning Electron Microscope: images surfaces at nanometre resolution with an electron beam; used for defect review.

Siamese network
Two weight-sharing networks that embed two inputs so that their similarity can be compared.

Spatial Transformer Network (STN)
A differentiable network module that learns to warp its input to a canonical pose.

Test-time adaptation (TTA)
Adapting a model on unlabelled test data during inference (for example, updating normalisation statistics).

Threshold (τ)
The score above which a die or pixel is called anomalous.

Unknown anomaly
An anomalous region that does not match any known defect class; it is sent for engineer investigation.

Vision Transformer (ViT)
A network that splits an image into patches and processes them with self-attention.

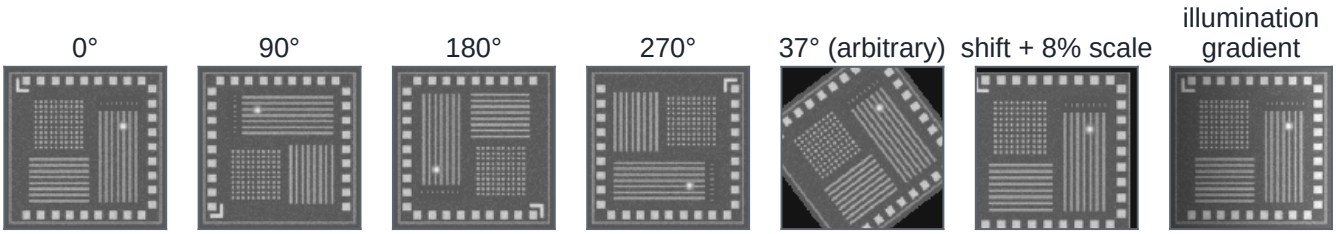
Wafer
A thin disc of semiconductor material (usually silicon) on which many identical dies are fabricated.

Wafer map
A grid showing each die's position and result (pass/fail or bin code); shows spatial patterns across the wafer.

Yield
Fraction of dies on a wafer that work correctly.

30.2 Visual example gallery (Artifact 11)

The gallery collects the conceptual examples. All panels are synthetic renderings produced by `src/figures.py` and `prototype/toy_reference_memory.py`; none is a real SEM or optical measurement, and none is an experimental result.



Synthetic conceptual illustration — not real SEM / optical data

Figure 19. Gallery: geometric and photometric variation of one die (with a particle defect): 0°, 90°, 180°, 270°, an arbitrary 37°, a shift with 8% scale change, and an illumination gradient. Every variation must be handled without hiding the particle. SYNTHETIC

Table 28. Index of the visual examples in this dossier.

Example requested	Where
Full wafer with die divisions; magnified die	Figure 2
Normal vs defective die; microscopic defect examples	Figure 3
Same die under different rotations	Figure 19, Diagram 5 (Figure 9)
Reference dies given to the model; unseen dies inspected	Figure 1, Diagram 3 (Figure 7)
Reference matching and localised defect heat map	Diagram 6 (Figure 11)
Wafer map with suspicious die positions	Figure 15; data-type comparison in Figure 4
Engineer providing feedback	Figure 13, Diagram 7 (Figure 12)

WHY NO REAL SEM OR DATASET IMAGES ARE REPRODUCED

Real images from Carinthia-S, MVTec AD and the Wafer Surface Defect set would add authenticity, but their licences were not all verified (MVTec AD is CC BY-NC-SA 4.0; Carinthia and Carinthia-S licences are unconfirmed). Rather than reproduce images whose reuse terms are unclear, this dossier uses synthetic illustrations and links to the official dataset pages (Section 17). Once licences are confirmed in milestone 2, a real-image gallery can be added with full attribution.

30.3 Final quality-control review (research brief, Part 16)

Table 29. Quality-control checklist and how each item was addressed.

Check	Outcome
Does the document accurately explain the original idea?	Yes. Sections 4 and 8 restate it in the brief's own terms (sample-guided learning, cross-die transfer, rotation, localisation, unknowns, feedback), and Diagram 1 follows the requested flow exactly.
Is same-wafer vs cross-wafer vs cross-type clear?	Yes. Diagrams 3–4, Section 10.2, RQ3, Experiments C–D and separate split rules (Section 20.2). Cross-type is expected to need re-enrolment.
Are raw die images and wafer maps distinguished?	Yes. Section 5.1 (figure and table), Stage 9, and dataset rows that mark WM-811K/MixedWM38 as maps.
Are all papers real and correctly cited?	All 34 analysed papers are cited with official links. 29 were checked against official records in this session (code S) and 5 against an earlier verification log in this repository (code P). Some background references (backbones, registration, continual learning) are cited from standard bibliographic knowledge (code K). The codes are printed with every reference. Full texts could not be opened, so quoted numbers are limited to abstracts and official records, and the reader is asked to verify them.
Are dataset counts and annotation types verified?	Counts for Carinthia, Carinthia-S, WM-811K, Wafer Surface Defect, MVTec AD and VisA come from official records. Unverified items (licences, resolutions, class names, MixedWM38 and MPDD counts) are marked as such.
Is the novelty defensible?	The novelty is framed as a gap in evaluated combinations and protocols (Section 15.2), with each closest paper's overlap stated (Section 15.1). No "first ever" claims are made.

Check	Outcome
Are diagrams consistent with the methodology?	Yes. Diagram 2 matches Stages 1–10; Diagram 5 and Figure 10 match the "warp the reference" design of Stage 3; Diagram 7 matches the workflow of Stage 10.5; Diagram 8 matches the frozen-encoder design.
Are the experiments feasible with available data?	Partly. A, B, E and F run on public data; C and D need synthetic or partner data; G runs on public data with simulated feedback. Stated in Section 17.7.
Are the metrics appropriate?	Yes. Metrics are matched to tasks and annotation levels (Sections 17.8, 21), with AUPRC/AUPRO preferred for rare and small defects.
Are all important assumptions stated?	Key assumptions: references are normal (checked by memory QC); dies of one design share a layout; feedback errors occur at the simulated rates; synthetic data is realistic enough for relative comparisons. Each has a mitigation in Section 26.
Can a beginner follow the pipeline?	Sections 4–5 explain 19 concepts with the simple → example → technical → why pattern; Section 19.8 works through the maths by hand; the toy prototype runs in seconds.
Can a faculty member evaluate the contribution?	Sections 13–16 separate evidence from hypothesis; Section 12 gives falsifiable hypotheses; Sections 20–22 give the protocol.
Can a student start implementing?	Yes. Section 23 gives the tool stack, directory layout, interfaces, pseudocode and a config; Section 25 gives the order of work.
Inconsistencies corrected during review	Worked-example distances recomputed (Section 19.8); diagram labels and layouts fixed after rendering; Diagram 1 and the workflow gates aligned with the roadmap; dataset limitations moved into every relevant table.

30.4 How this document was produced

The document is generated from source: `src/sections/*.html` (text), `src/papers.py`, `src/tables.py` and `src/refs.py` (structured data used for tables, the spreadsheet and references), `src/figures.py` (all figures), and `build.py` (assembly, figure and citation numbering, PDF rendering with headless Chromium, spreadsheet export). Rebuild with `python3 build.py`. The same data feeds the document and the spreadsheets, so they cannot disagree.